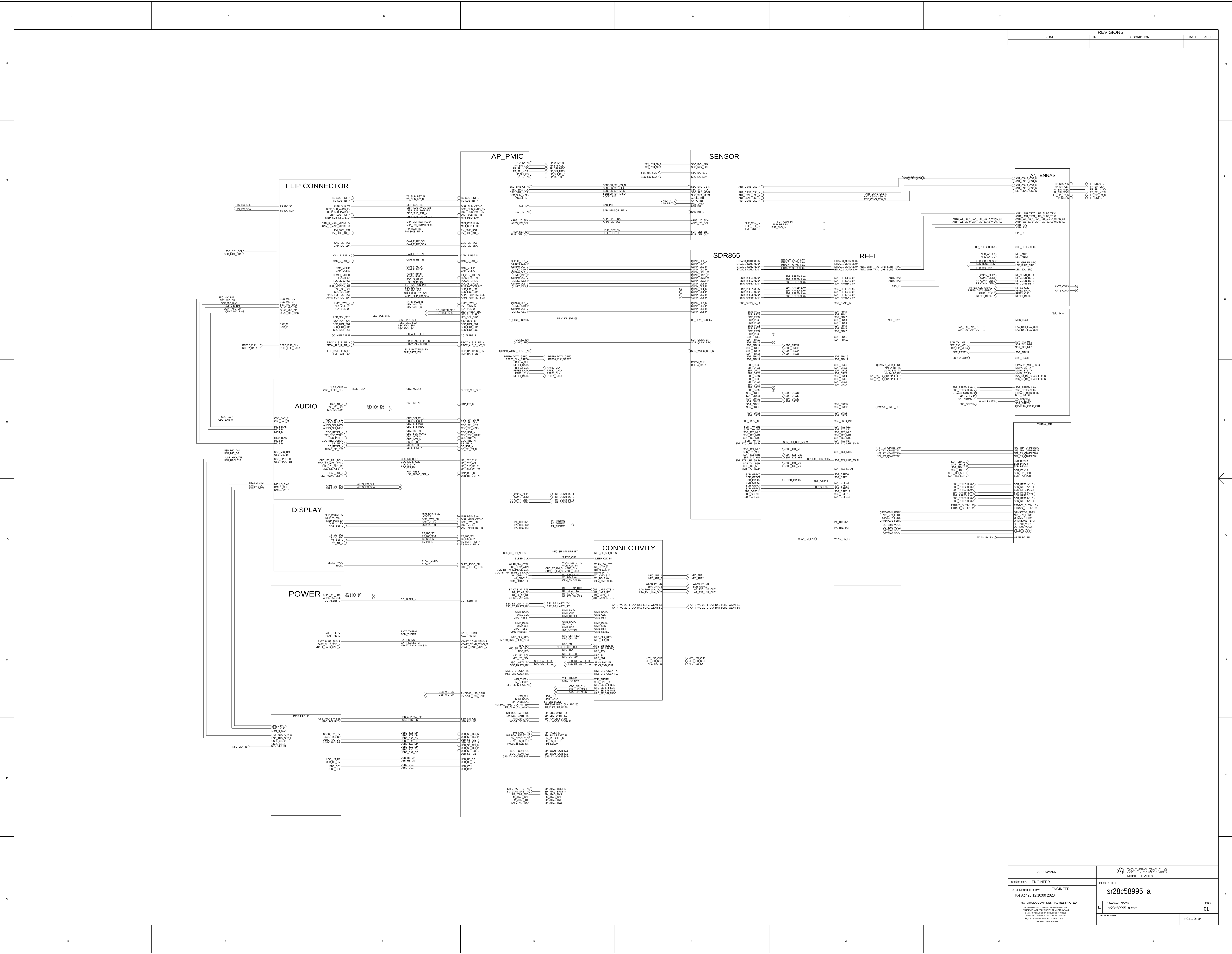
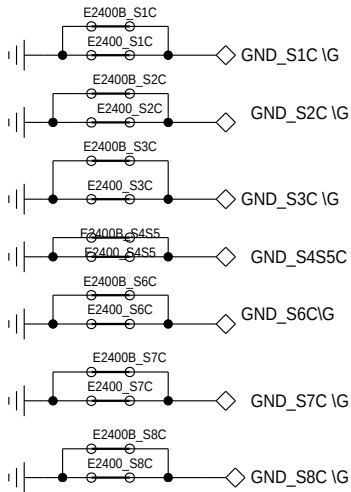
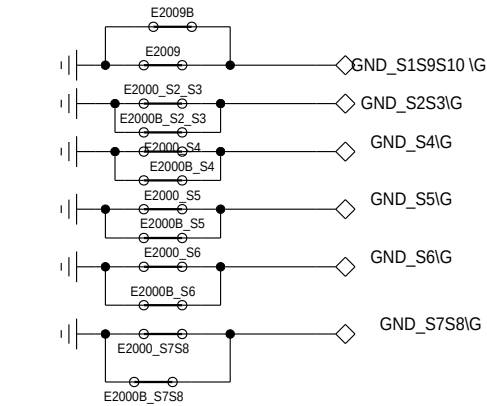
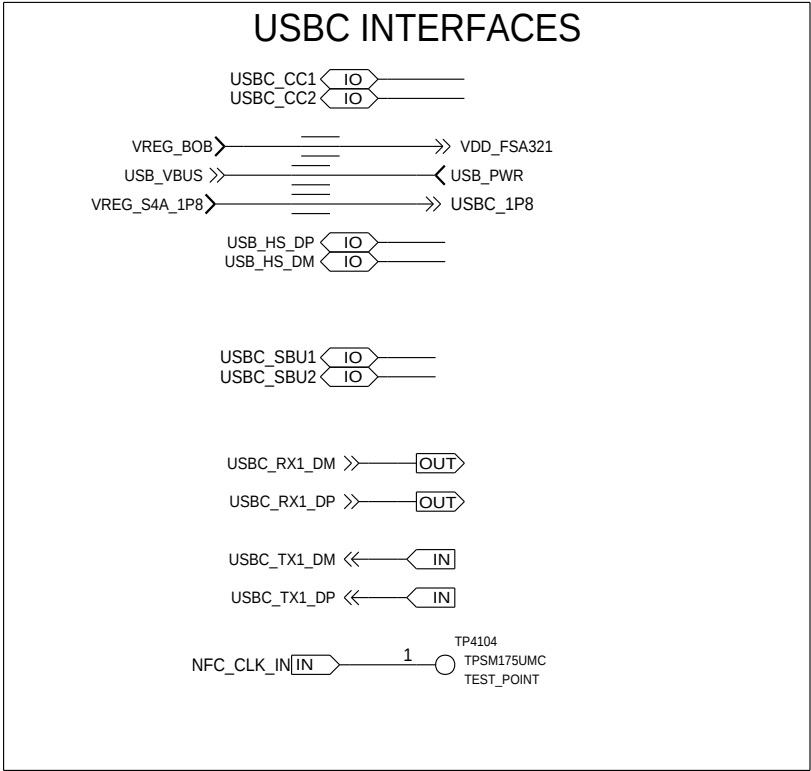
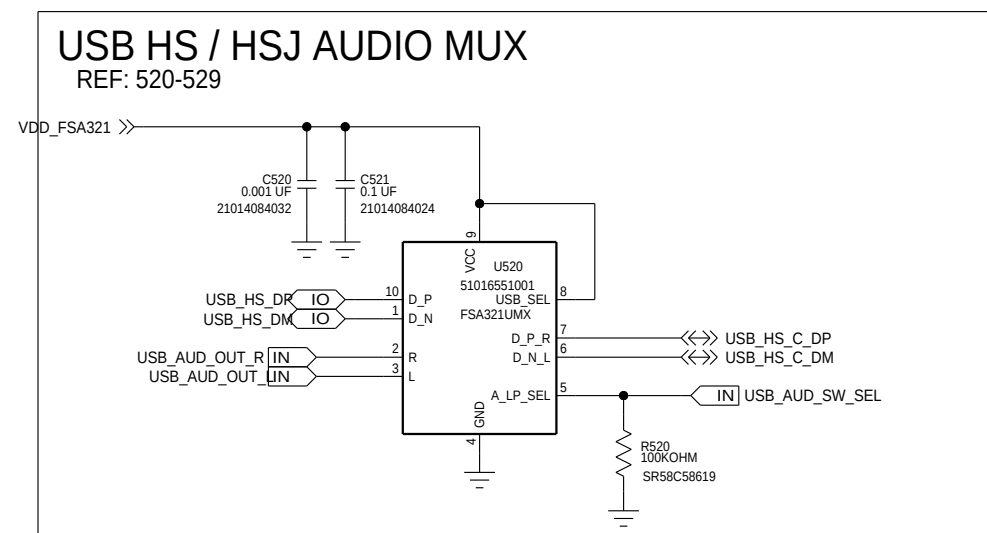
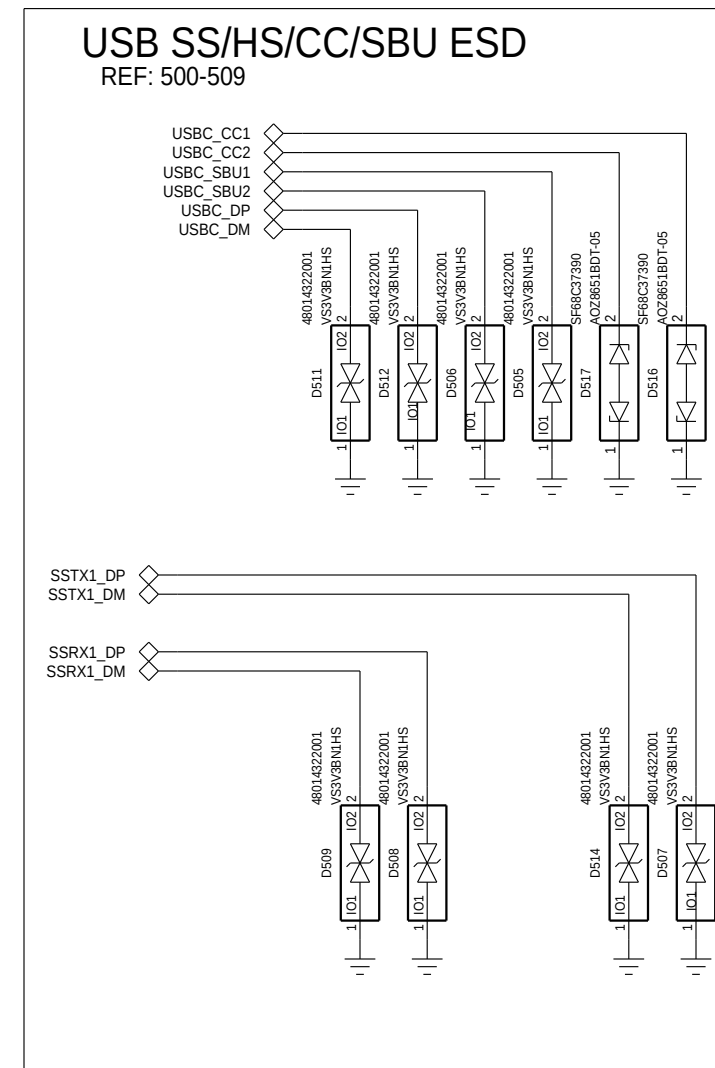
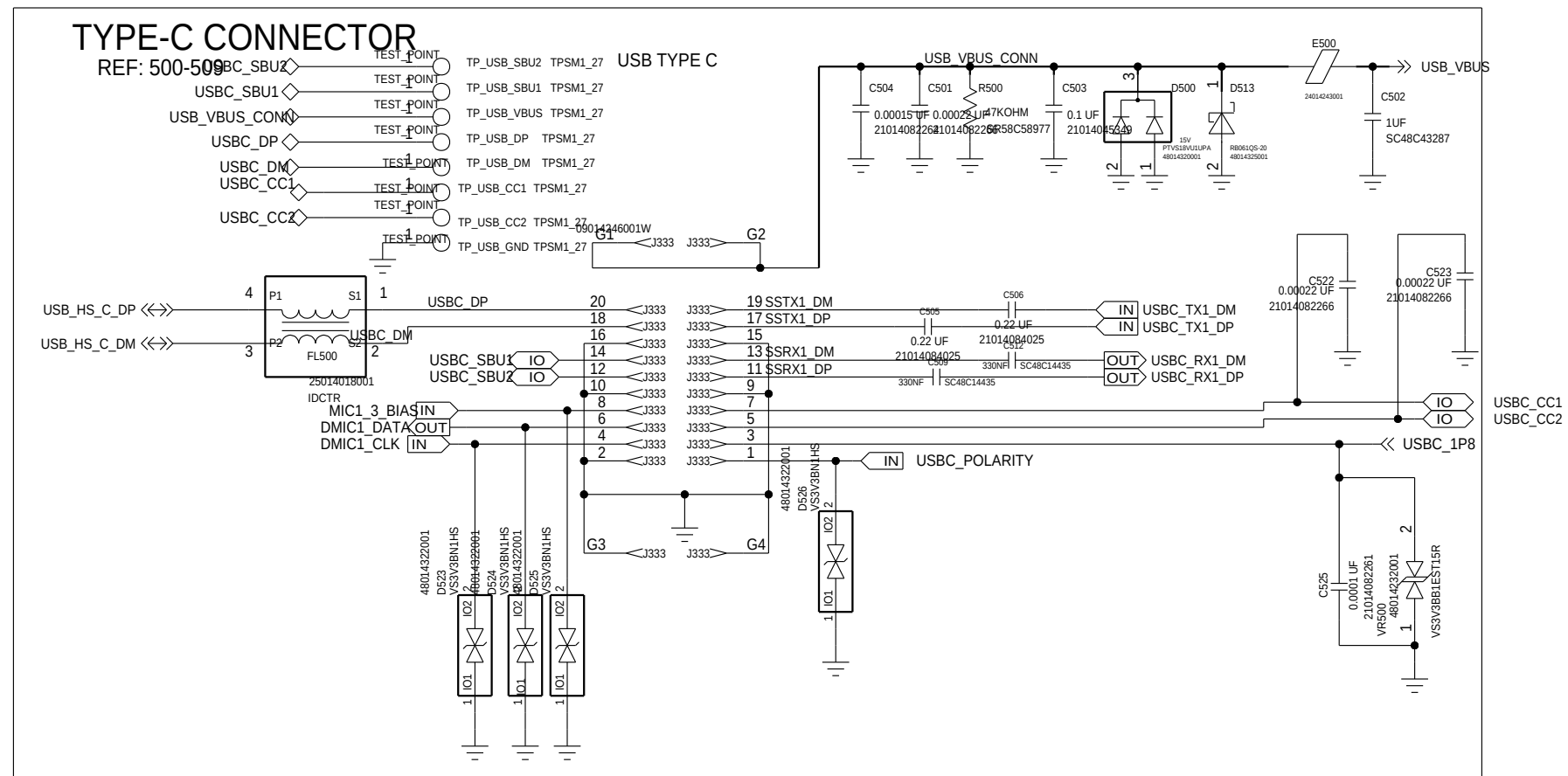




APPROVALS		BLOCK TITLE	
ENGINEER	ENGINEER	sr28c58995_a	
LAST MODIFIED BY: ENGINEER		PROJECT NAME	
Tur Apr 28:10:00 2020		E	
MOTOROLA CONFIDENTIAL, RESTRICTED		REV	
The drawings on this sheet and its supplement are the property of Motorola and are not to be reproduced without the written permission of Motorola. This document is for internal use only and is not to be distributed outside of Motorola.		01	
CADD FILE NAME:		PAGE 1 OF 84	



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MHG634		BLOCK TITLE: portable	
LAST MODIFIED BY: MHG634 Thu Apr 30 13:37:06 2020		PROJECT NAME sr28c58995_a.cpm	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		B	REV 01
		CAD FILE NAME: PAGE 2 OF 84	

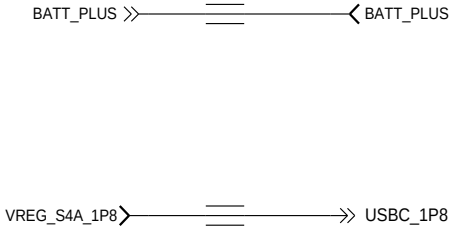


APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MHG634	BLOCK TITLE: portable		
LAST MODIFIED BY: MHG634 Thu Apr 30 13:36:58 2020			
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		B PROJECT NAME sr28c58995_a.cpm	REV 01
		CAD FILE NAME:	PAGE 4 OF 84

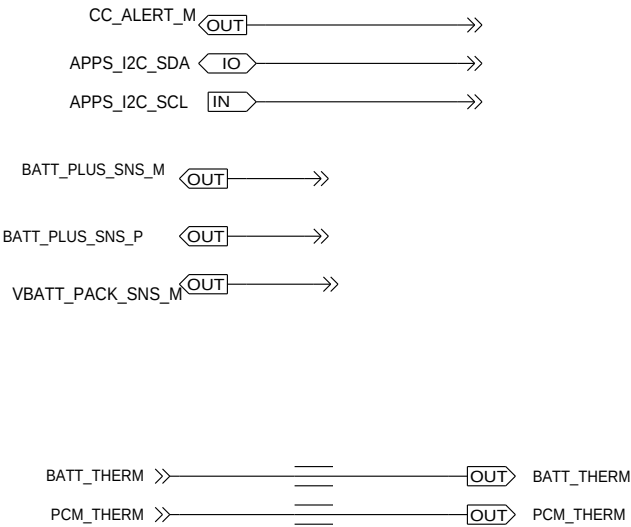
PWR: PORTS

REF 500-699

POWER



PORTS



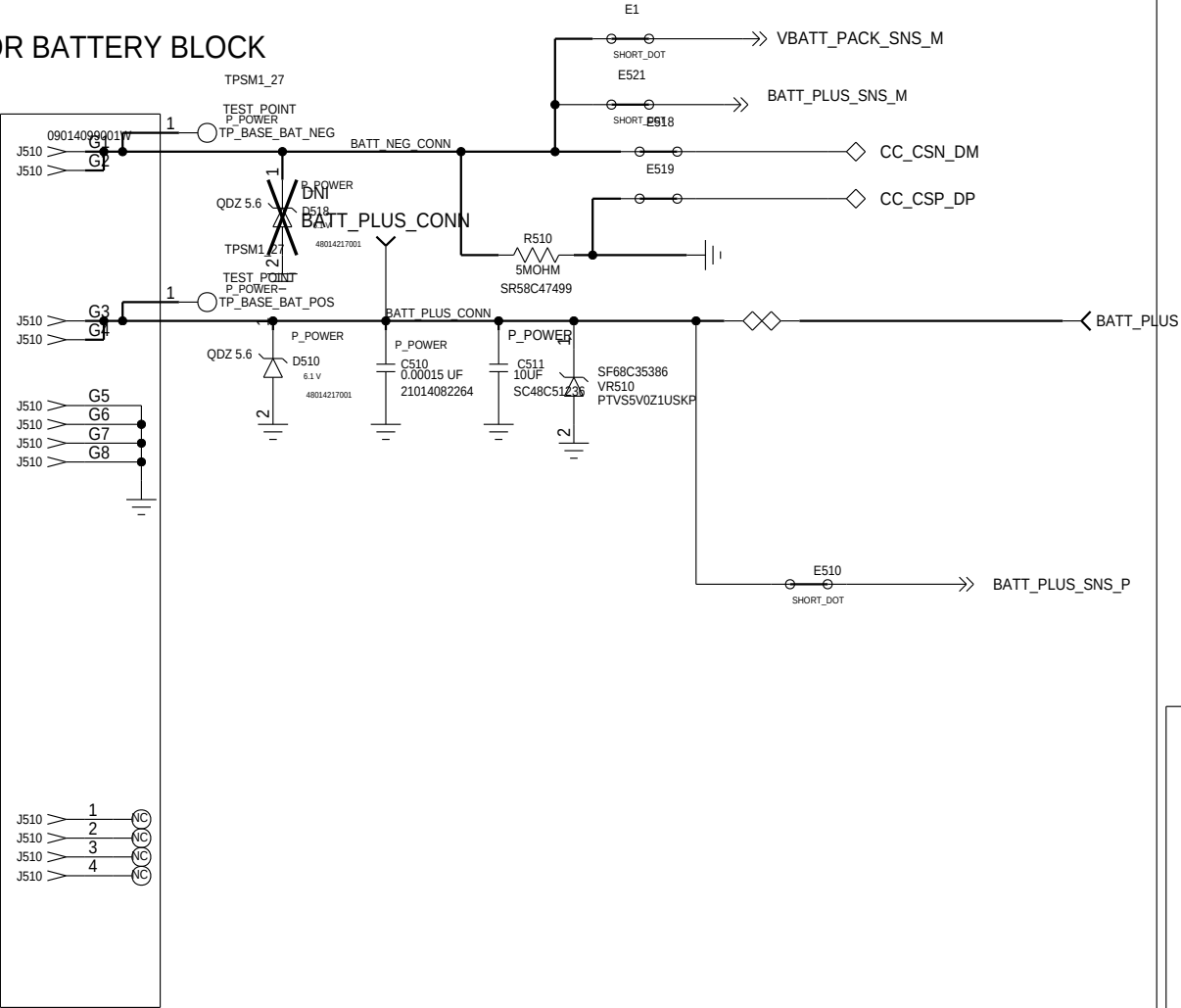
APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER: MOTOROLA		BLOCK TITLE: power		
LAST MODIFIED BY: MOTOROLA Wed Apr 15 10:39:07 2020				
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME		REV
		B sr28c58995_a.cpm		01
		CAD FILE NAME:		PAGE 5 OF 84

PWR: BATT

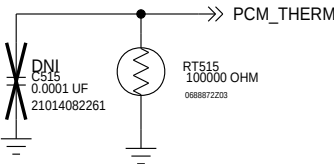
REF: 510-519

BATTERY BLOCK

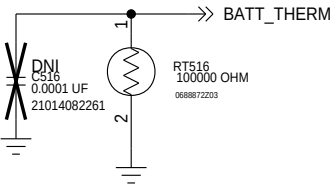
POR BATTERY BLOCK



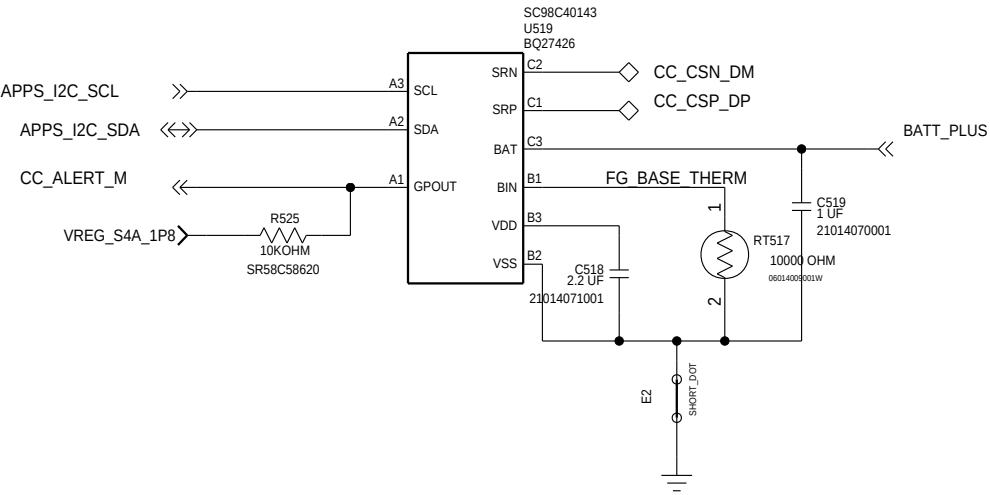
AUX_THERM



BATT_THERM



FUEL GUAGE

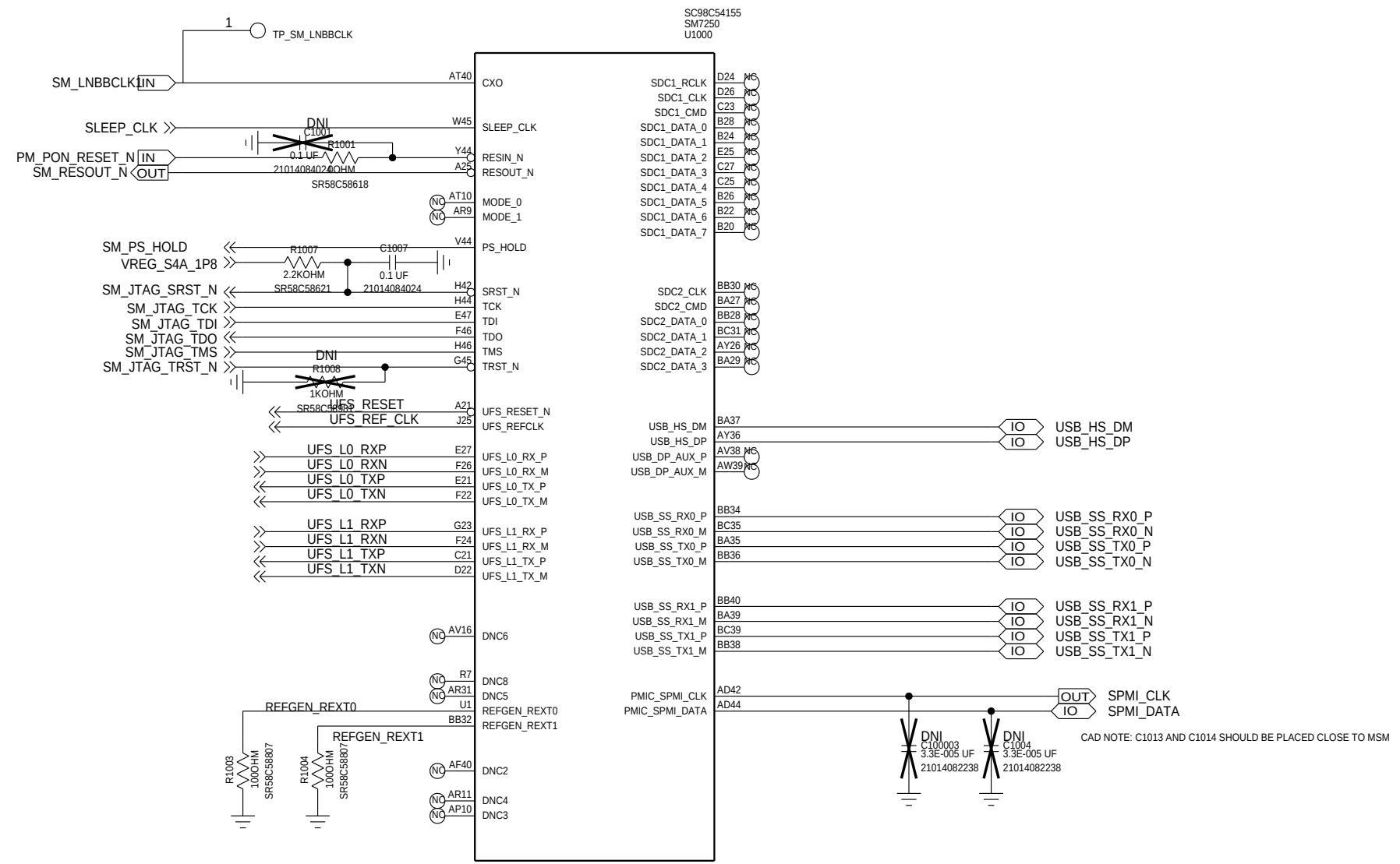


MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Apr 30 13:37:11 2020		power	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	
		B sr28c58995_a.cpm	
		REV 01	
		CAD FILE NAME:	
		PAGE 6 OF 84	

SM7250: CTRL

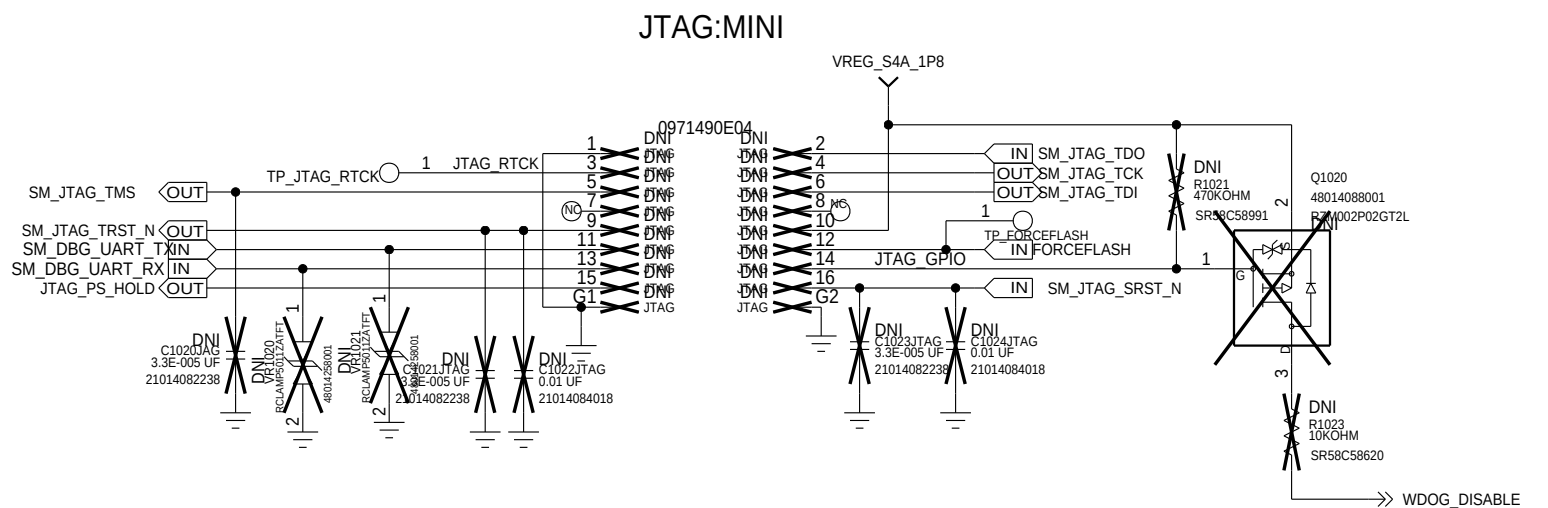
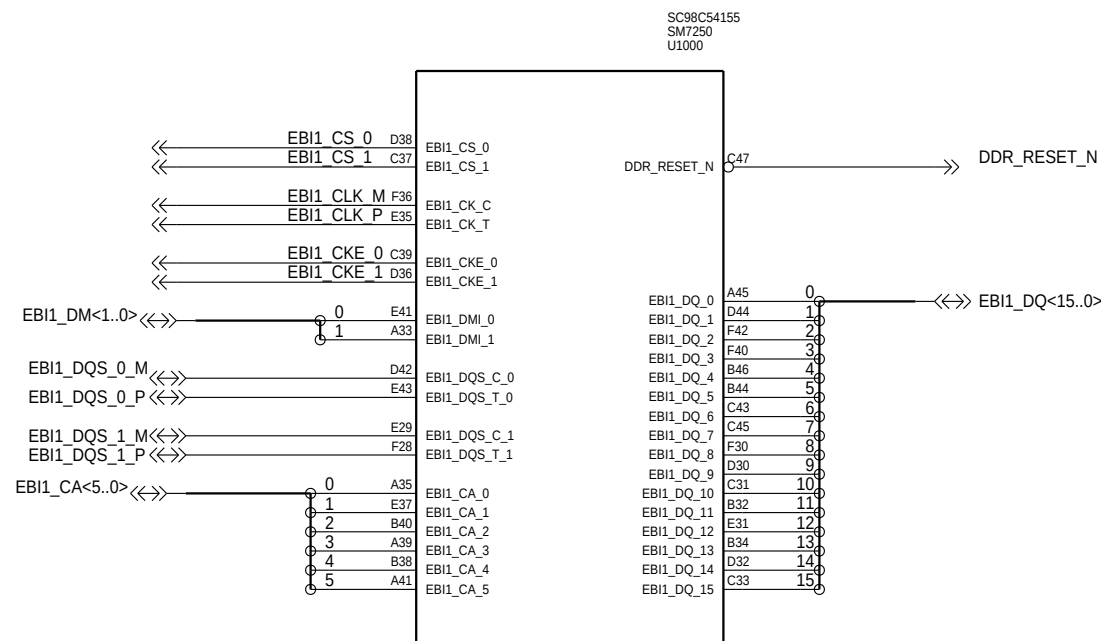
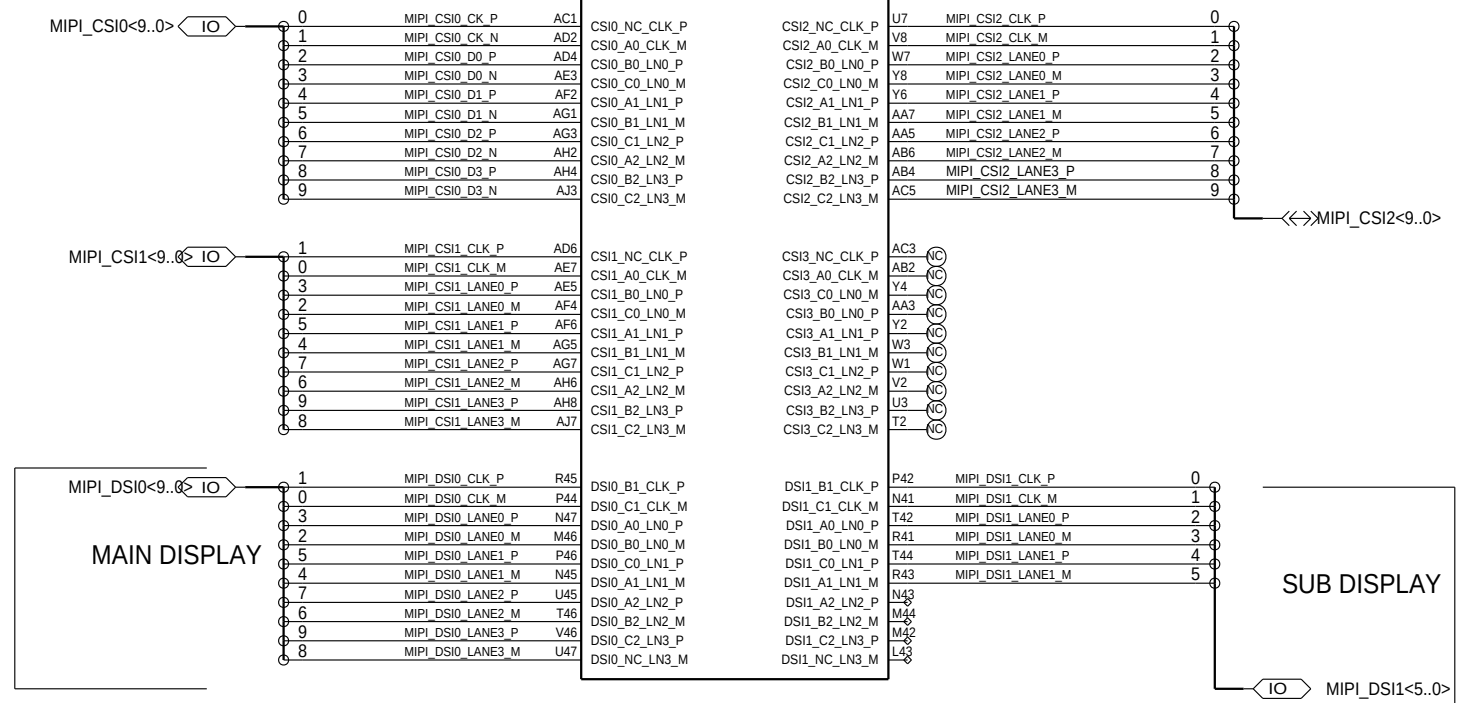
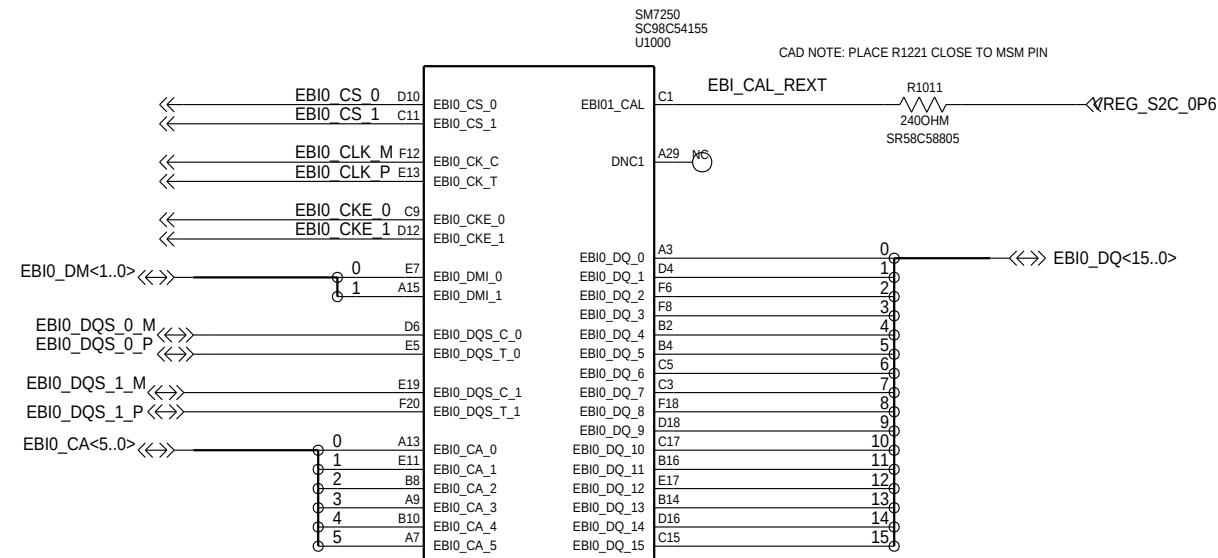
REF 1000-1010



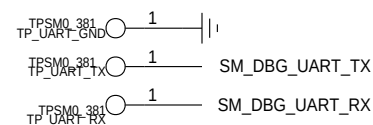
APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Apr 30 13:37:16 2020		ap_pmic	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	
		B sr28c58995_a.cpm	
		REV 01	
		CAD FILE NAME:	
		PAGE 7 OF 84	

SM7250: MIPI/DSI/CSI/EBI0

REF 1011-1029

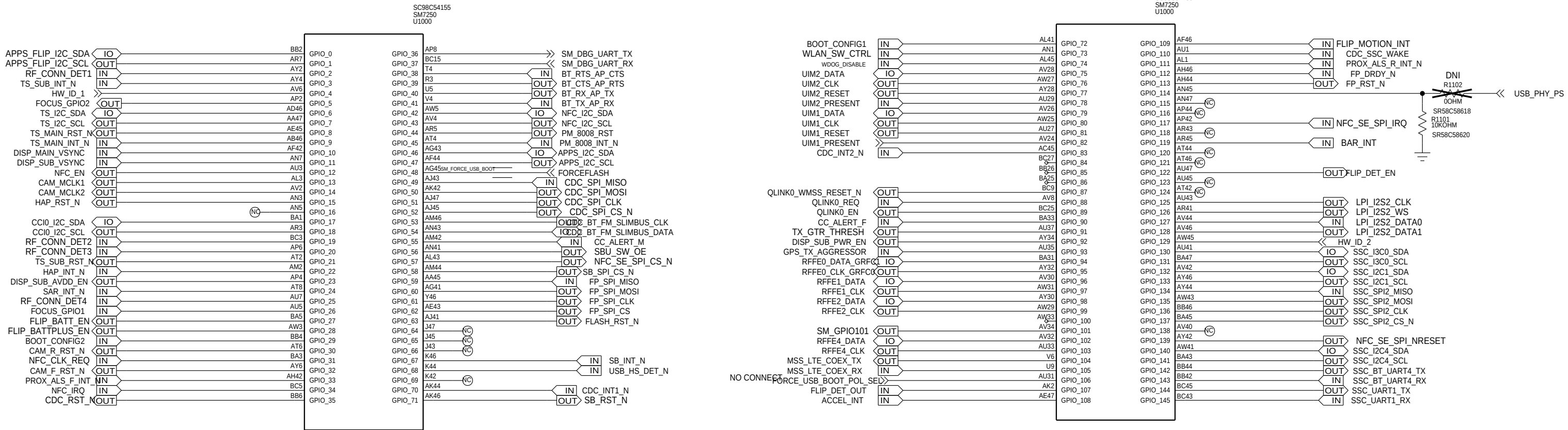


PLACE DEBUG TP OUTSIDE OF
BATTERY AREA



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:	BLOCK TITLE: ap_pmic		
LAST MODIFIED BY: Thu Apr 30 13:37:20 2020			
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	
THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		B	sr28c58995_a.cpm
		CAD FILE NAME:	
		PAGE 8 OF 84	

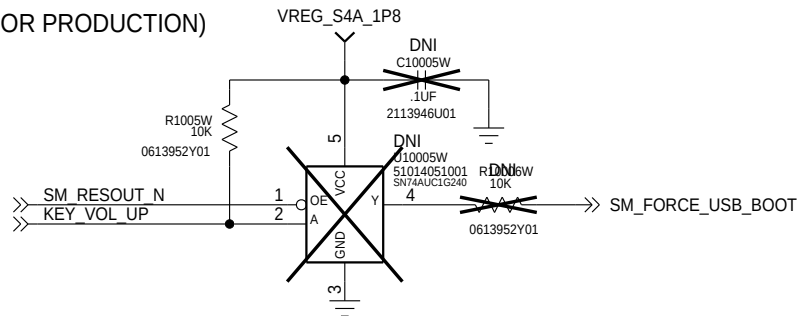
REF 1030-1049



NOTE: DO NOT HAVE PULL-UPS ON THE FOLLOWING GPIOs UNLESS INTEND FOR BOOT OR SECURE BOOT RELATED CONFIGURATIONS:
GPIO_8, 9, 91, 107, 108, 109

FORCE FLASH SUPPORT
WITH VOL UP KEY

(DNI FOR PRODUCTION)



APPROVALS

ENGINEER:

LAST MODIFIED BY: MOTOROLA
Thu Apr 30 13:37:37 2020

MOTOROLA CONFIDENTIAL RESTRICTED

THE DRAWING ON THIS PRINT AND INFORMATION
THEREWITH ARE PROPRIETARY TO MOTOROLA AND
SHALL NOT BE USED OR DISCUSSED IN WHOLE
OR IN PART WITHOUT MOTOROLA'S CONSENT.
© COPYRIGHT, MOTOROLA. THIS DOES
NOT IMPLY PUBLICATION



MOTOROLA

MOBILE DEVICES

BLOCK TITLE:

ap_pmic

B

PROJECT NAME

sr28c58995_a.cpm

REV

01

CAD FILE NAME:

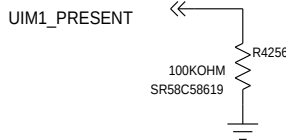
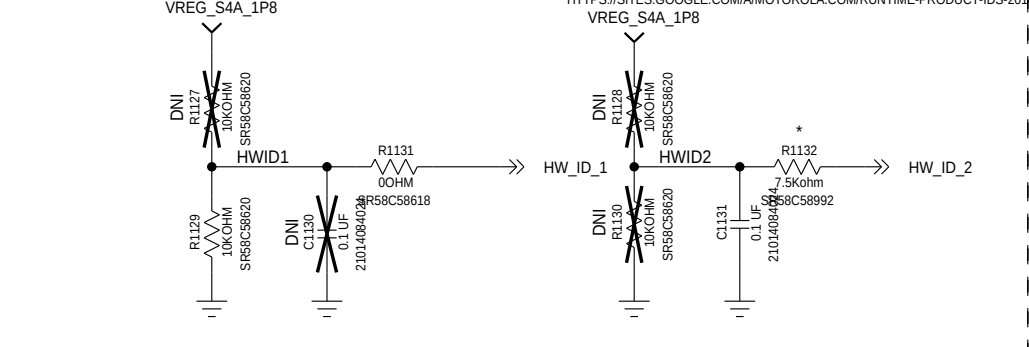
PAGE 9 OF 84

SM7250: CONFIG

I2C PULL-UP RESISTORS



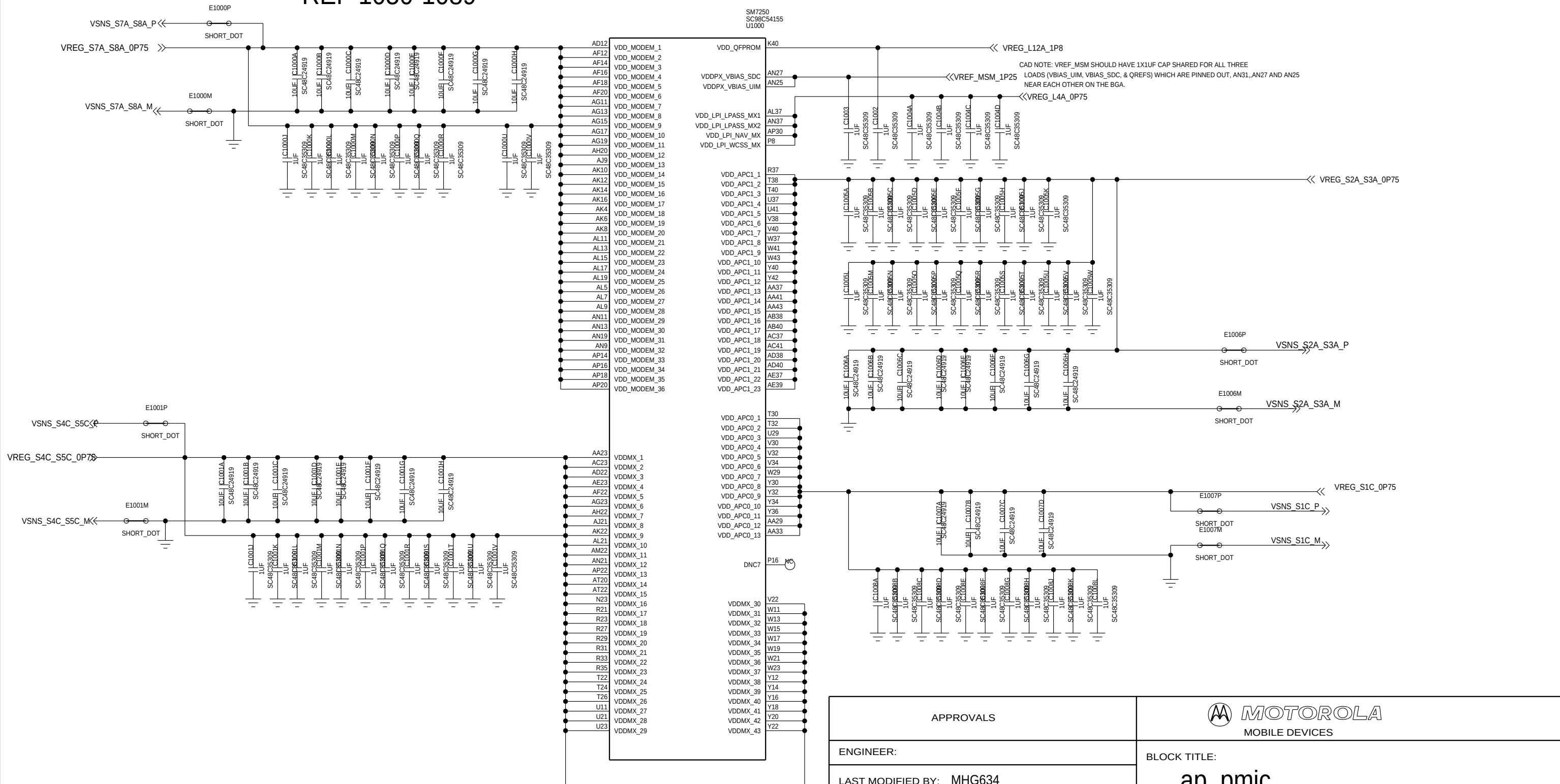
MOT BOARD ID



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FAN021 Thu Apr 30 13:37:59 2020		ap_pmic	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	
		B sr28c58995_a.cpm	
		REV 01	
		PAGE 10 OF 84	

SM7250: PWR1

REF 1050-1089



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: MHG634 Thu Apr 30 13:38:14 2020		ap_pmic	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	
		B sr28c58995_a.cpm	
		REV 01	
		CAD FILE NAME:	
		PAGE 11 OF 84	

SM7250: PWR2
REF 1000-1099

D

C

B

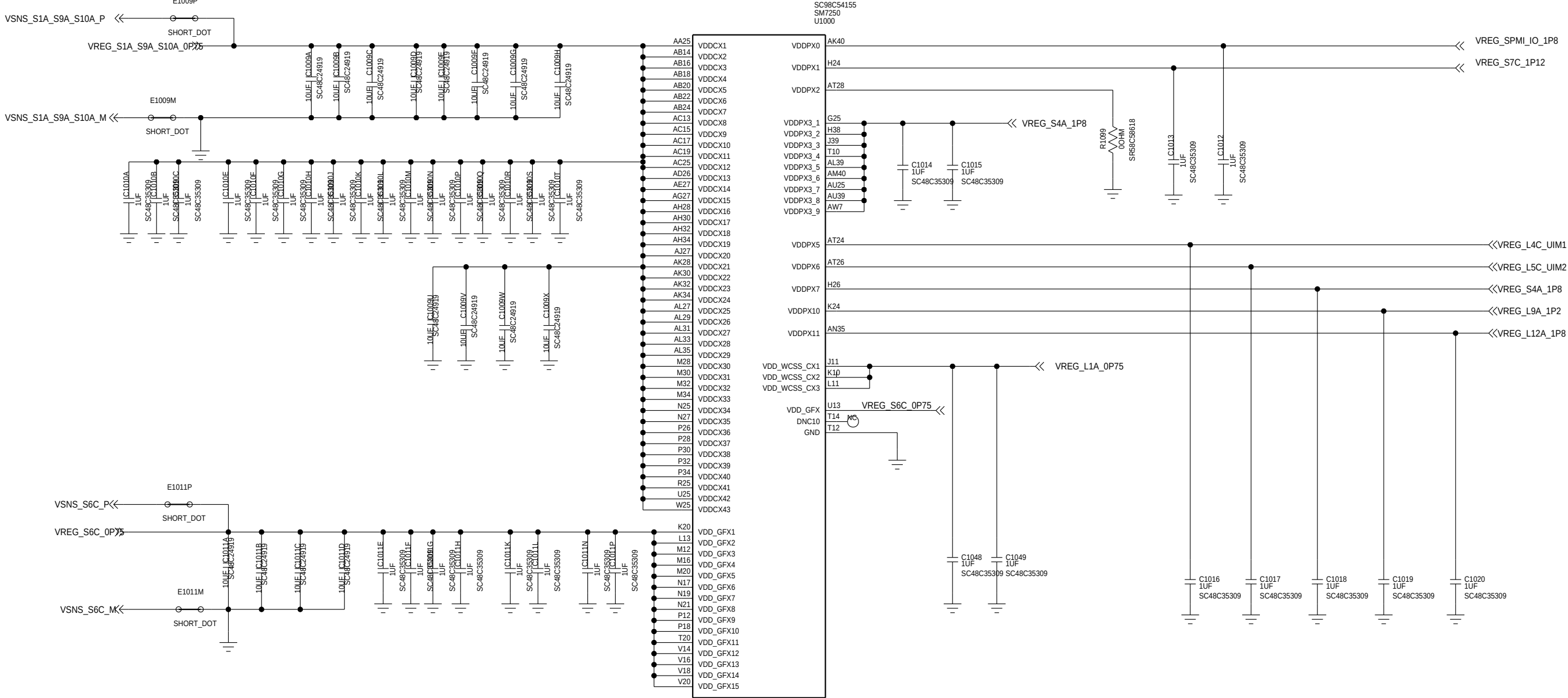
A

D

C

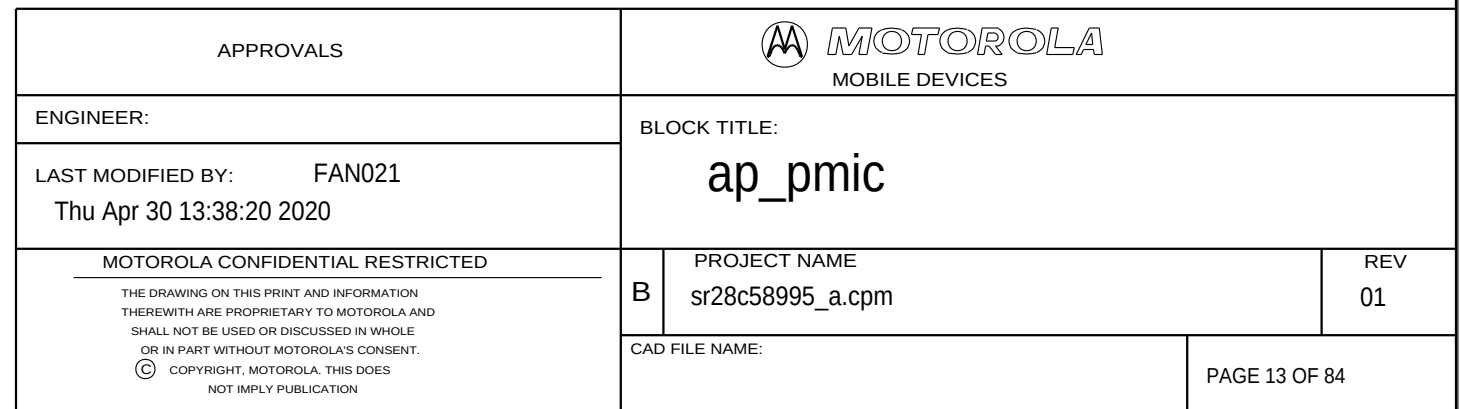
B

A



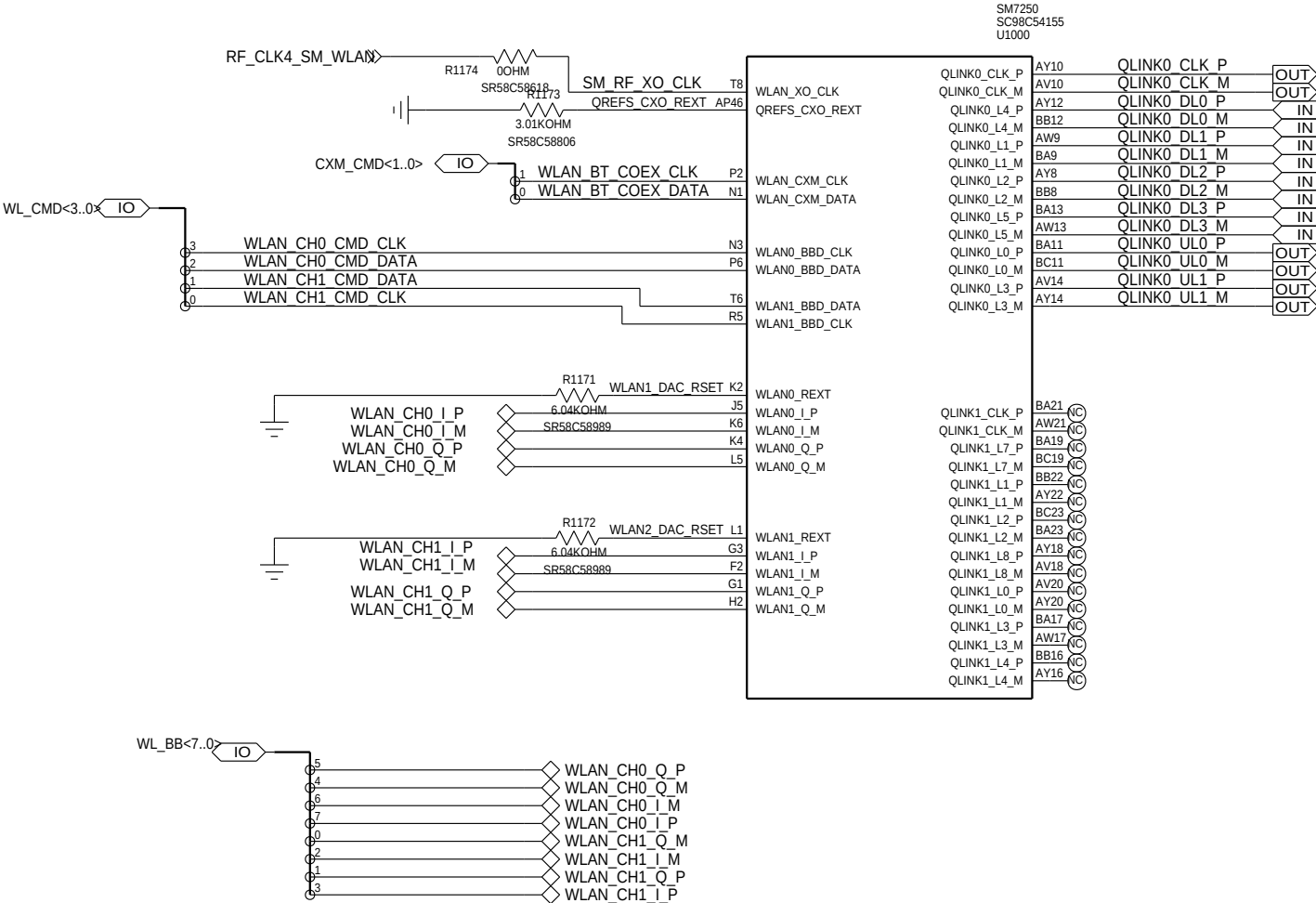
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FAN021 Thu Apr 30 13:37:47 2020		ap_pmic	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	
		B sr28c58995_a.cpm	
		REV 01	
		CAD FILE NAME:	
		PAGE 12 OF 84	

REF 1090-1169



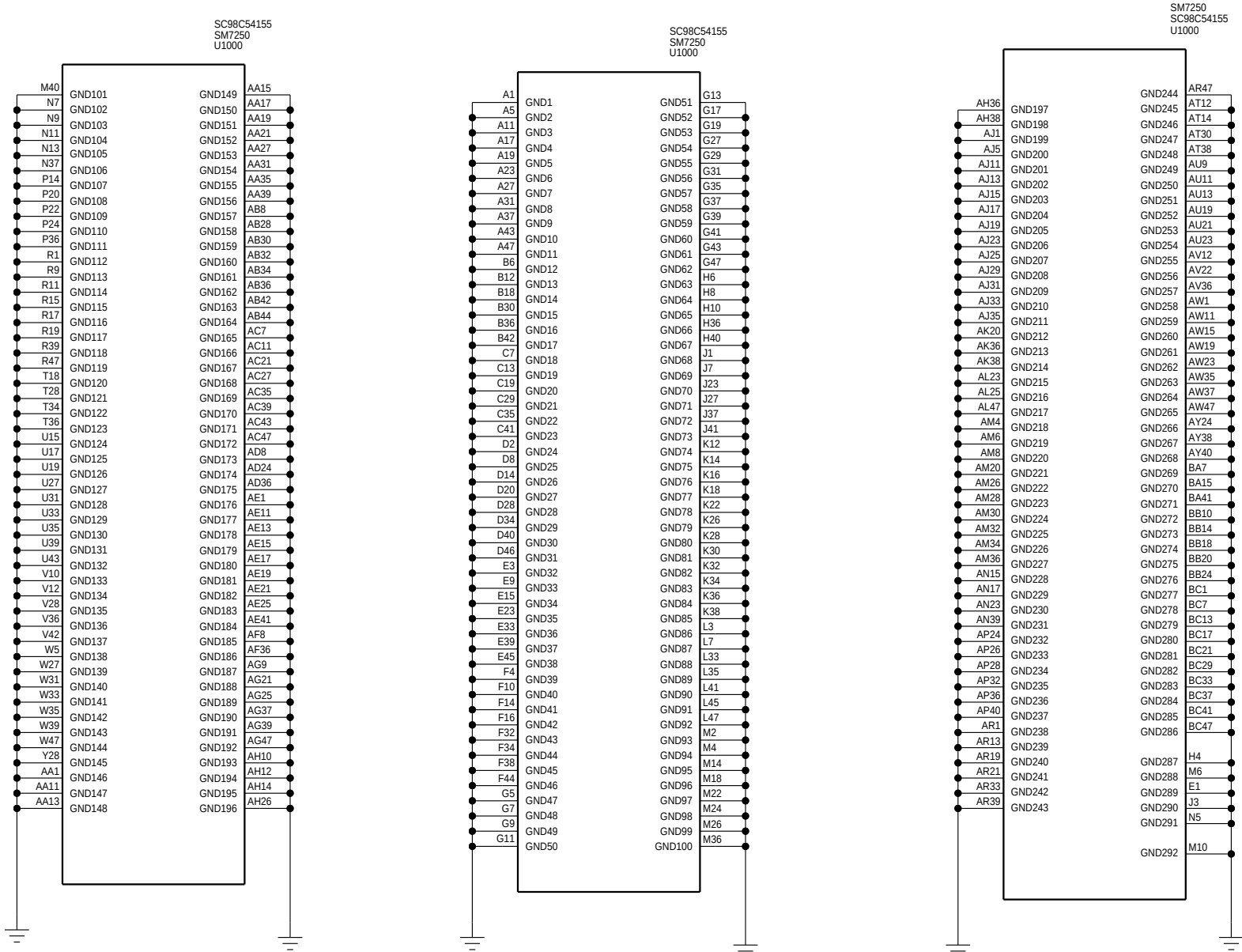
SM7250: RF INTF

REF 1170-1199



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FAN021 Thu Apr 30 13:37:57 2020		ap_pmic	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		B PROJECT NAME sr28c58995_a.cpm	REV 01
		CAD FILE NAME:	
		PAGE 14 OF 84	

SM7250: GROUND



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FAN021 Thu Apr 30 13:38:24 2020		ap_pmic	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	
		sr28c58995_a.cpm	
		REV 01	
		PAGE 15 OF 84	

PM7250: POWER

INPUT POWER

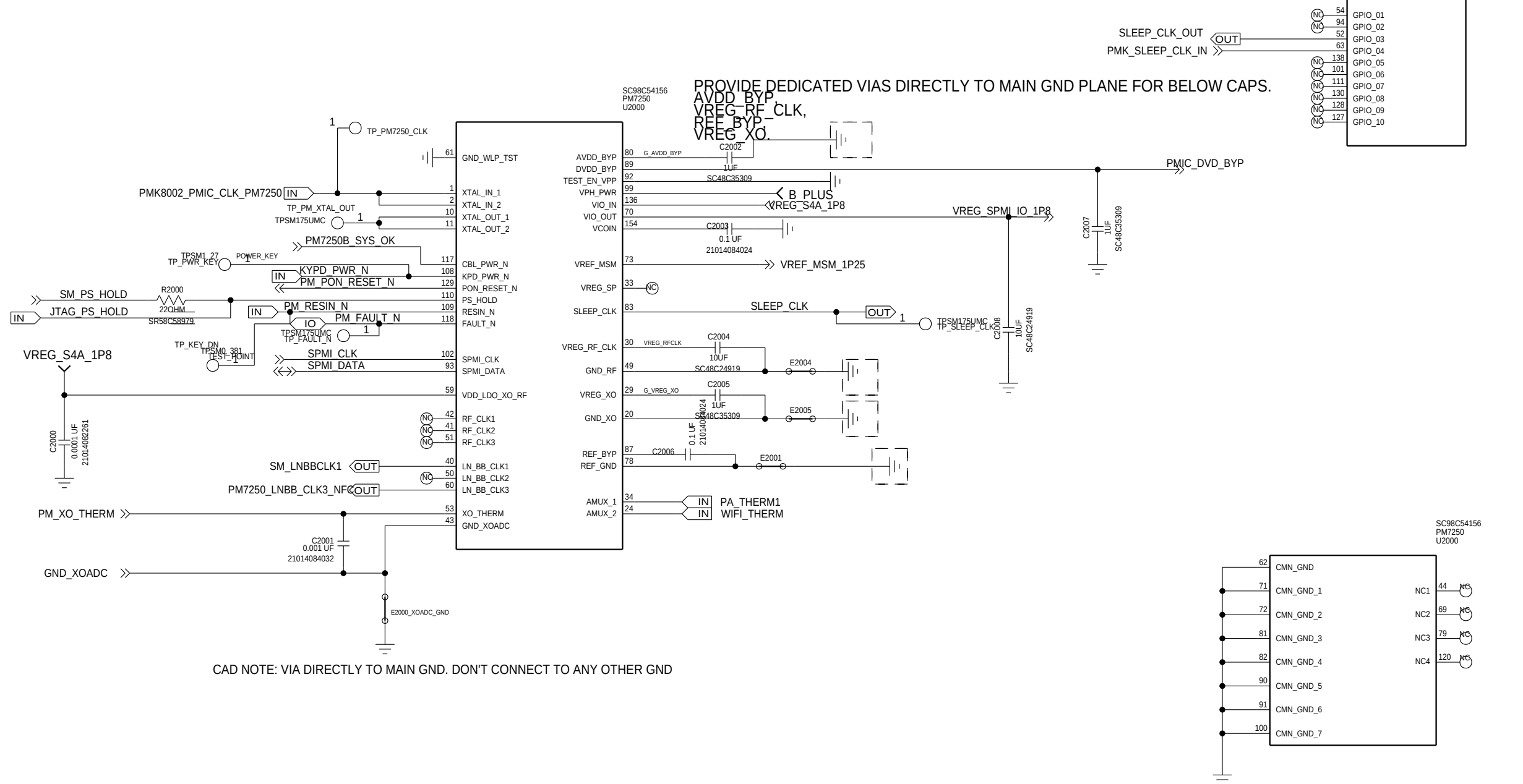
CURRENT DRAIN

OUTPUT POWER

VREG_L1A_0P75	>>>	VREG_L1A_0P75
VREG_L2A_3P1	>>>	VREG_L2A_3P1
VREG_L3A_0P8	>>>	VREG_L3A_0P8
VREG_L4A_0P75	>>>	VREG_L4A_0P75
VREG_L5A_0P88	>>>	VREG_L5A_0P88
VREG_L6A_1P2	>>>	VREG_L6A_1P2
VREG_L7A_1P8	>>>	VREG_L7A_1P8
VREG_L8A_0P75	>>>	VREG_L8A_0P75
VREG_L9A_1P2	>>>	VREG_L9A_1P2
VREG_L10A_3P0	>>>	VREG_L10A_3P0
VREG_L11A_0P75	>>>	VREG_L11A_0P75
VREG_L12A_1P8	>>>	VREG_L12A_1P8
VREG_L13A_3P1	>>>	VREG_L13A_3P1
VREG_L14A_1P8	>>>	VREG_L14A_1P8
VREG_L15A_1P7	>>>	VREG_L15A_1P7
VREG_L16A_3P0	>>>	VREG_L16A_3P0
VREG_L17A_1P8	>>>	VREG_L17A_1P8
VREG_L18A_0P75	>>>	VREG_L18A_0P75

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FAN021 Wed Apr 15 10:39:12 2020		ap_pmic	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		B	PROJECT NAME sr28c58995_a.cpm
		REV 01	
		CAD FILE NAME:	
		PAGE 17 OF 84	

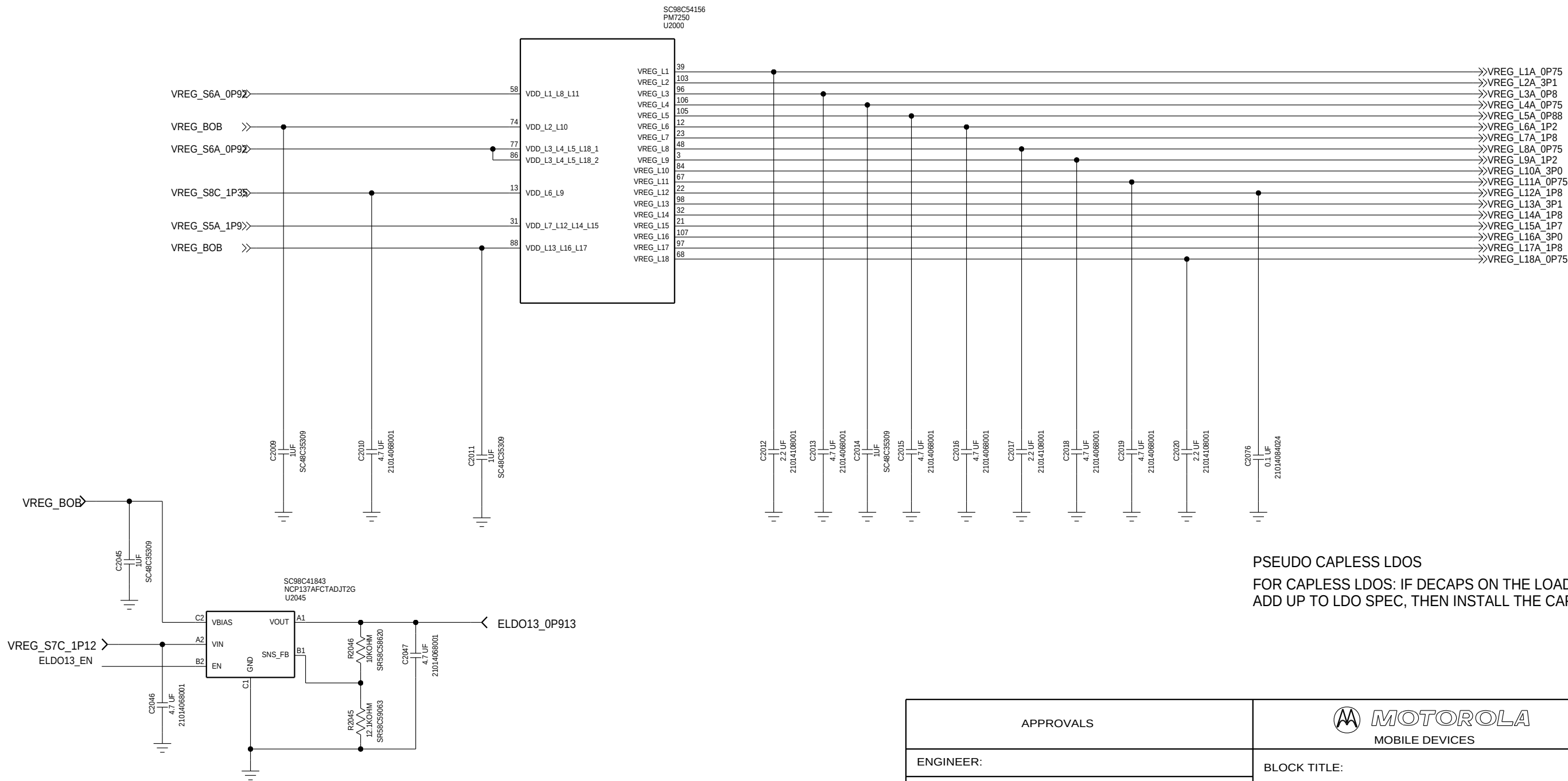
PM7250: HK/GHRG/LED/GPIO



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FAN021 Fri May 08 12:13:40 2020		ap_pmic	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	
		B	sr28c58995_a.cpm
		CAD FILE NAME:	
		PAGE 18 OF 84	
REV 01			

PM7250: VREGS

REF: 2000-2099

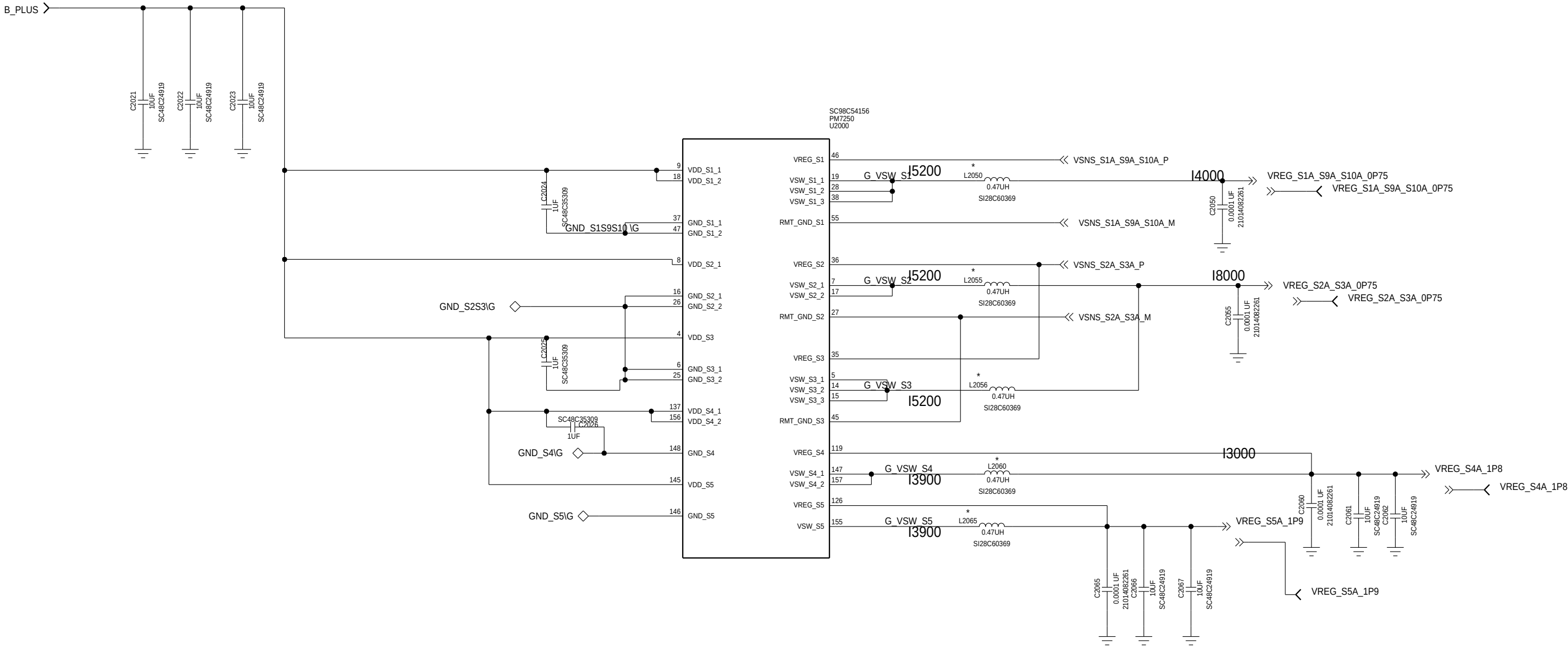


PSEUDO CAPLESS LDOS
FOR CAPLESS LDOS: IF DECAPS ON THE LOAD SIDE DO NOT
ADD UP TO LDO SPEC, THEN INSTALL THE CAP CLOSE TO THE PMIC

PLACE HOLDER FOR NCP137AFCTADJT2G

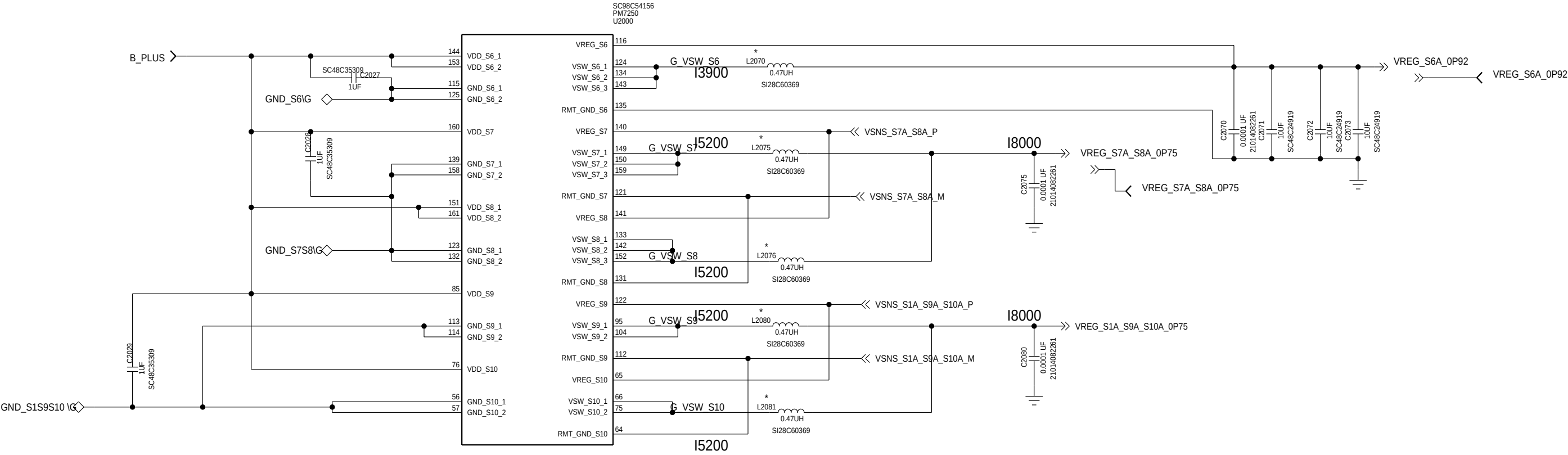
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: MHG634 Thu Apr 30 13:37:44 2020		ap_pmic	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	
		B sr28c58995_a.cpm	
		CAD FILE NAME:	
		PAGE 19 OF 84	
		REV 01	

PM7250: SMPS 1



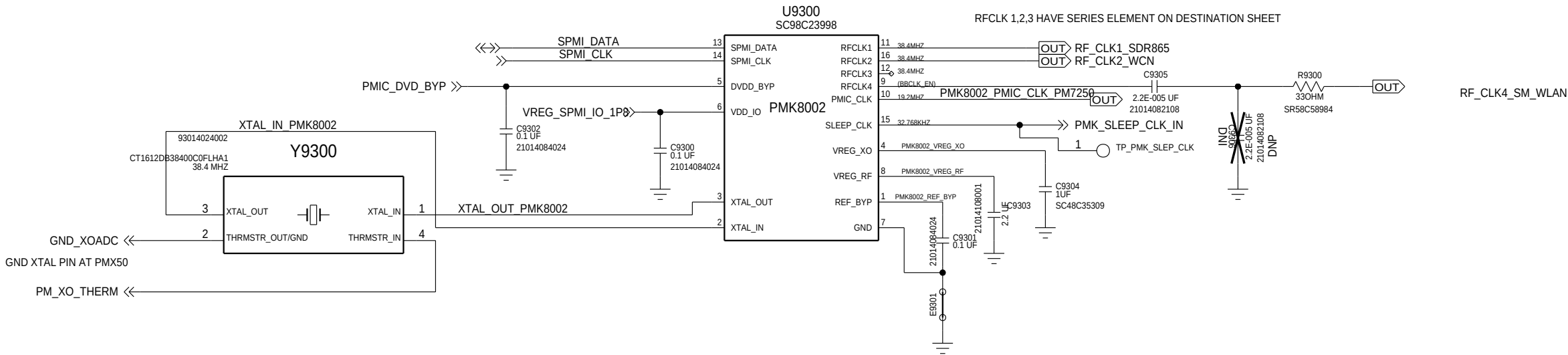
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FAN021 Thu Apr 30 13:38:02 2020		ap_pmic	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	
		B sr28c58995_a.cpm	
		REV 01	
		CAD FILE NAME:	
		PAGE 20 OF 84	

PM7250: SMPS 2



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FAN021 Fri May 22 18:09:15 2020		ap_pmic	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	
		B sr28c58995_a.cpm	REV 01
		CAD FILE NAME:	
		PAGE 21 OF 84	

PMK8002



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FAN021 Fri May 08 12:15:01 2020		ap_pmic	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	
		B sr28c58995_a.cpm	
		REV 01	
		CAD FILE NAME:	
		PAGE 22 OF 84	

D



2

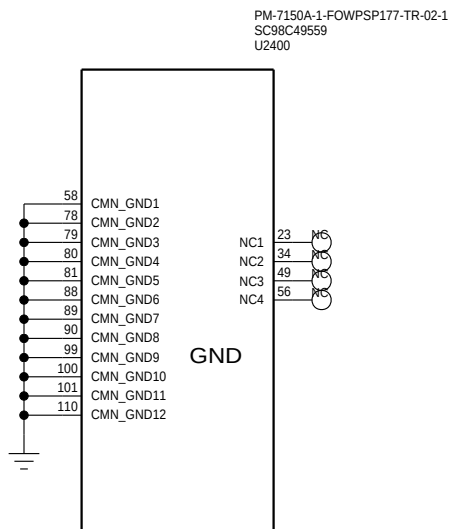
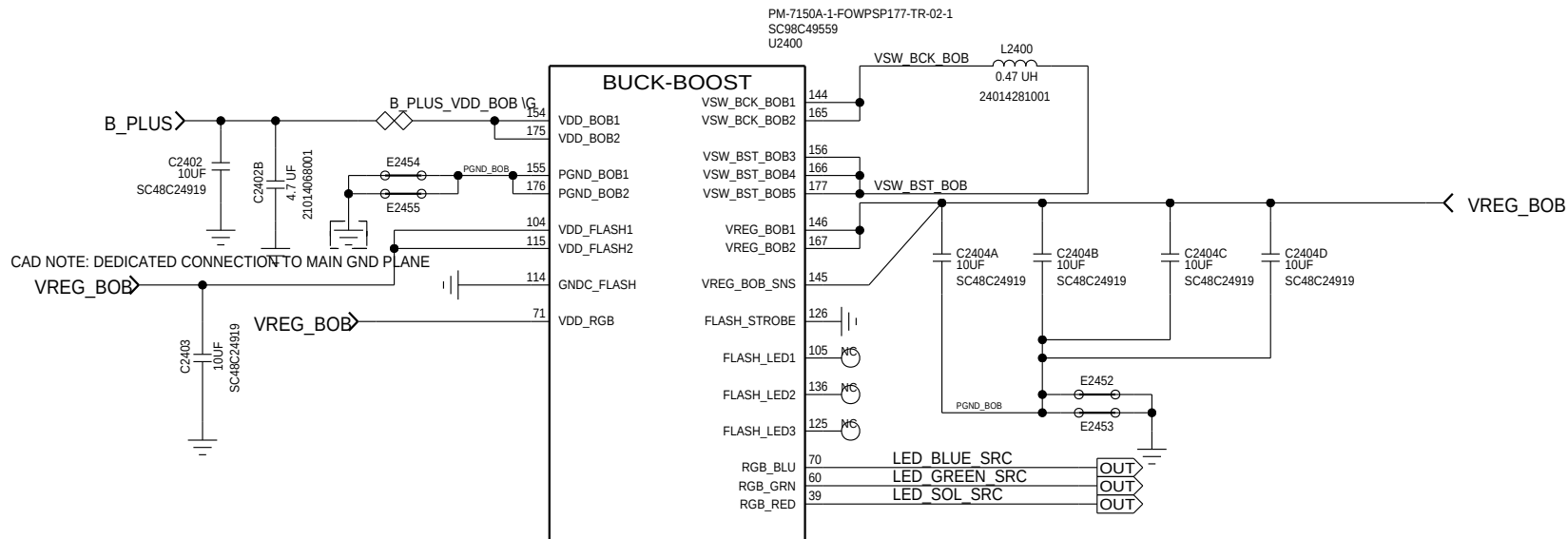
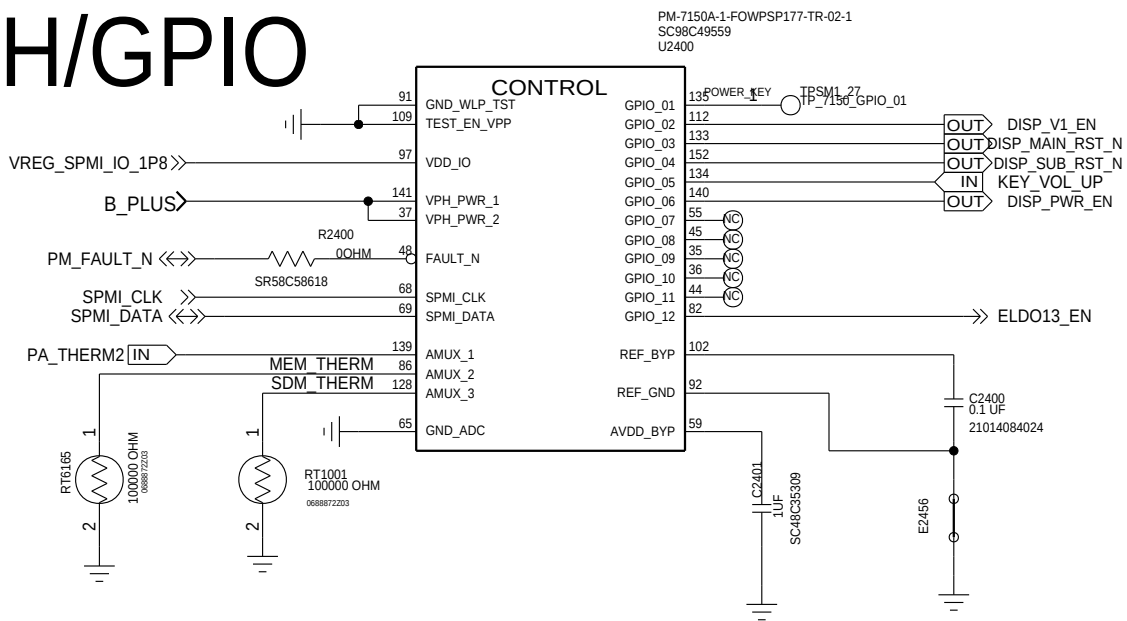


8	7	6	5	4	3	2	1
PM7150A/PMIC3: POWER INPUT POWER OUTPUT POWER CURRENT DRAIN							
D							D
C							C
B							B
A							A
8	7	6	5	4	3	2	1

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FAN021 Wed Jun 12 12:22:29 2019		ap_pmic	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		B	PROJECT NAME sr28c58995_a.cpm
		CAD FILE NAME: PAGE 24 OF 84	

PM7150A: CNTL/BOB/FLASH/GPIO

REF: 2450 - 2499



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FAN021 Thu Apr 30 13:37:31 2020		ap_pmic	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	
		B sr28c58995_a.cpm	REV 01
		CAD FILE NAME:	
		PAGE 25 OF 84	

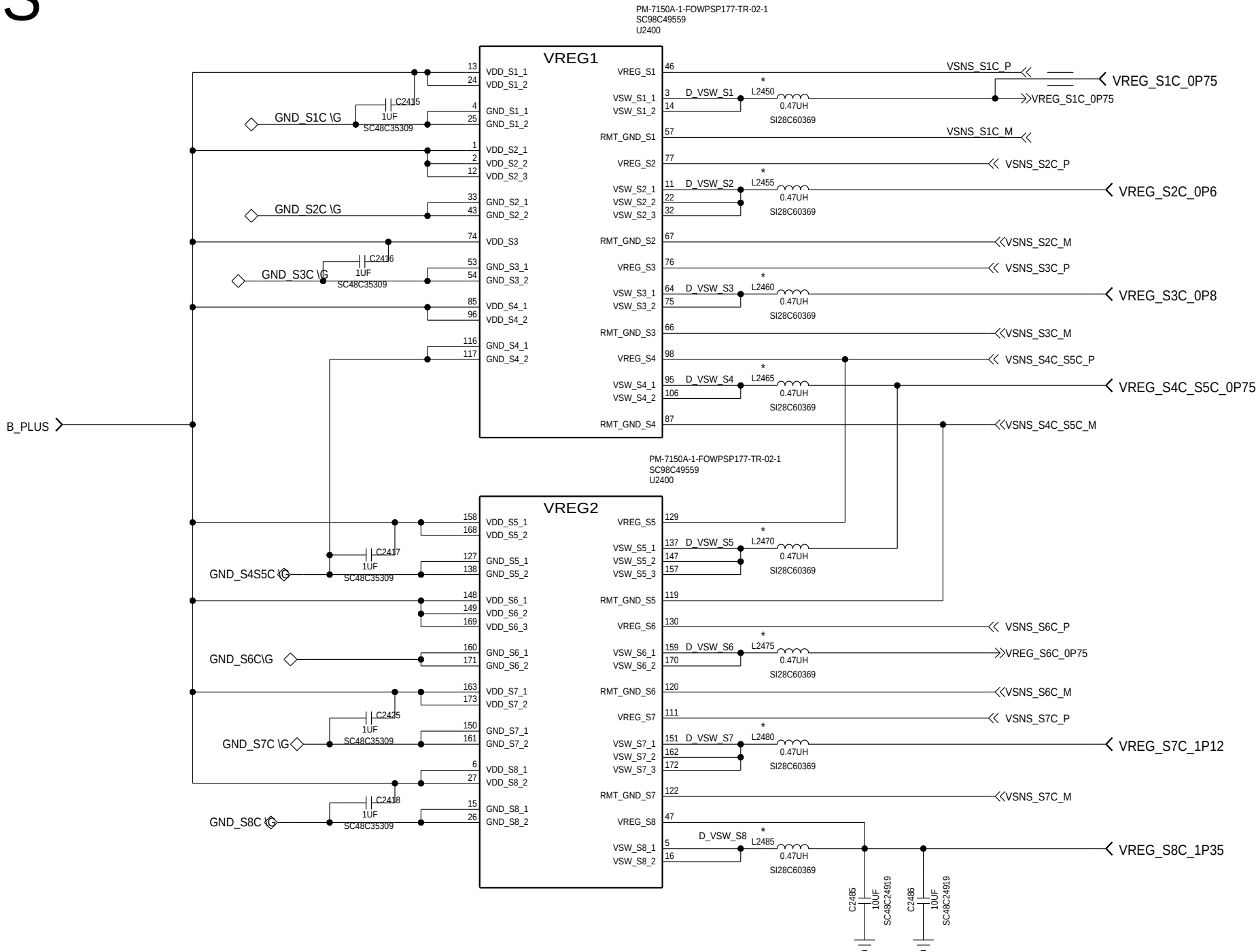
2

C

B

A

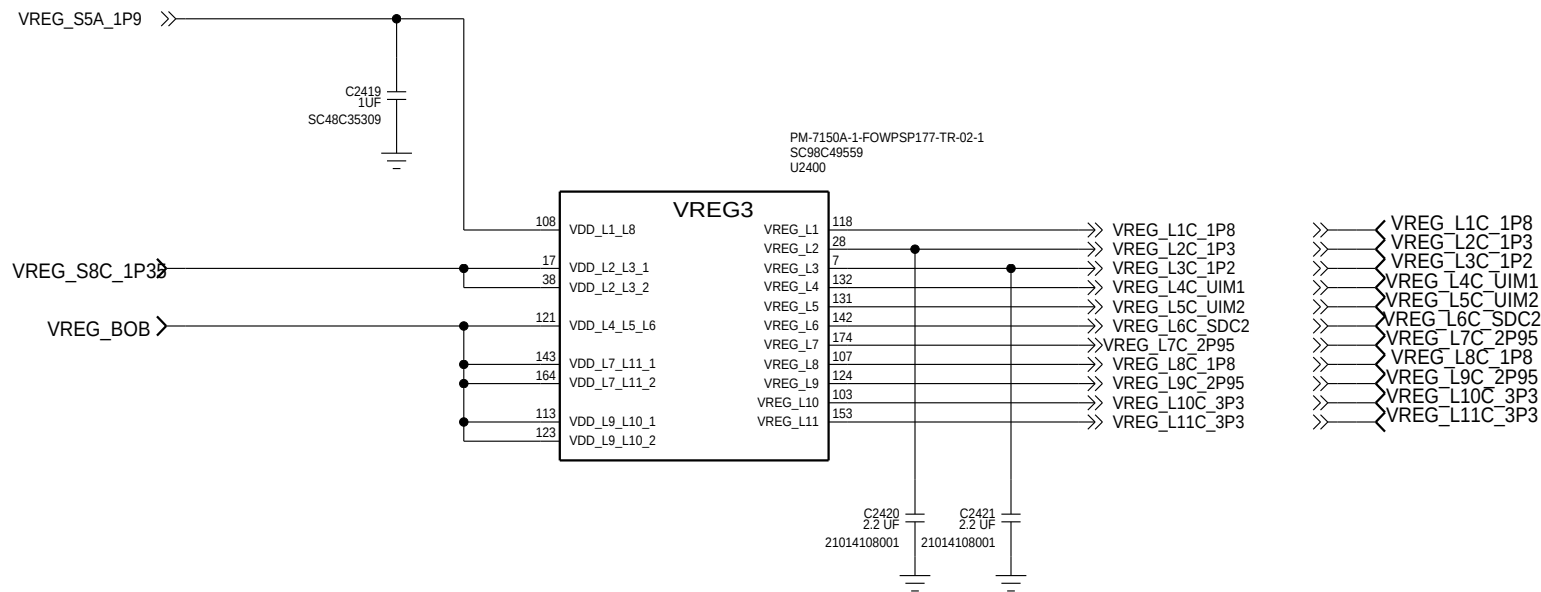
PM7150A: SMPS



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FAN021 Thu Apr 30 13:38:08 2020		ap_pmic	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	
		B sr28c58995_a.cpm	
		CAD FILE NAME:	
		PAGE 27 OF 84	
REV 01			

PM7150A: REGS

REF: 2560-2599

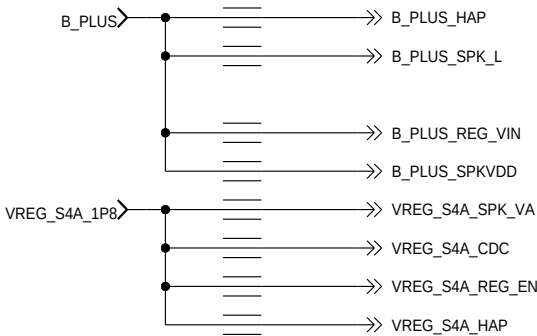


PSEUDO CAPLESS LDOS
FOR CAPLESS LDOS: IF DECAPS ON THE LOAD SIDE DO NOT
ADD UP TO LDO SPEC, THEN INSTALL THE CAP CLOSE TO THE PMIC

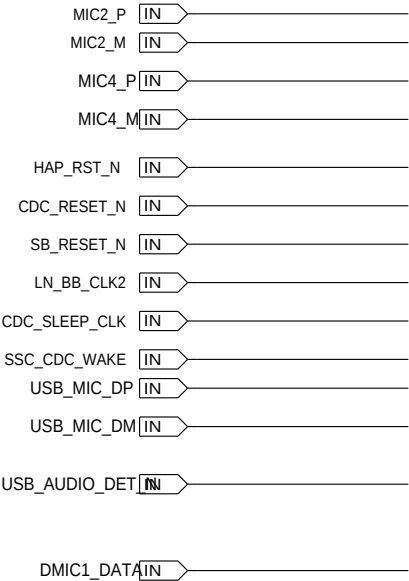
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FAN021 Thu Apr 30 13:38:10 2020		ap_pmic	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	
		B sr28c58995_a.cpm	
		REV 01	
		CAD FILE NAME:	
		PAGE 28 OF 84	

AUDIO: PORTS

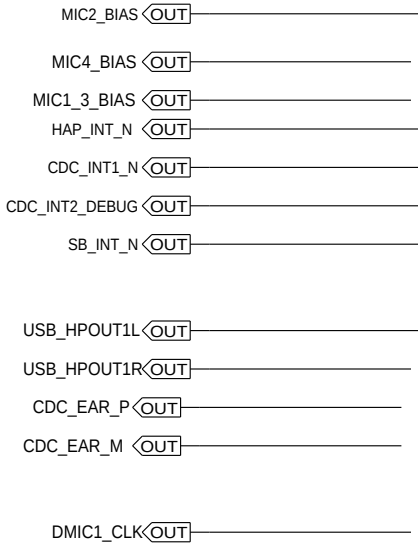
POWER



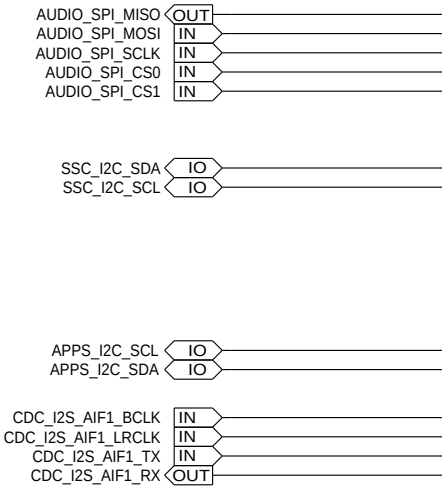
INPUTS



OUTPUTS



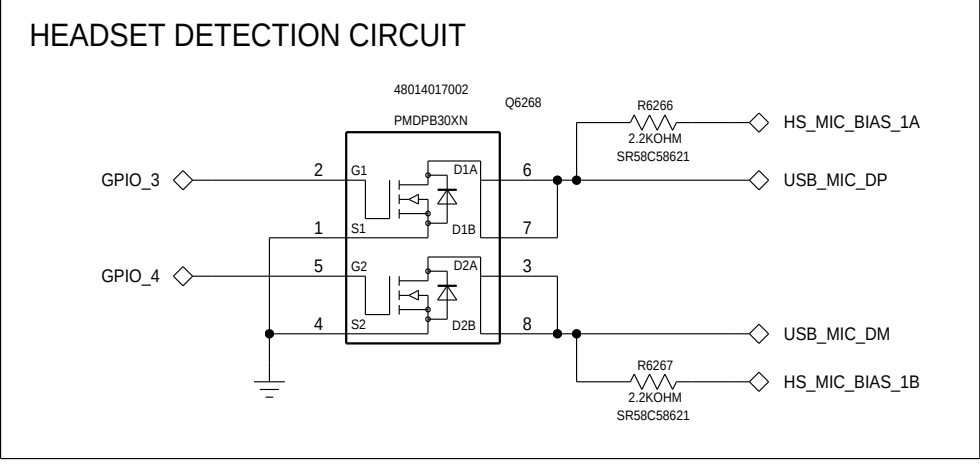
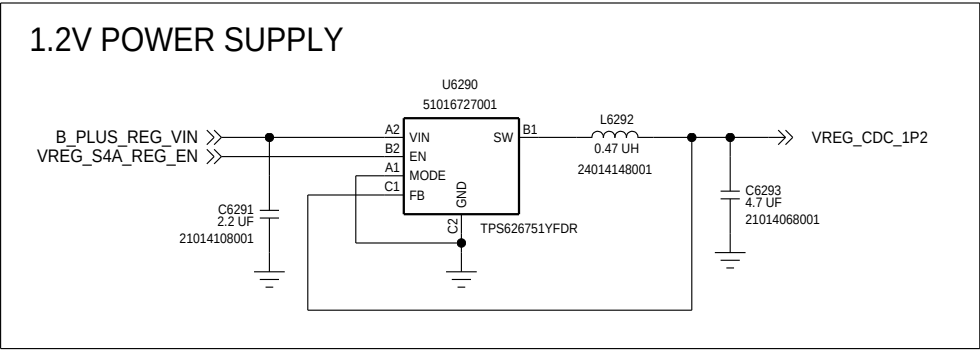
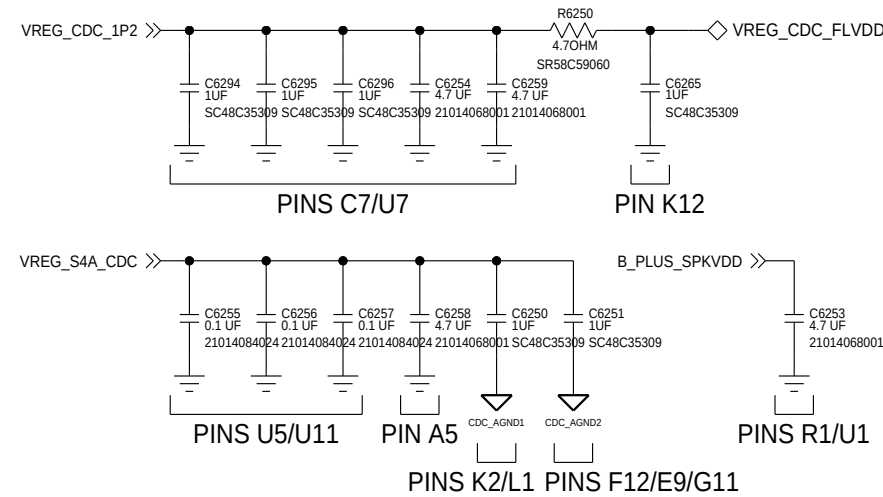
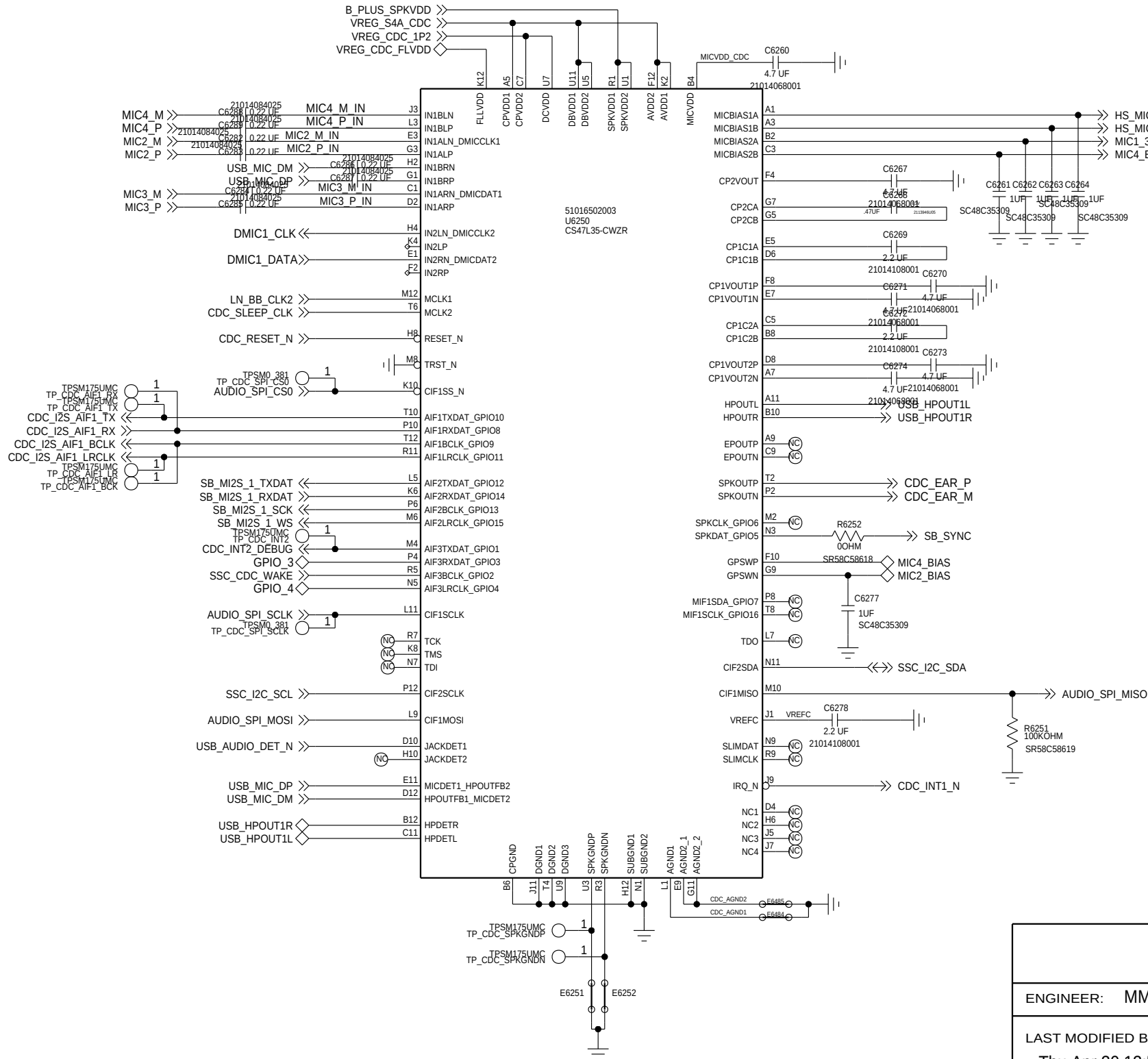
INTERFACES



APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER: MML		BLOCK TITLE: audio		
LAST MODIFIED BY: MML Fri May 22 18:10:44 2020				
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME		REV
		B sr28c58995_a.cpm		01
		CAD FILE NAME:		PAGE 29 OF 84

AUDIO: CIRRUS CODEC

REF 6240-6299

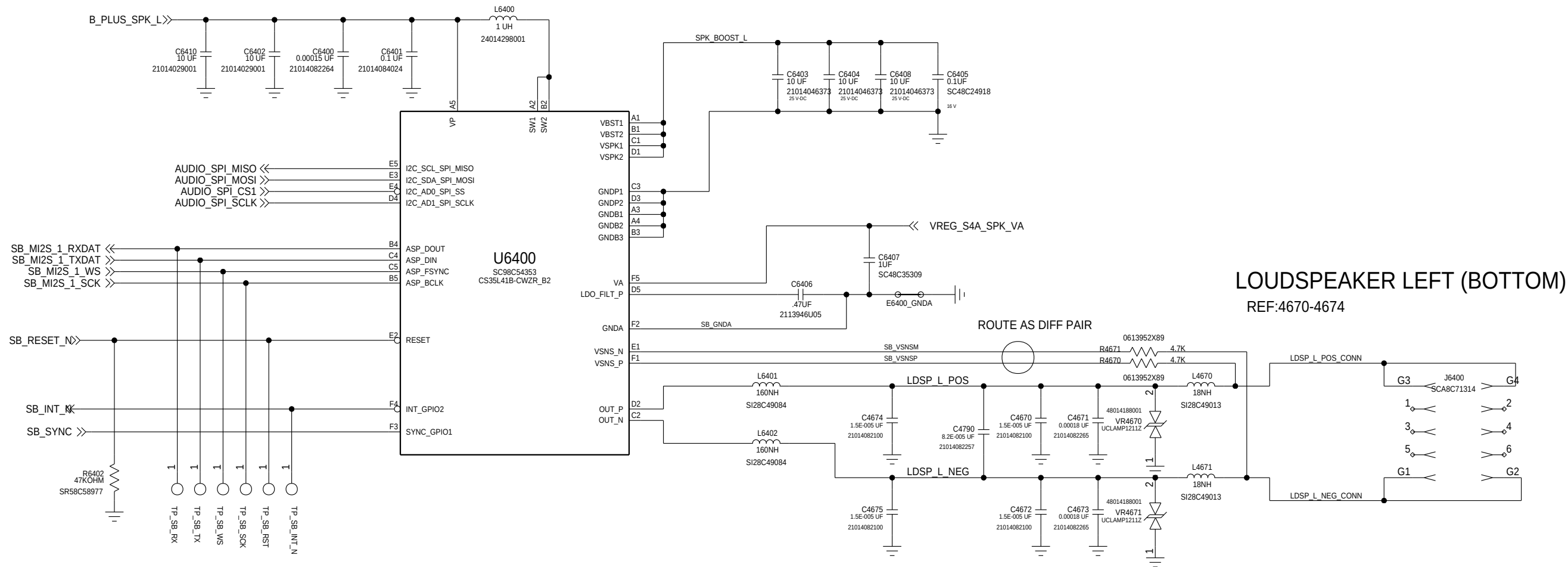


APPROVALS		BLOCK TITLE:	
ENGINEER: MML		audio	
LAST MODIFIED BY: MML Thu Apr 30 13:53:09 2020		PROJECT NAME sr28c58995_a.cpm	
MOTOROLA CONFIDENTIAL RESTRICTED		REV 01	
THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PAGE 30 OF 84	

AUDIO: SMARTBOOST
REF 6400-6499

LEFT SPEAKER AMP

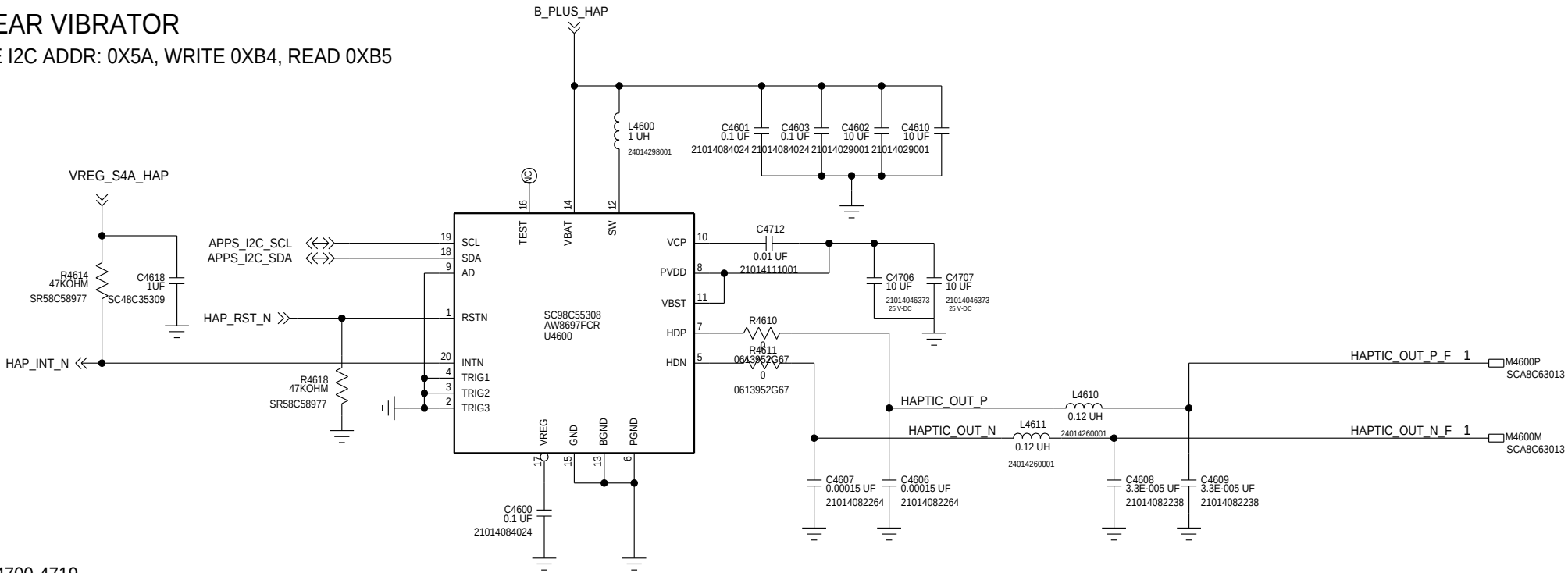
BASE I2C ADDR: 0X40, WRITE 0X80, READ 0X81



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: audio	
LAST MODIFIED BY: MOTOROLA Thu Apr 30 13:38:32 2020			
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		B PROJECT NAME sr28c58995_a.cpm	REV 01
		CAD FILE NAME: PAGE 31 OF 84	

AUDIO: HAPTICS

LINEAR VIBRATOR
BASE I2C ADDR: 0X5A, WRITE 0XB4, READ 0XB5



REFDES: 4700-4719

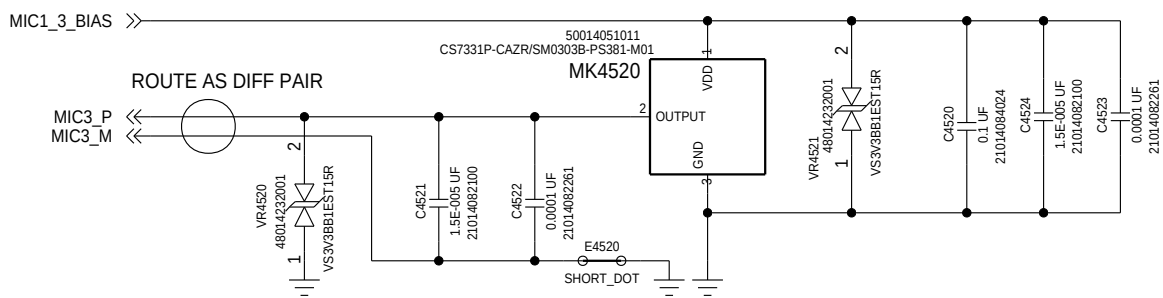
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Apr 30 13:38:41 2020		audio	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	
		B sr28c58995_a.cpm	
		REV 01	
		CAD FILE NAME:	
		PAGE 32 OF 84	

AUDIO: MICS

REF: 4500-4599

MIC 3

REF: 4520-4529



APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER: DENNIS BUDNICK		BLOCK TITLE: audio		
LAST MODIFIED BY: NATE CONNELL Thu Apr 30 13:38:43 2020				
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME		REV
		B sr28c58995_a.cpm		01
		CAD FILE NAME:		PAGE 33 OF 84

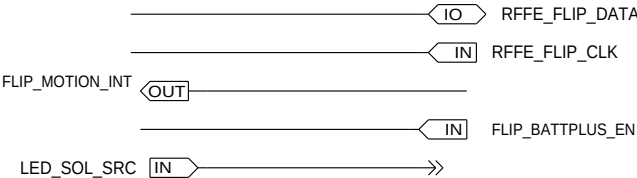
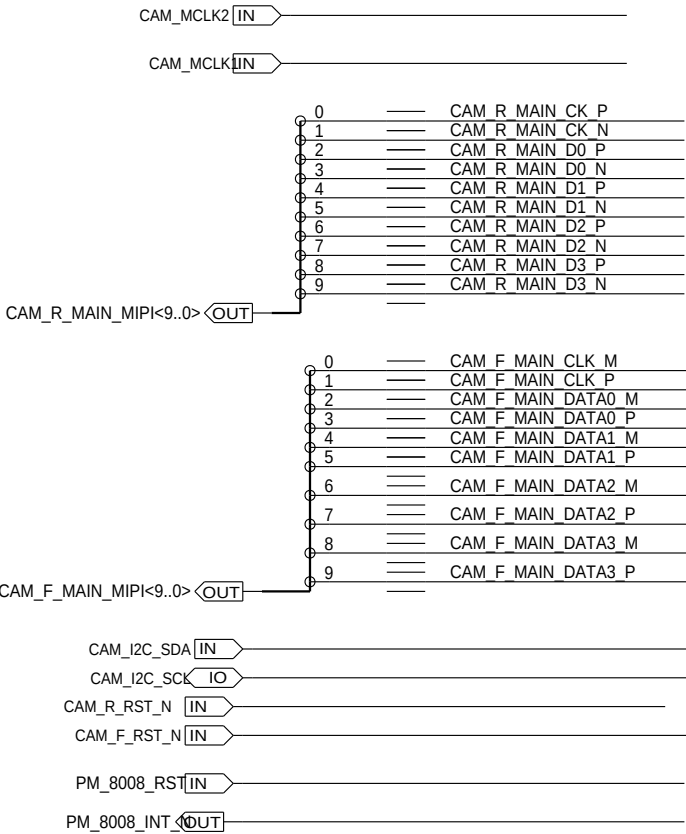
FLIP: PORTS

REF: 4100-4199

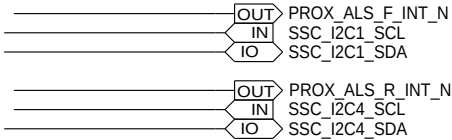
POWER



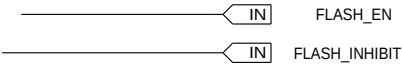
CAMERA



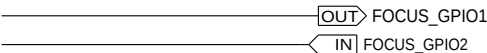
PROX/ALS



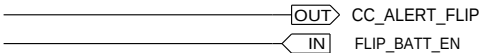
FLASH



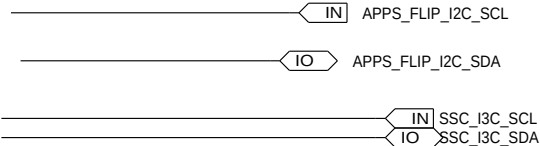
TOF



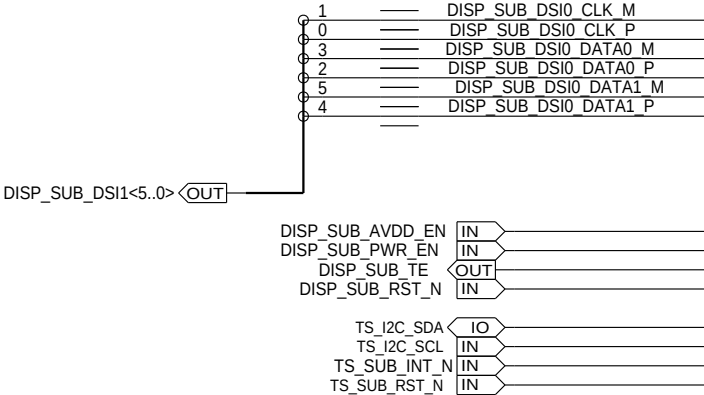
FUEL GUAGE



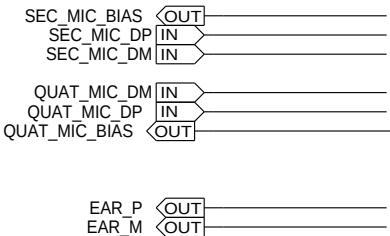
I2C



CLI DISPLAY/TOUCH



AUDIO



SIDE KEYS



APPROVALS

ENGINEER: MOTOROLA

LAST MODIFIED BY: MOTOROLA
Wed Apr 15 10:39:06 2020

MOTOROLA CONFIDENTIAL RESTRICTED

THE DRAWING ON THIS PRINT AND INFORMATION
THEREWITH ARE PROPRIETARY TO MOTOROLA AND
SHALL NOT BE USED OR DISCUSSED IN WHOLE
OR IN PART WITHOUT MOTOROLA'S CONSENT.
© COPYRIGHT, MOTOROLA. THIS DOES
NOT IMPLY PUBLICATION



MOTOROLA

MOBILE DEVICES

BLOCK TITLE:

flip_connector

B

PROJECT NAME

sr28c58995_a.cpm

REV

01

CAD FILE NAME:

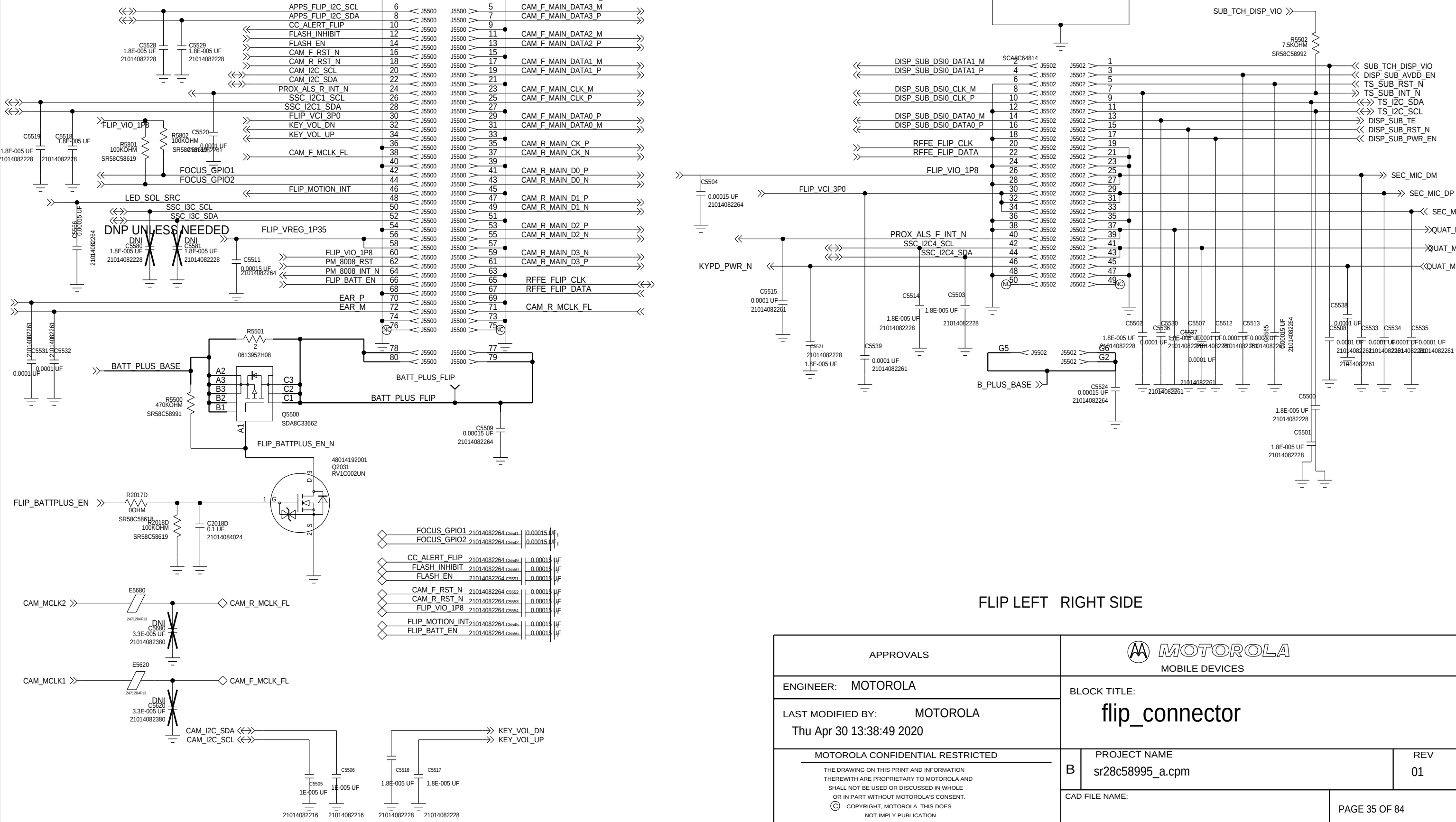
PAGE 34 OF 84

BASE: CONNECTOR

FLIP RIGHT

FLIP LEFT

REF:5500-5549 POWER

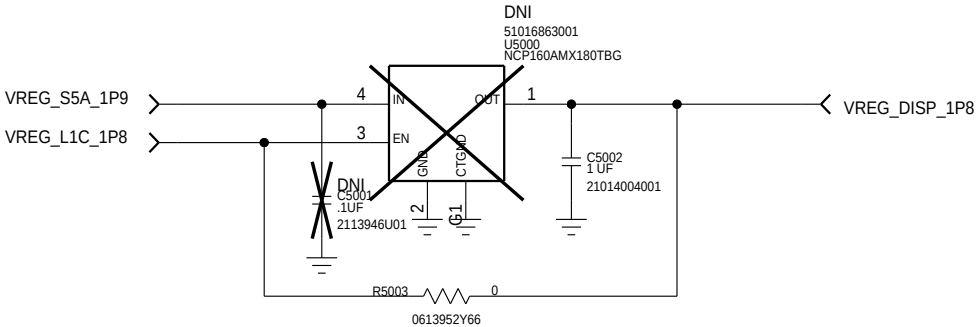


APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Apr 30 13:38:49 2020		flip_connector	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	REV
		B sr28c58995_a.cpm	01
CAD FILE NAME:			PAGE 35 OF 84

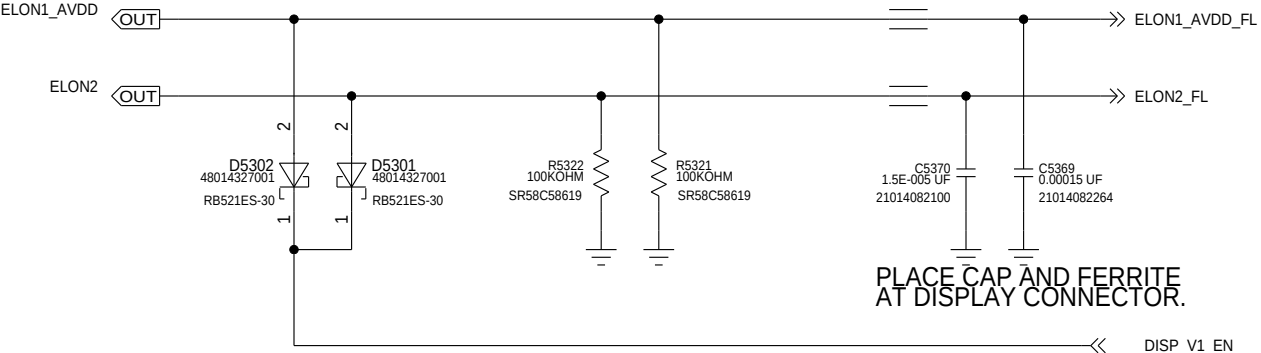
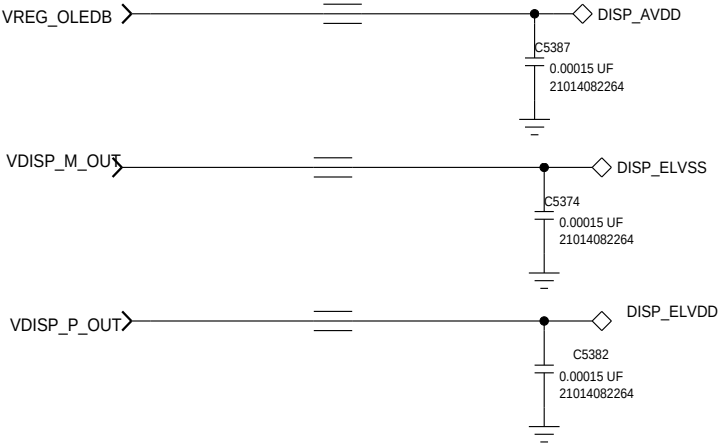
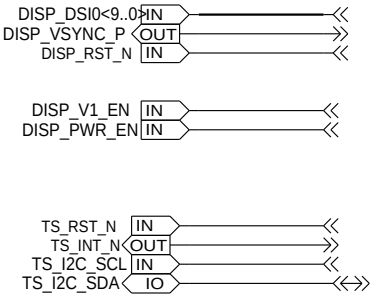
DISP: POWER

REF 5000-5499

DISPLAY TOUCH VIO REG



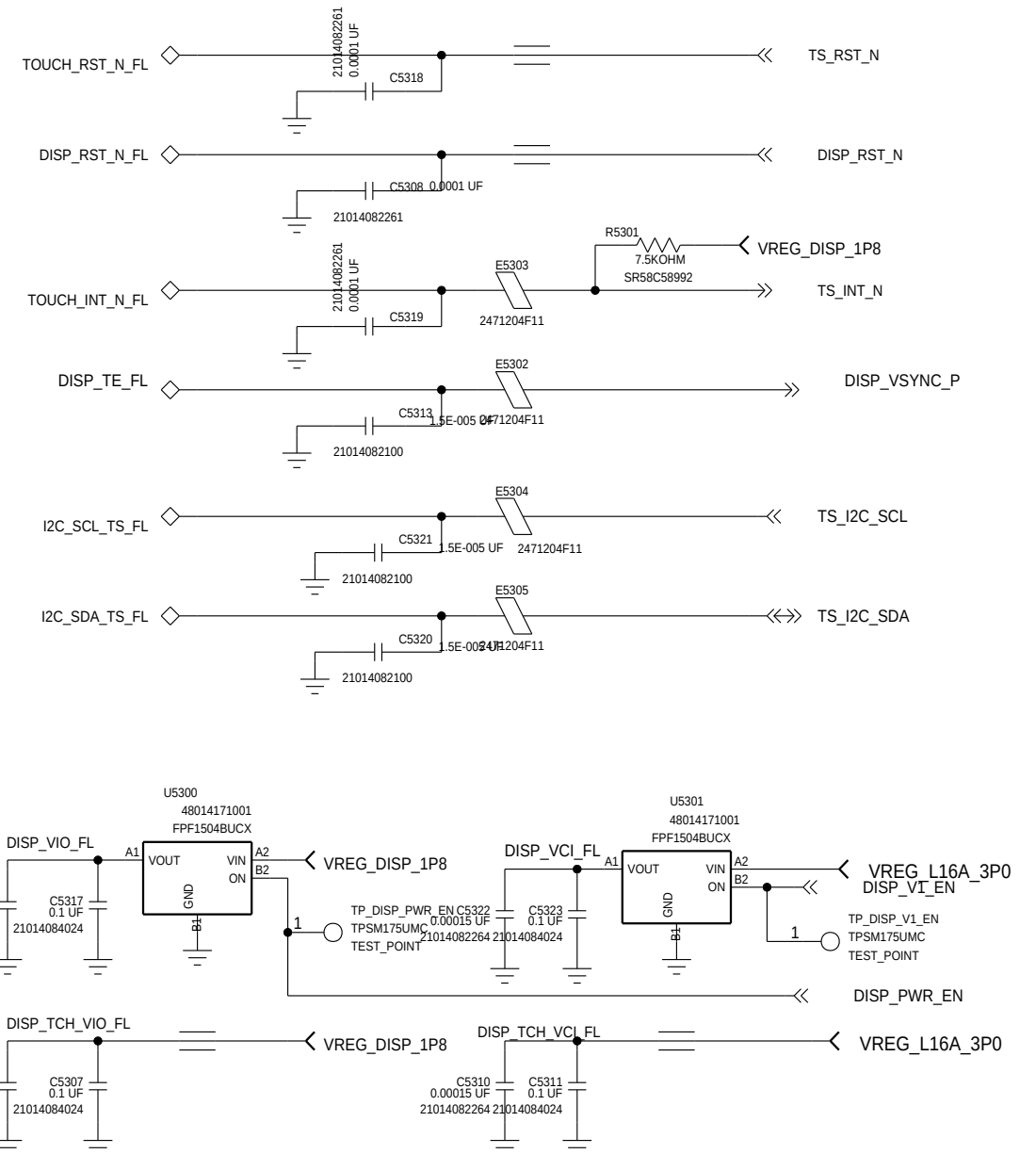
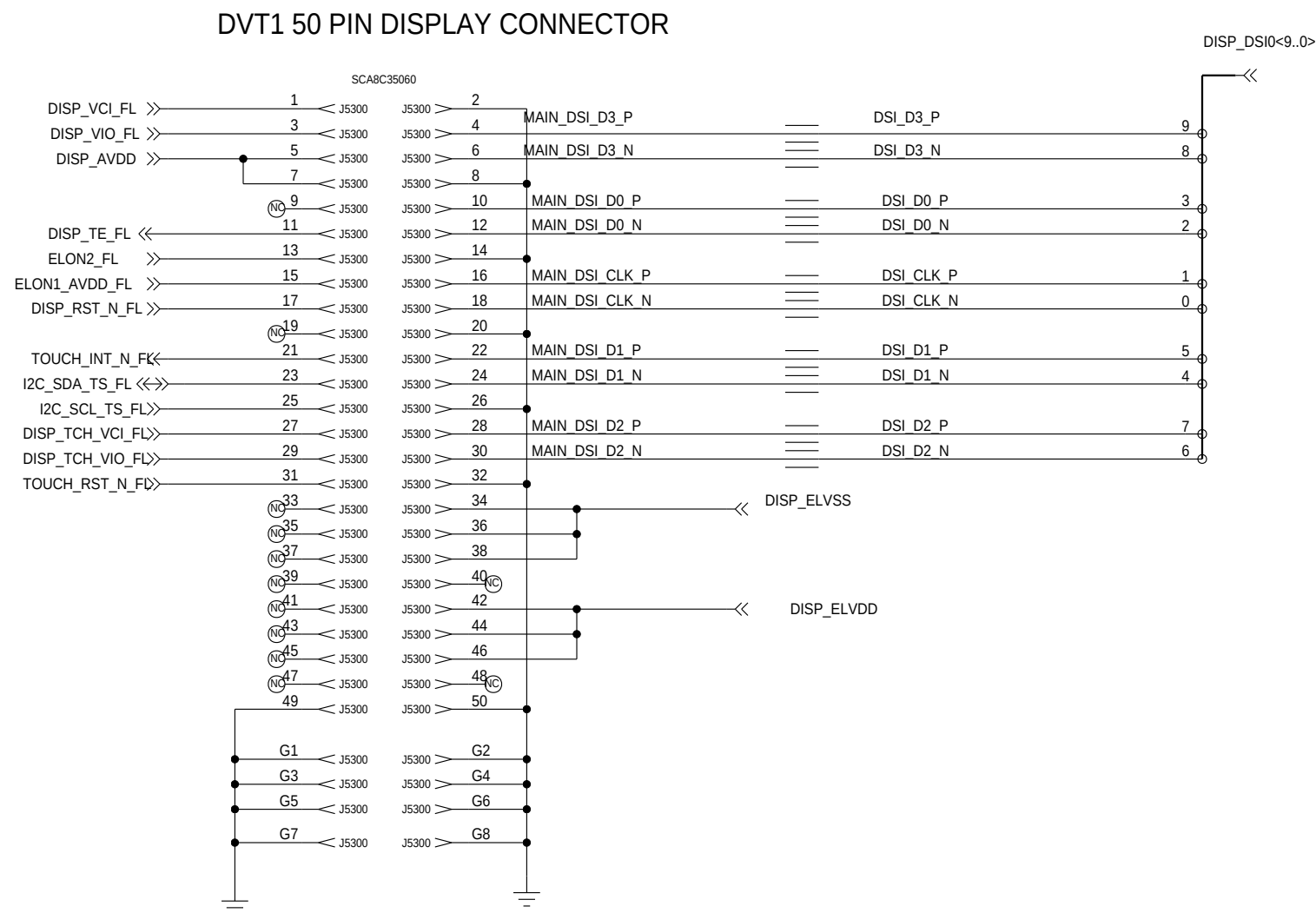
DISPLAY PORTS



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: display	
LAST MODIFIED BY: MOTOROLA Thu Apr 30 13:38:55 2020		PROJECT NAME B sr28c58995_a.cpm	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		CAD FILE NAME: PAGE 36 OF 84	
		REV 01	

DISP: CONNECTOR

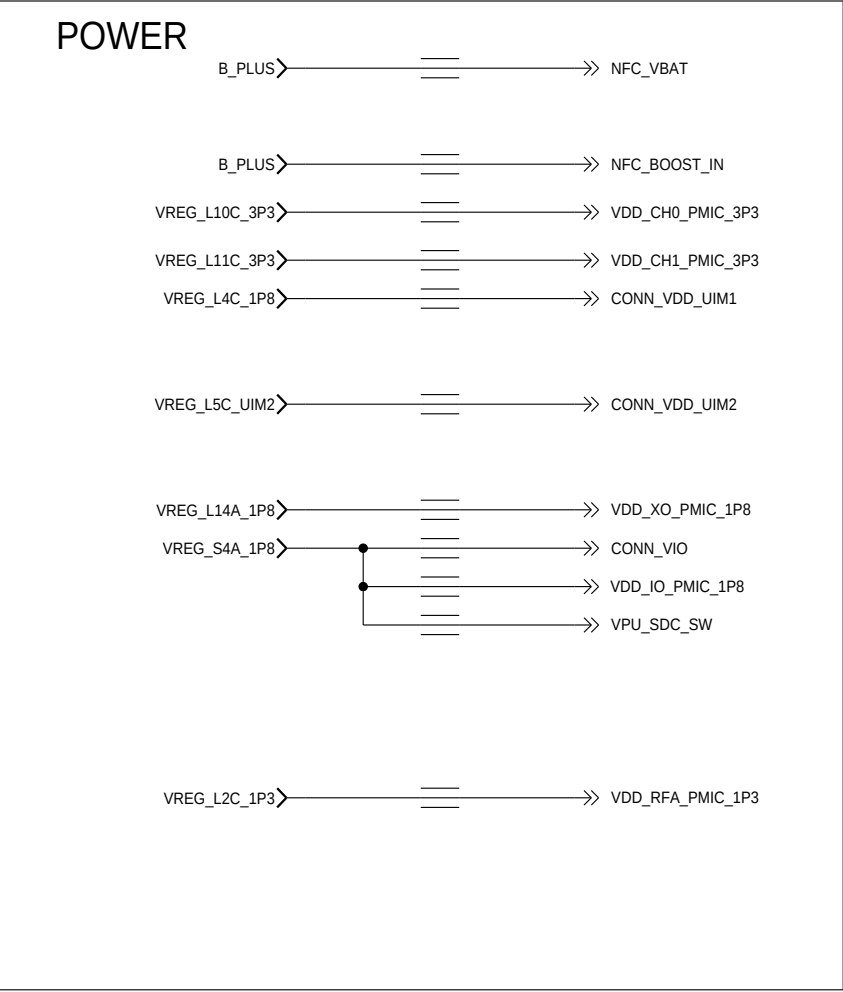
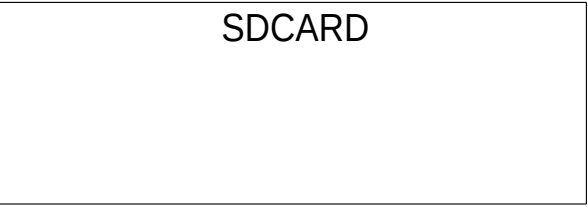
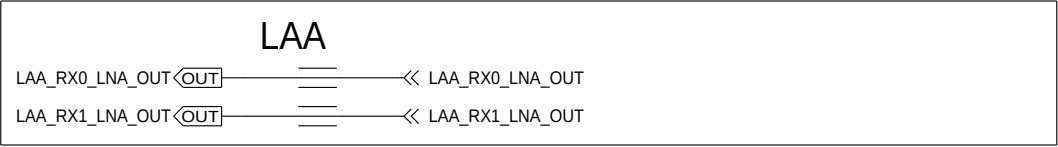
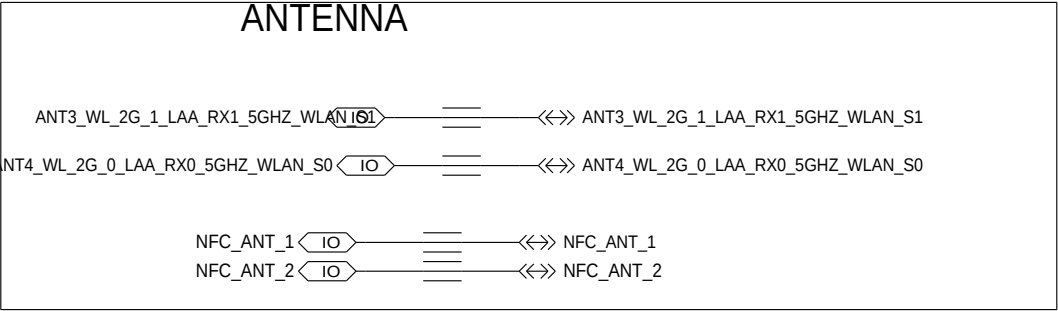
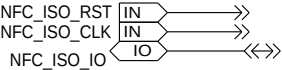
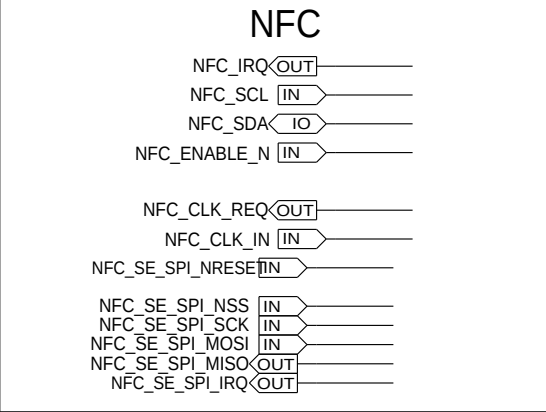
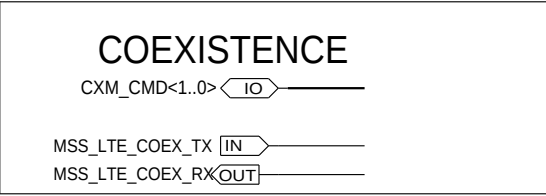
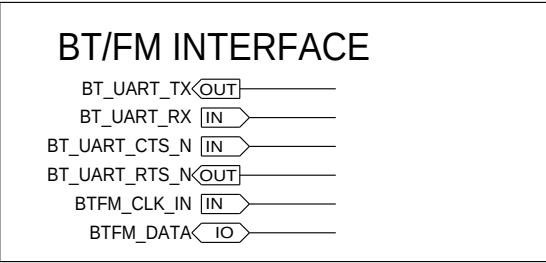
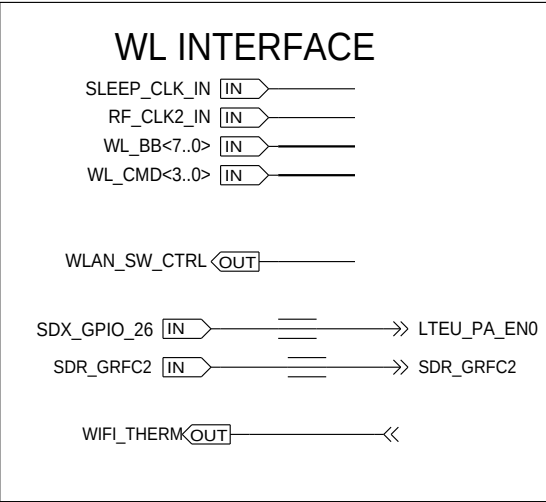
REF 5300-5349



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: NATE CONNELL	BLOCK TITLE: display		
LAST MODIFIED BY: NATE CONNELL Thu Apr 30 13:38:58 2020			
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	
THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		B	sr28c58995_a.cpm
		CAD FILE NAME:	
		PAGE 37 OF 84	

CONN: PORTS/POWER

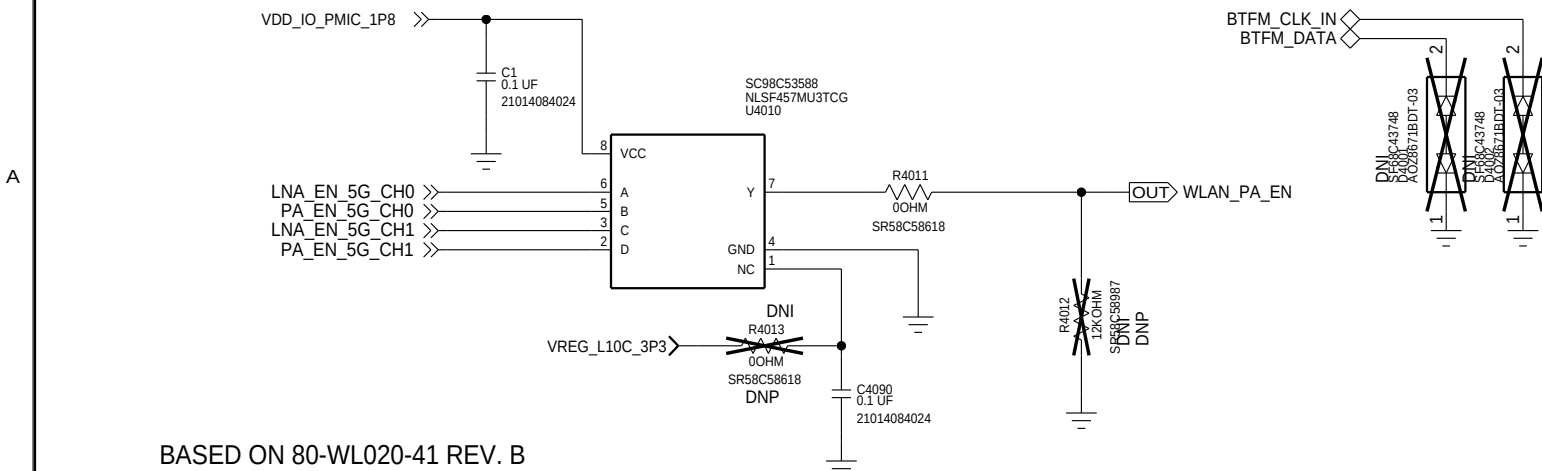
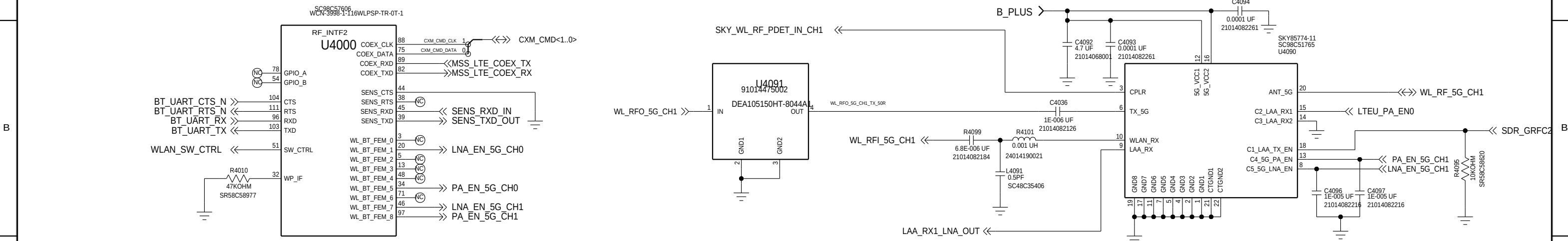
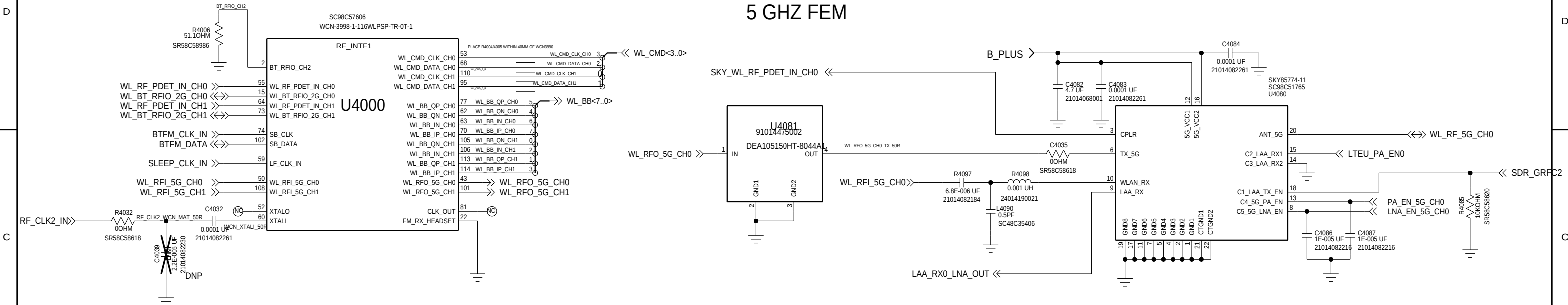
REF 4000-4499



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Wed Apr 15 10:38:59 2020		connectivity	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	
		B sr28c58995_a.cpm	REV 01
		CAD FILE NAME:	
		PAGE 38 OF 84	

CONN: WCN3998 RF/INTERFACES

REF: 4000-4099



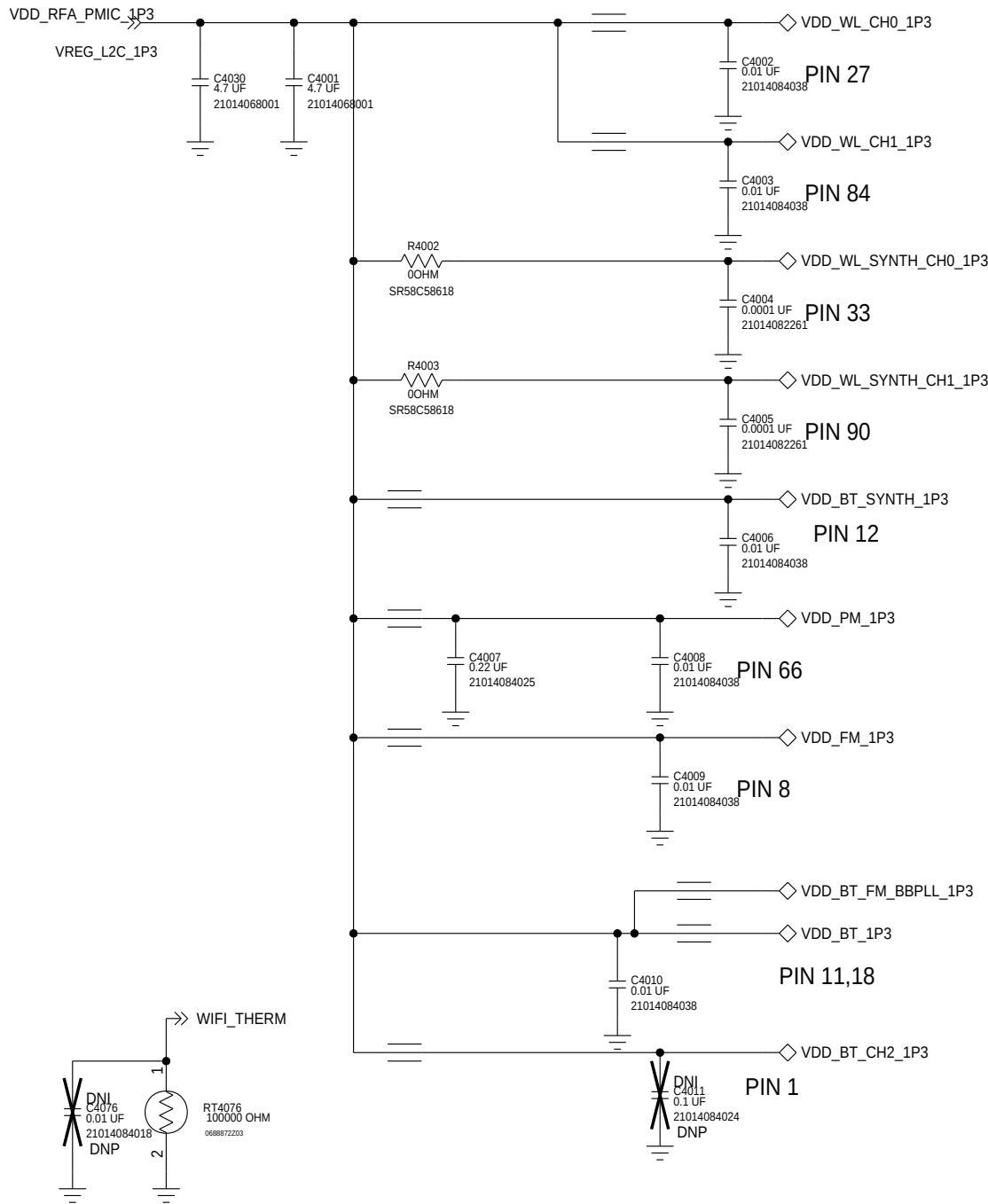
BASED ON 80-WL020-41 REV. B

MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

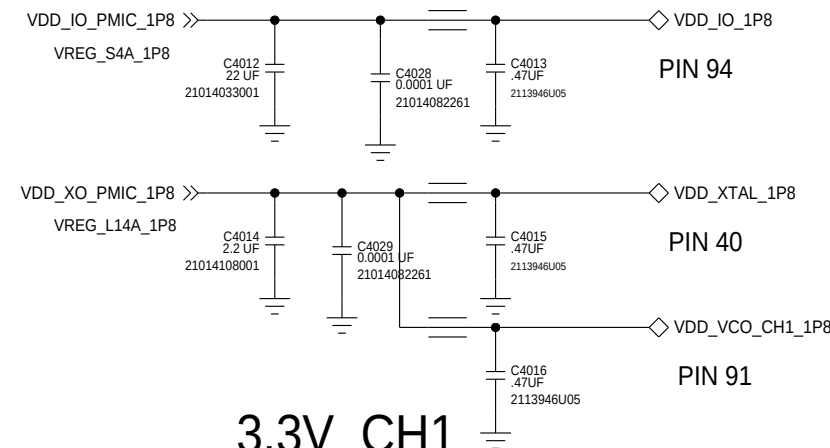
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: connectivity	
LAST MODIFIED BY: MOTOROLA Fri May 01 16:12:05 2020			
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME B sr28c58995_a.cpm	REV 01
		CAD FILE NAME:	PAGE 39 OF 84

CONN: WCN3998 POWER/GND
REF 4000-4029

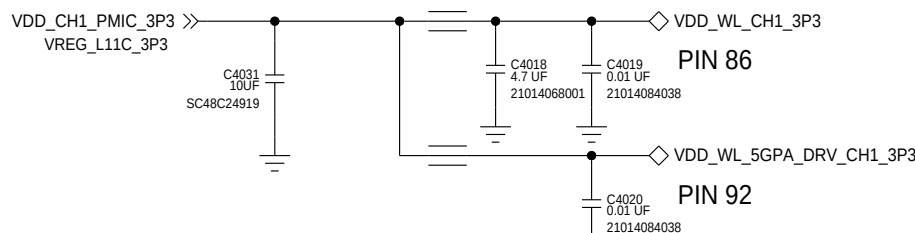
BT_FM_WLAN_1P3



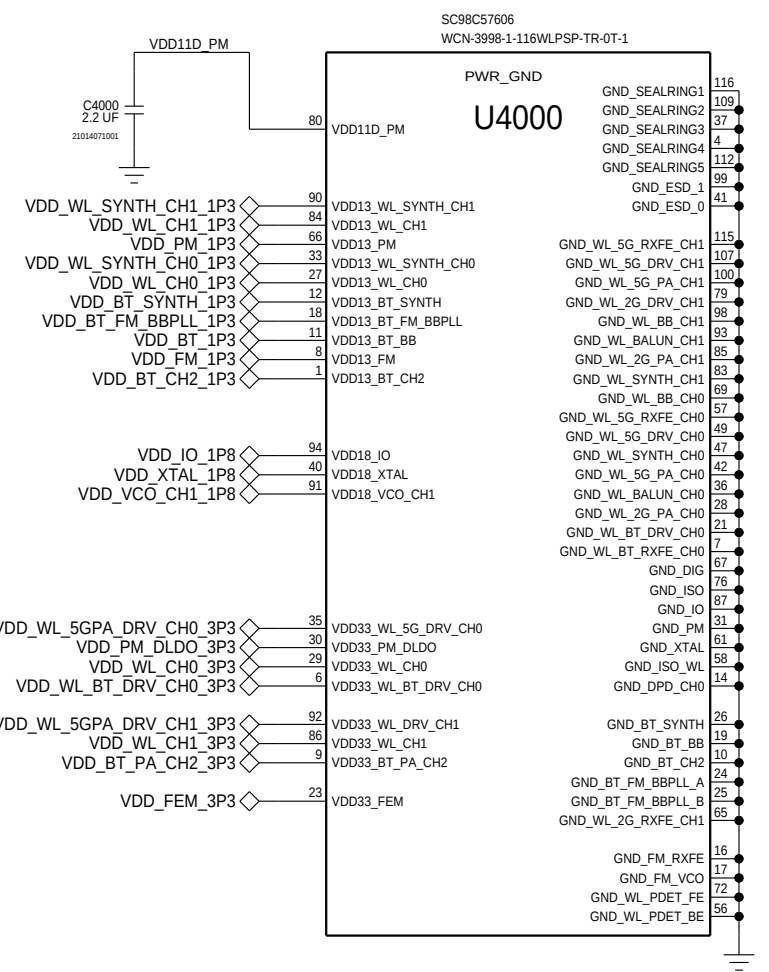
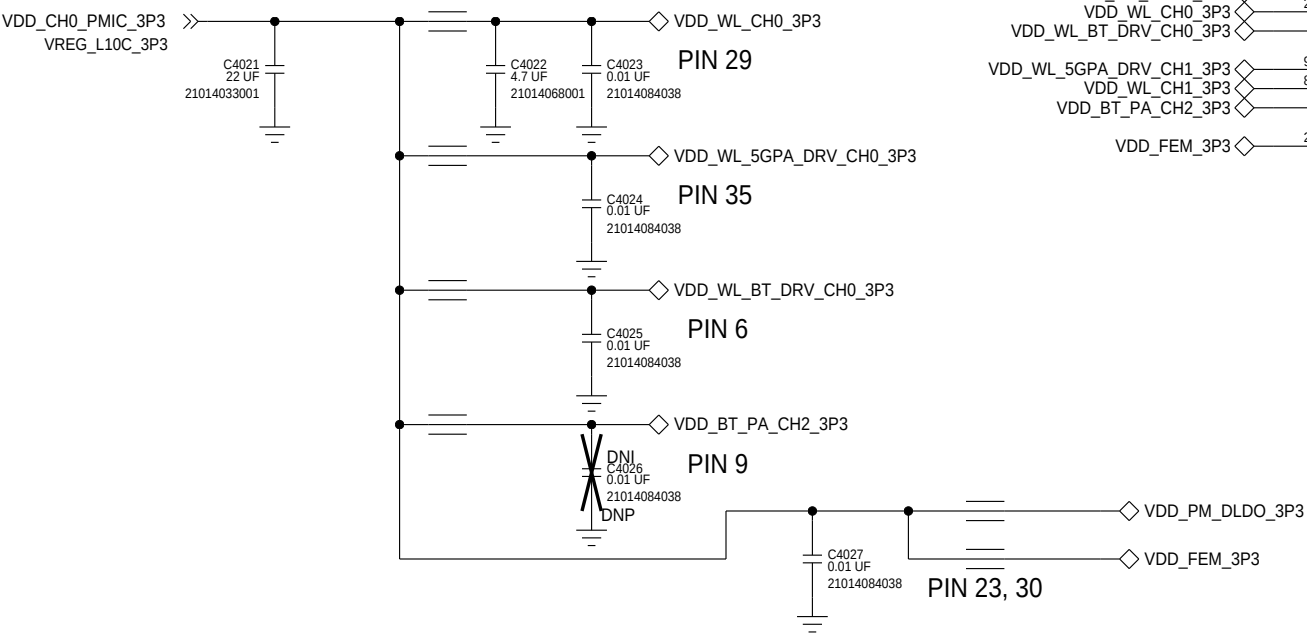
1.8V



3.3V_CH1



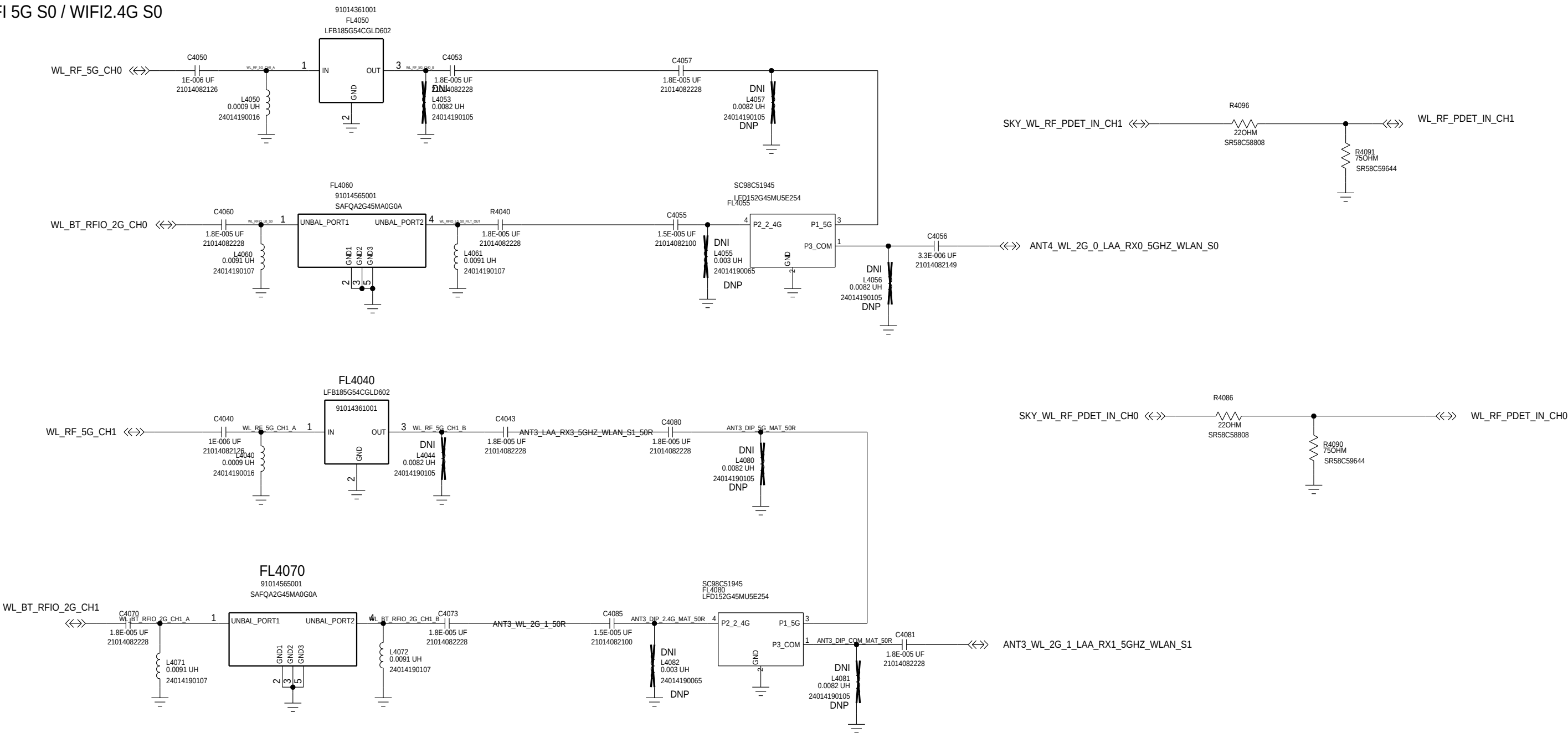
3.3V_CH0



BASED ON 80-WL020-41 REV. B

APPROVALS		BLOCK TITLE:	
ENGINEER: MOTOROLA		connectivity	
LAST MODIFIED BY: MOTOROLA Thu Apr 30 13:39:19 2020		PROJECT NAME sr28c58995_a.cpm	
MOTOROLA CONFIDENTIAL RESTRICTED		REV 01	
THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PAGE 40 OF 84	

CONN: WCN3998 RADIO
REF 4040-4099
WIFI 5G S0 / WIFI2.4G S0



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: connectivity	
LAST MODIFIED BY: MOTOROLA Thu Apr 30 13:39:11 2020		PROJECT NAME sr28c58995_a.cpm	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		REV 01	
		PAGE 41 OF 84	

REF: 4100-4199



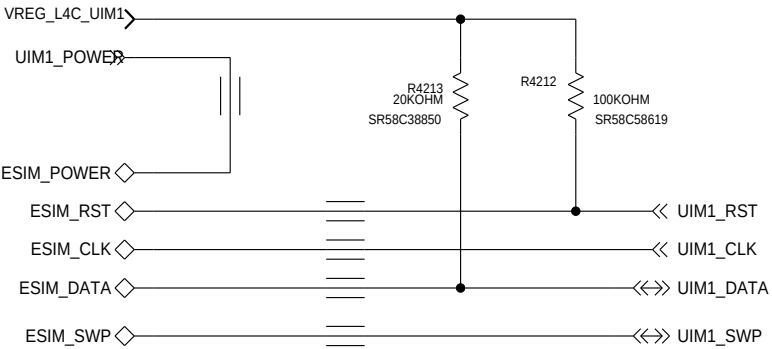
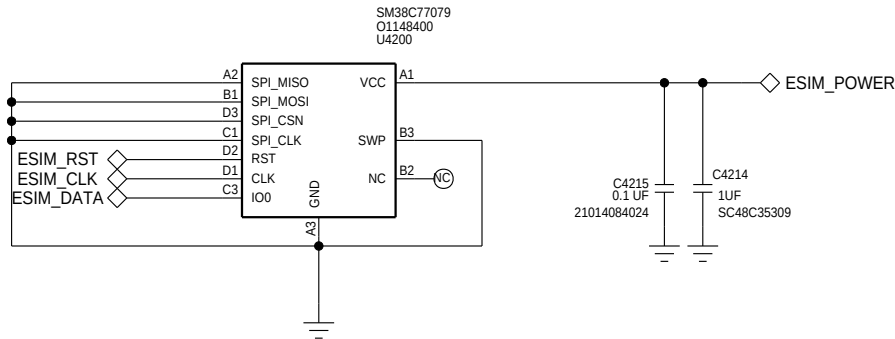
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA	BLOCK TITLE: connectivity		
LAST MODIFIED BY: MOTOROLA Thu Apr 30 13:39:03 2020			
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME B sr28c58995_a.cpm	REV 01
		CAD FILE NAME:	PAGE 42 OF 84

A



8	7	6	5	4	3	2
---	---	---	---	---	---	---

CONN: ESIM
REF:4200-4299

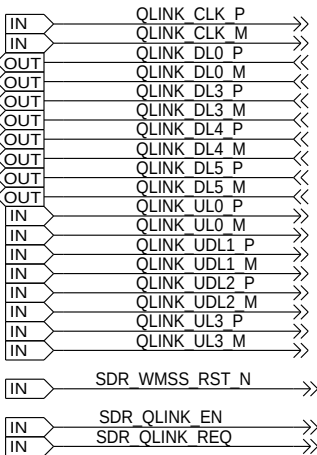
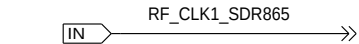
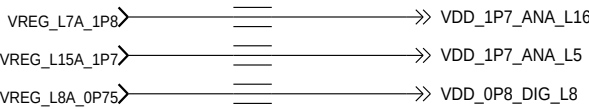


APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Tue May 12 15:00:25 2020		connectivity	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	
		B sr28c58995_a.cpm	
		REV 01	
		CAD FILE NAME:	
		PAGE 44 OF 84	

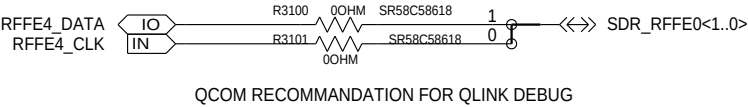
SDR865: PORTS

REF 3100-3149

SUPPLIES



RFFE MIPI

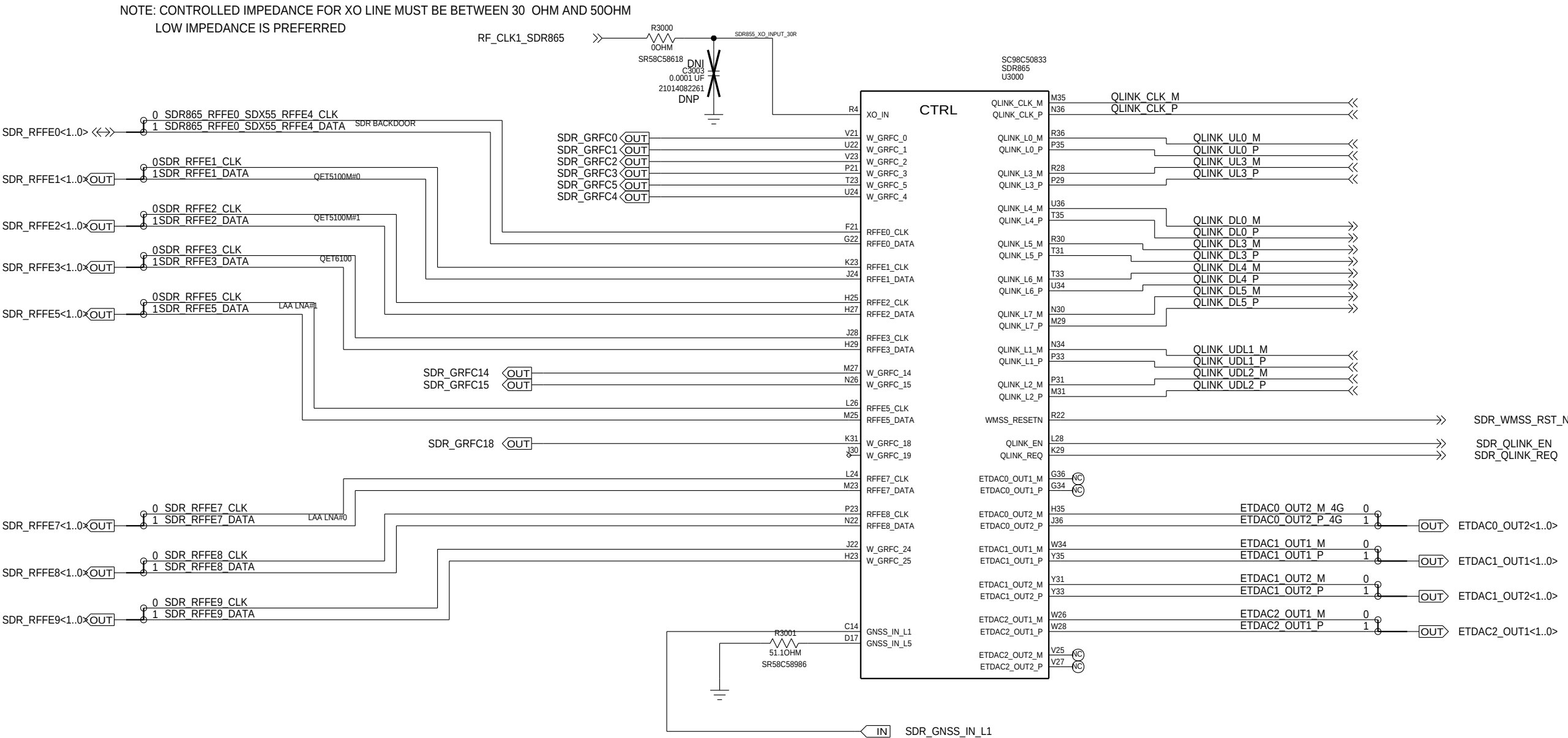


CURRENT DRAIN

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Apr 30 13:39:45 2020		blk_sdr660	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	
		sr28c58995_a.cpm	
		REV 01	
		PAGE 45 OF 84	

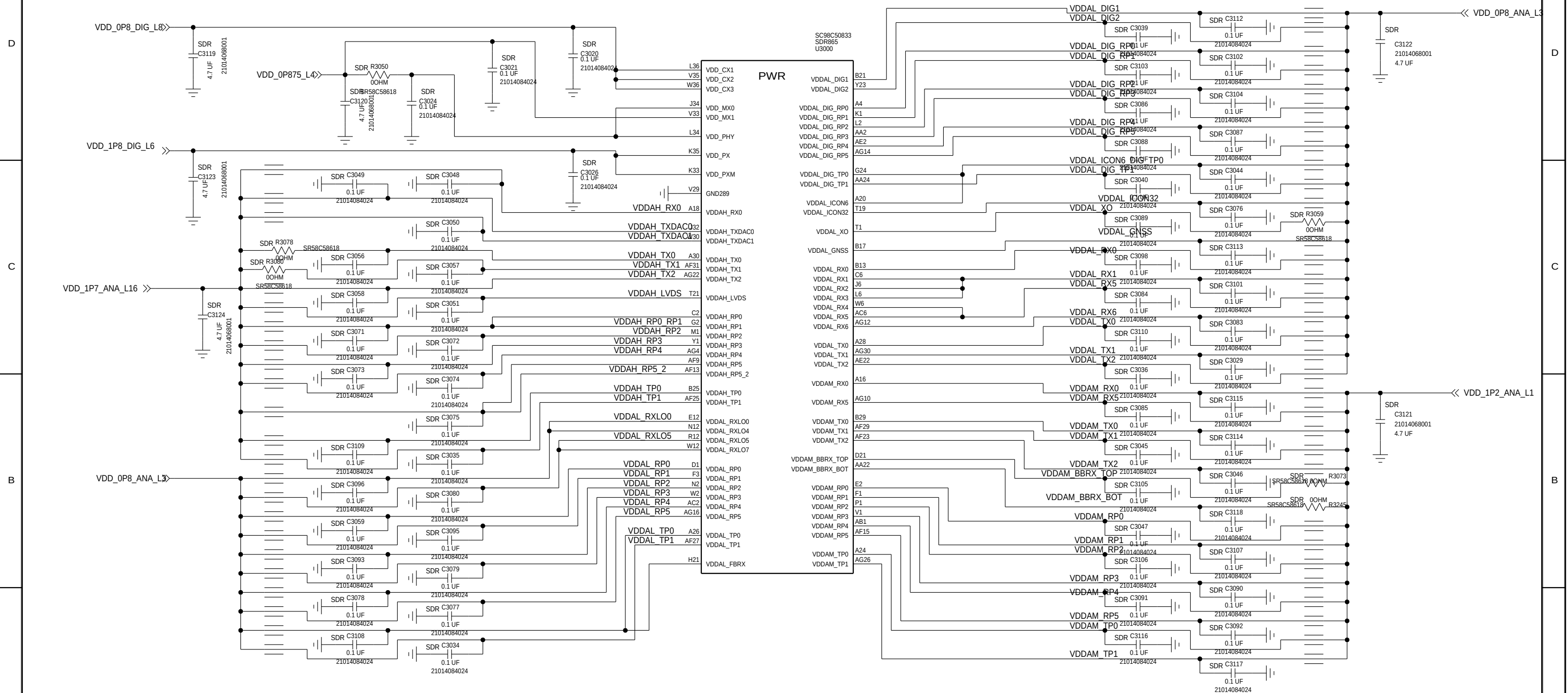
SDR865: CTRL / QLINK

REF 3000-3009



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Apr 30 13:39:38 2020		blk_sdr660	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	
		B 828458095_a.cpm	
		REV 01	
		CAD FILE NAME:	
		PAGE 46 OF 84	

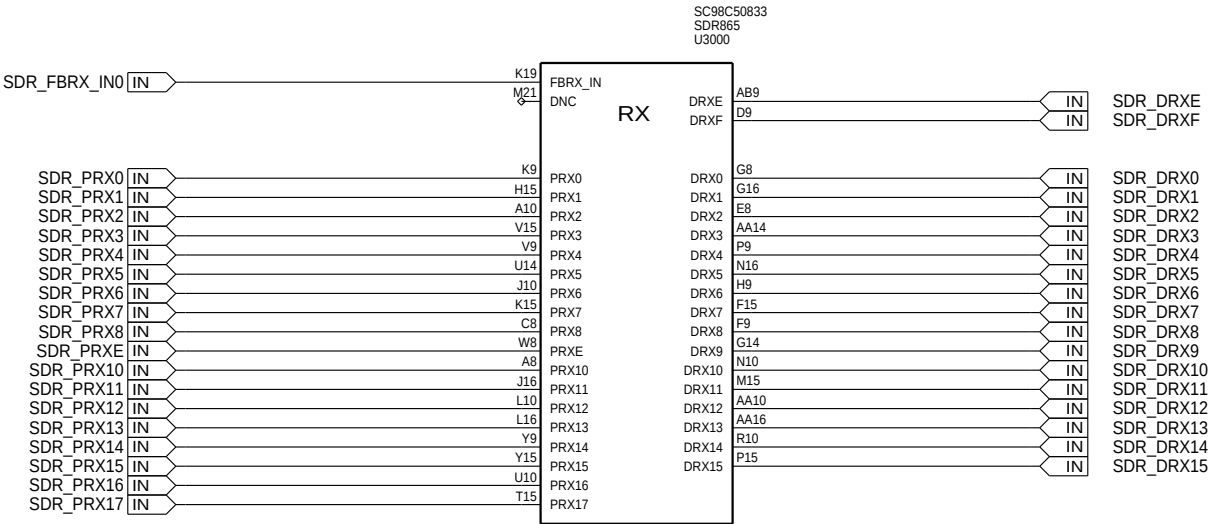
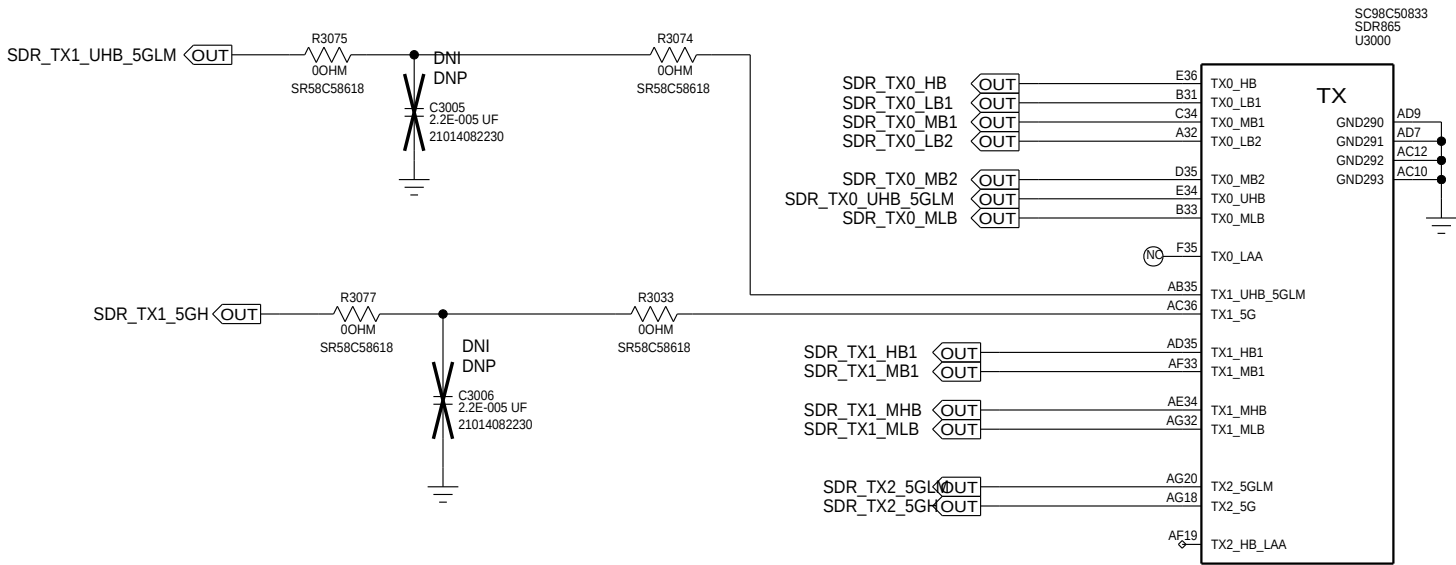
SDR865: POWER



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:	BLOCK TITLE: blk_sdr660		
LAST MODIFIED BY: Thu Apr 30 13:39:28 2020			
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME B 807150005_a.cpm	REV 01
		CAD FILE NAME:	PAGE 47 OF 84

SDR865: TX & RX

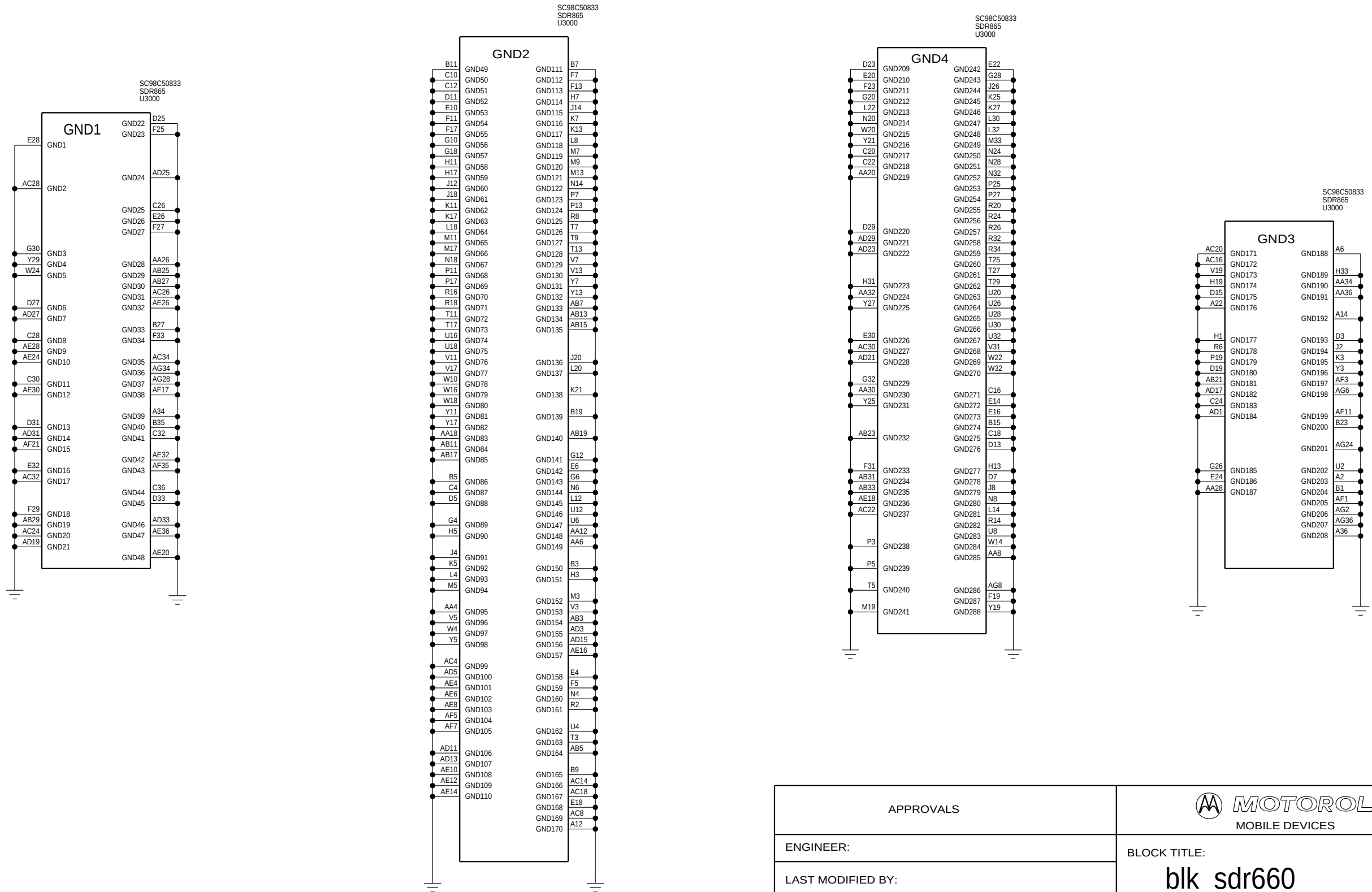
REF 3050-3099



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: Thu Apr 30 13:39:34 2020		blk_sdr660	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME B 6020-50005_a.cpm	
		CAD FILE NAME:	
		REV 01	
		PAGE 48 OF 84	

SDR865: GND

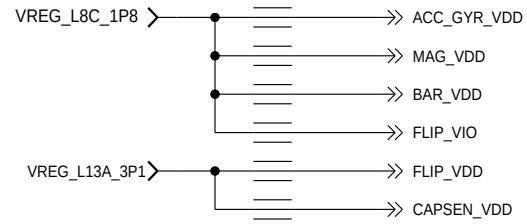
REF 3100-3149



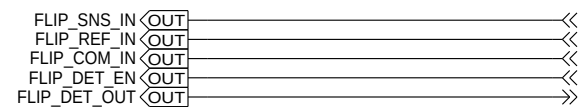
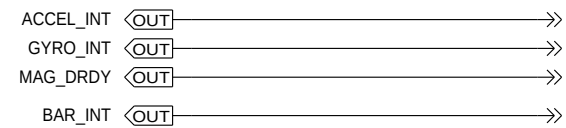
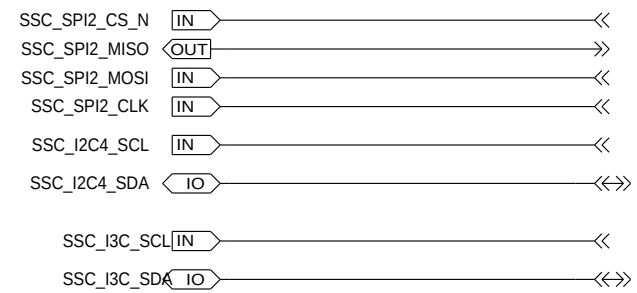
APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY:		blk_sdr660	
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	REV
THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		B 6028150095_a.cpm	01
		CAD FILE NAME:	
		PAGE 49 OF 84	

SENSOR: PORTS

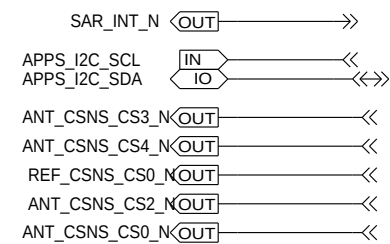
REF 000-499



INTERFACES - SENSOR



INTERFACES - CAPSENSE

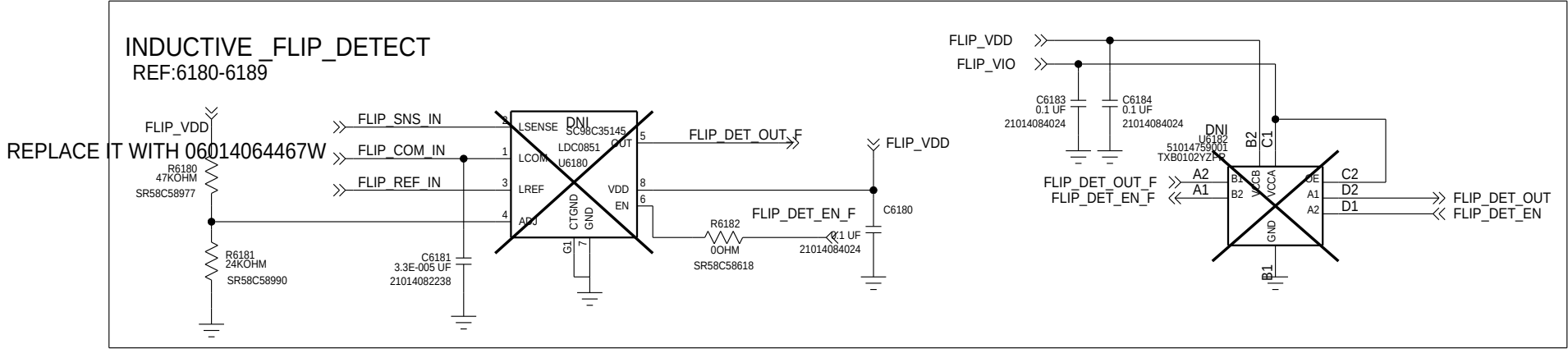


APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:	BLOCK TITLE: sensor		
LAST MODIFIED BY: Wed Apr 15 10:38:55 2020			
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME B sr28c58995_a.cpm	REV 01
		CAD FILE NAME:	PAGE 50 OF 84

D

A

SENSOR: INDUCTIVE
REF: 6100-6199



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Apr 30 13:39:55 2020		sensor	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	
		B sr28c58995_a.cpm	
		CAD FILE NAME:	
		PAGE 52 OF 84	
REV 01			

8

7

6

5

4

3

2

L

SENSOR: FINGERPRINT
REF 1500-1599

D

○ |

C

(

B

3 |

A

A |

APPROVALS



ENGINEER:

BLOCK TITLE:

LAST MODIFIED BY:
Mon Jun 10 14:11:36 2019

sensor

MOTOROLA CONFIDENTIAL RESTRICTED

THE DRAWING ON THIS PRINT AND INFORMATION
THEREWITH ARE PROPRIETARY TO MOTOROLA AND
SHALL NOT BE USED OR DISCUSSED IN WHOLE
OR IN PART WITHOUT MOTOROLA'S CONSENT.
© COPYRIGHT, MOTOROLA. THIS DOES
NOT IMPLY PUBLICATION

B	PROJECT NAME sr28c58995_a.cpm	REV 01
---	----------------------------------	-----------

CAD FILE NAME:	PAGE 53 OF 84
----------------	---------------

8

7

6

5

4

3

2

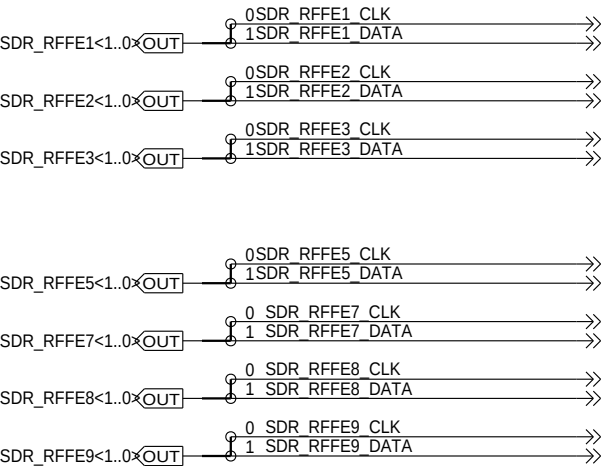
1

8	7	6	5	4	3	2	1																		
RFFE: BLOCK DIAGRAM REFDES: 7000-8999																									
D							D																		
C							C																		
B							B																		
A							A																		
8	7	6	5	4	3	2	1																		
				<table><tr><td colspan="2">APPROVALS</td><td colspan="2">  MOTOROLA MOBILE DEVICES </td></tr><tr><td colspan="2">ENGINEER: MOTOROLA</td><td colspan="2" rowspan="2">BLOCK TITLE: rffe_mtk6 </td></tr><tr><td colspan="2">LAST MODIFIED BY: MOTOROLA Thu Mar 01 17:16:19 2018</td></tr><tr><td colspan="2" rowspan="2"> MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION </td><td>B</td><td>PROJECT NAME sr28c58995_a.cpm</td><td>REV 01</td></tr><tr><td colspan="2">CAD FILE NAME:</td><td>PAGE 54 OF 84</td></tr></table>		APPROVALS		 MOTOROLA MOBILE DEVICES		ENGINEER: MOTOROLA		BLOCK TITLE: rffe_mtk6		LAST MODIFIED BY: MOTOROLA Thu Mar 01 17:16:19 2018		MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		B	PROJECT NAME sr28c58995_a.cpm	REV 01	CAD FILE NAME:		PAGE 54 OF 84		
APPROVALS		 MOTOROLA MOBILE DEVICES																							
ENGINEER: MOTOROLA		BLOCK TITLE: rffe_mtk6																							
LAST MODIFIED BY: MOTOROLA Thu Mar 01 17:16:19 2018																									
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		B	PROJECT NAME sr28c58995_a.cpm	REV 01																					
		CAD FILE NAME:		PAGE 54 OF 84																					

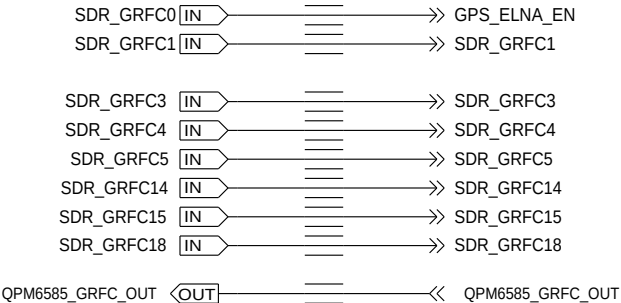
RFFE: PORTS

REFDES: 7000-8199

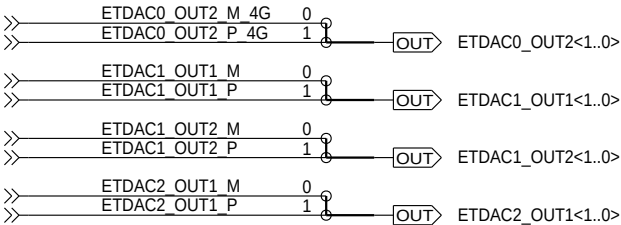
INTERFACES - RFFE MIPI



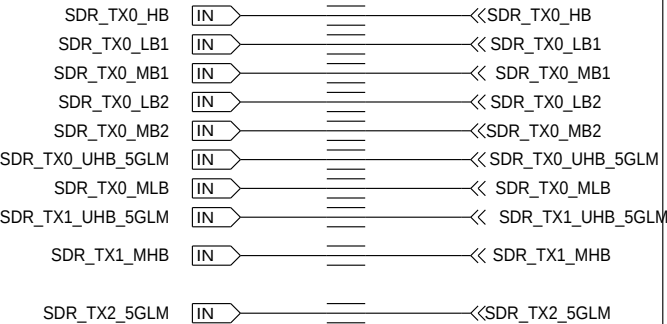
INTERFACES - GRFC



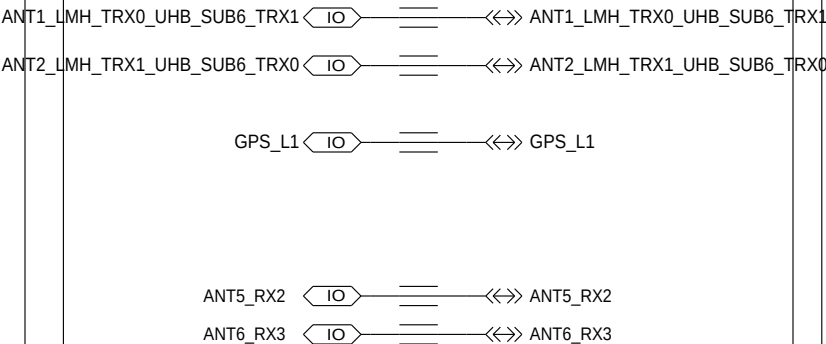
ET DAC



TRANSMITTER - RF

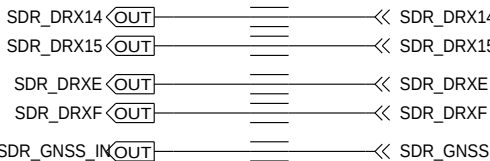
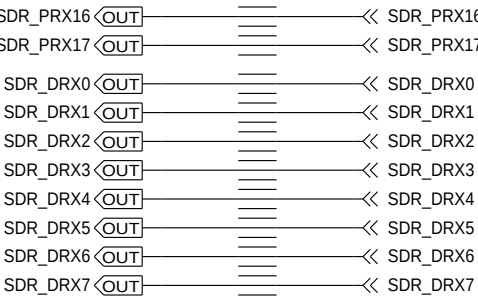
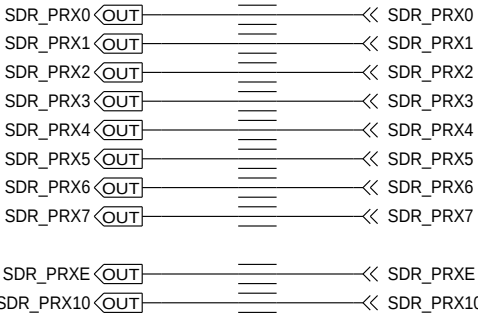


OUTPUTS - RF ANTENNA CONNECTORS

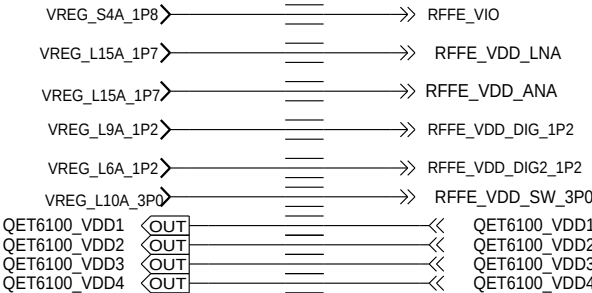


QUALCOMM RECOMMENDS PLACING BOTH THERMISTORS FOR FUTURE SW IMPLEMENTATION THEY ARE WORKING ON

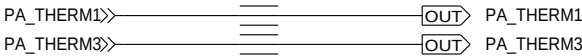
RECEIVER - RF



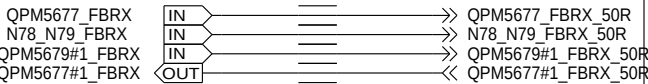
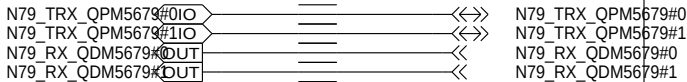
POWER



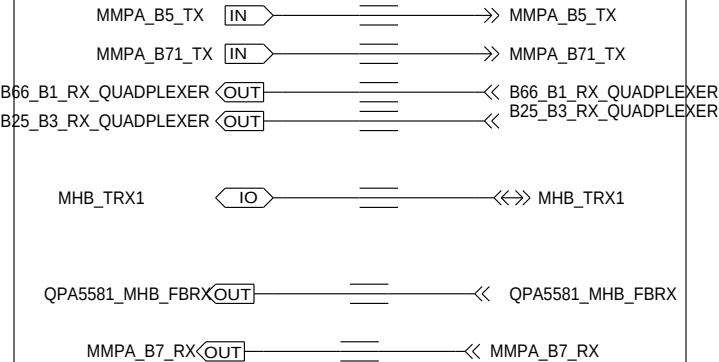
THERMISTORS



CHINA RF



NA RF



APPROVALS

ENGINEER: MOTOROLA

LAST MODIFIED BY: MOTOROLA
Wed Apr 15 10:38:46 2020

MOTOROLA CONFIDENTIAL RESTRICTED

THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT.
© COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION



MOTOROLA

MOBILE DEVICES

BLOCK TITLE:

rffe_mtk6

B

PROJECT NAME

sr28c58995_a.cpm

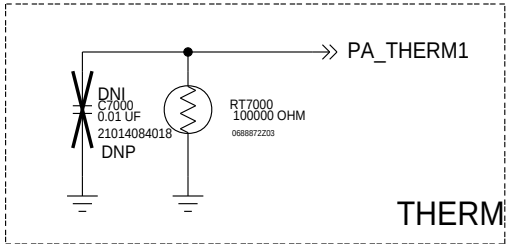
REV

01

CAD FILE NAME:

PAGE 55 OF 84

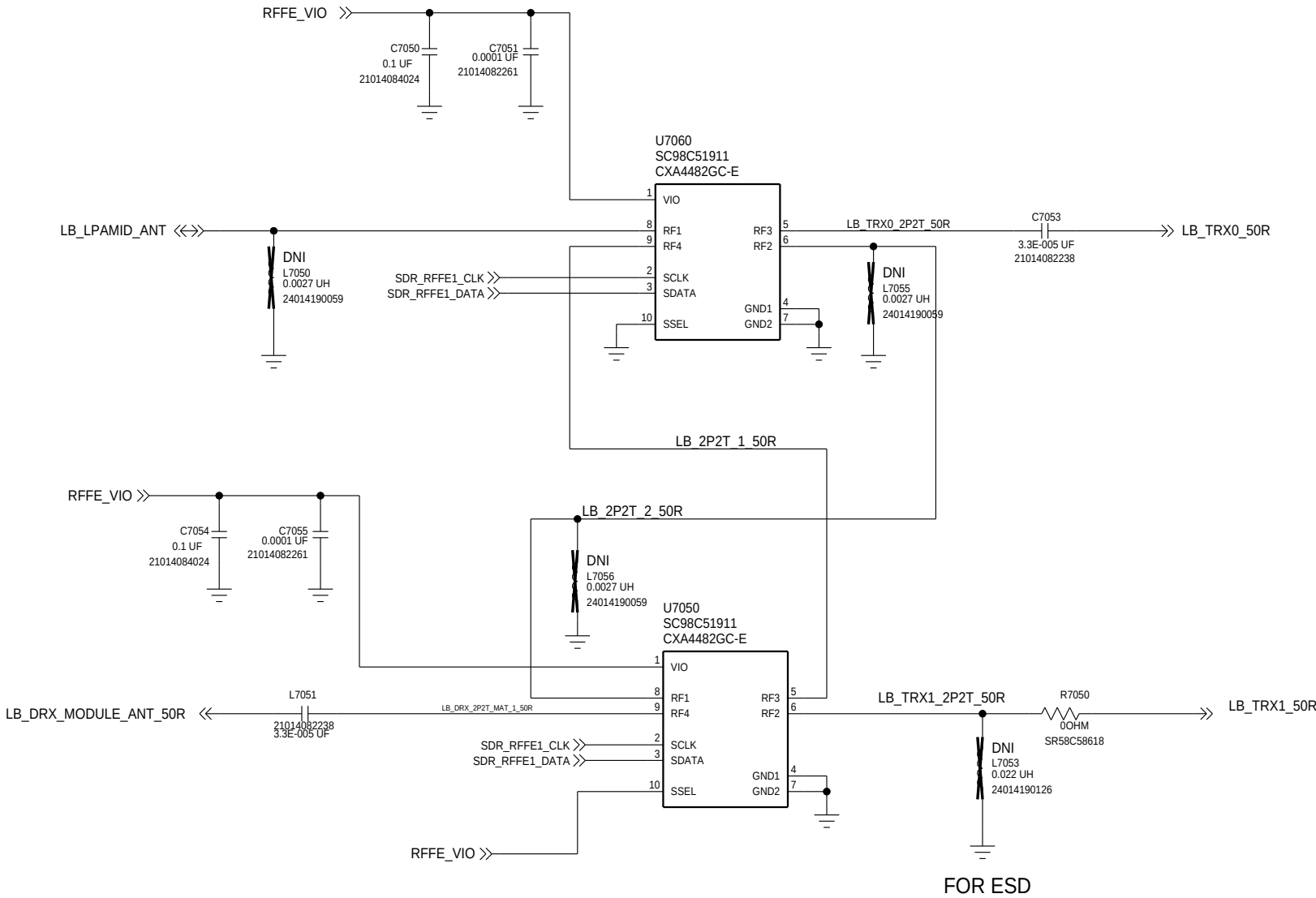
REFDES: 7000-7024



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:	BLOCK TITLE: rffe_mtk6		
LAST MODIFIED BY: Wed May 06 13:58:36 2020			
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME B sr28c58995_a.cpm	REV 01
		CAD FILE NAME:	PAGE 56 OF 84

RFFE: LB SWITCH

REFDES: 7050-7099

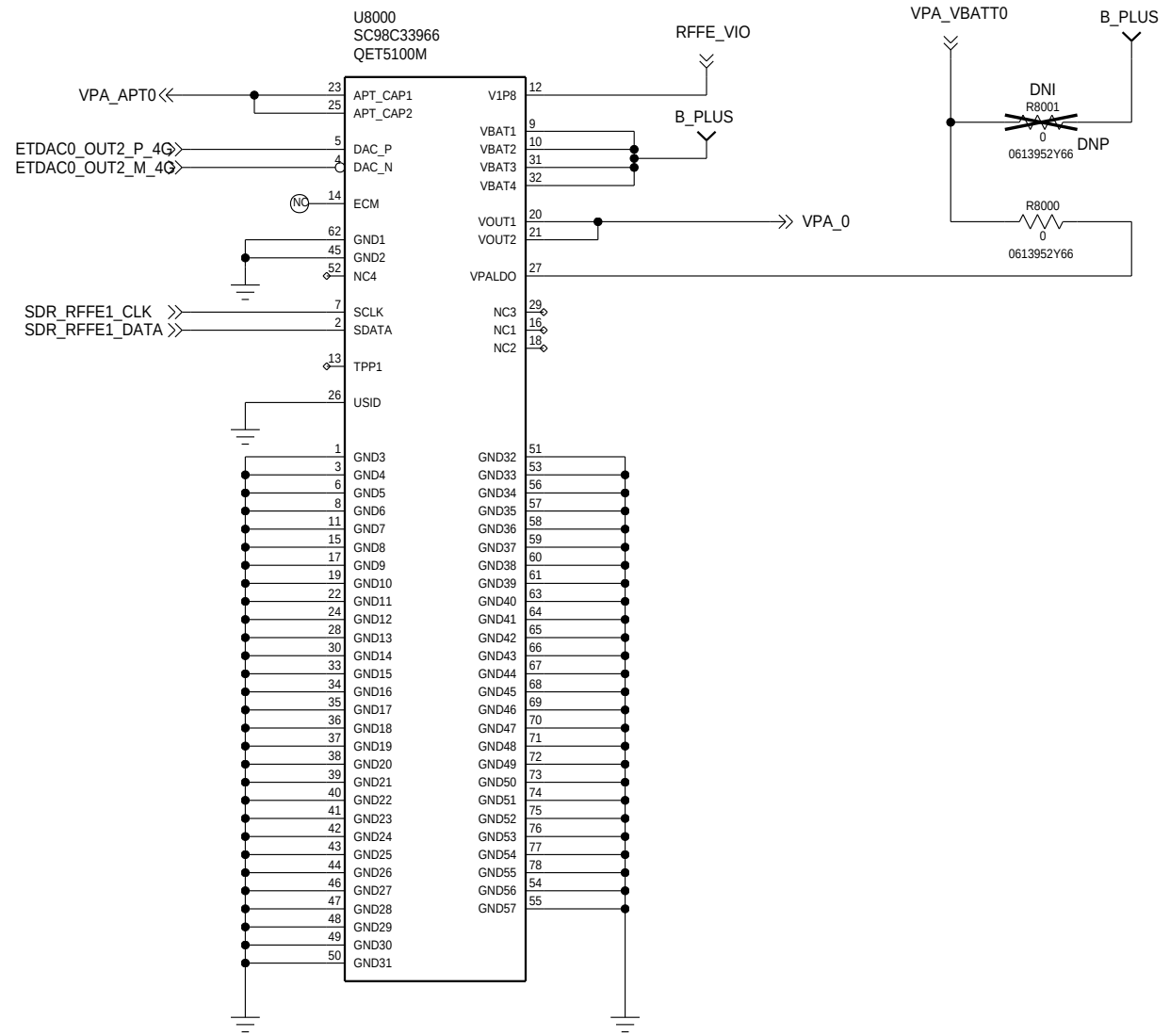


FOR ESD

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: Thu Apr 30 13:40:37 2020		rffe_mtk6	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	
		B sr28c58995_a.cpm	
		CAD FILE NAME:	
		PAGE 57 OF 84	
REV 01			

RFFE: QET5100M

REF 8000-8099



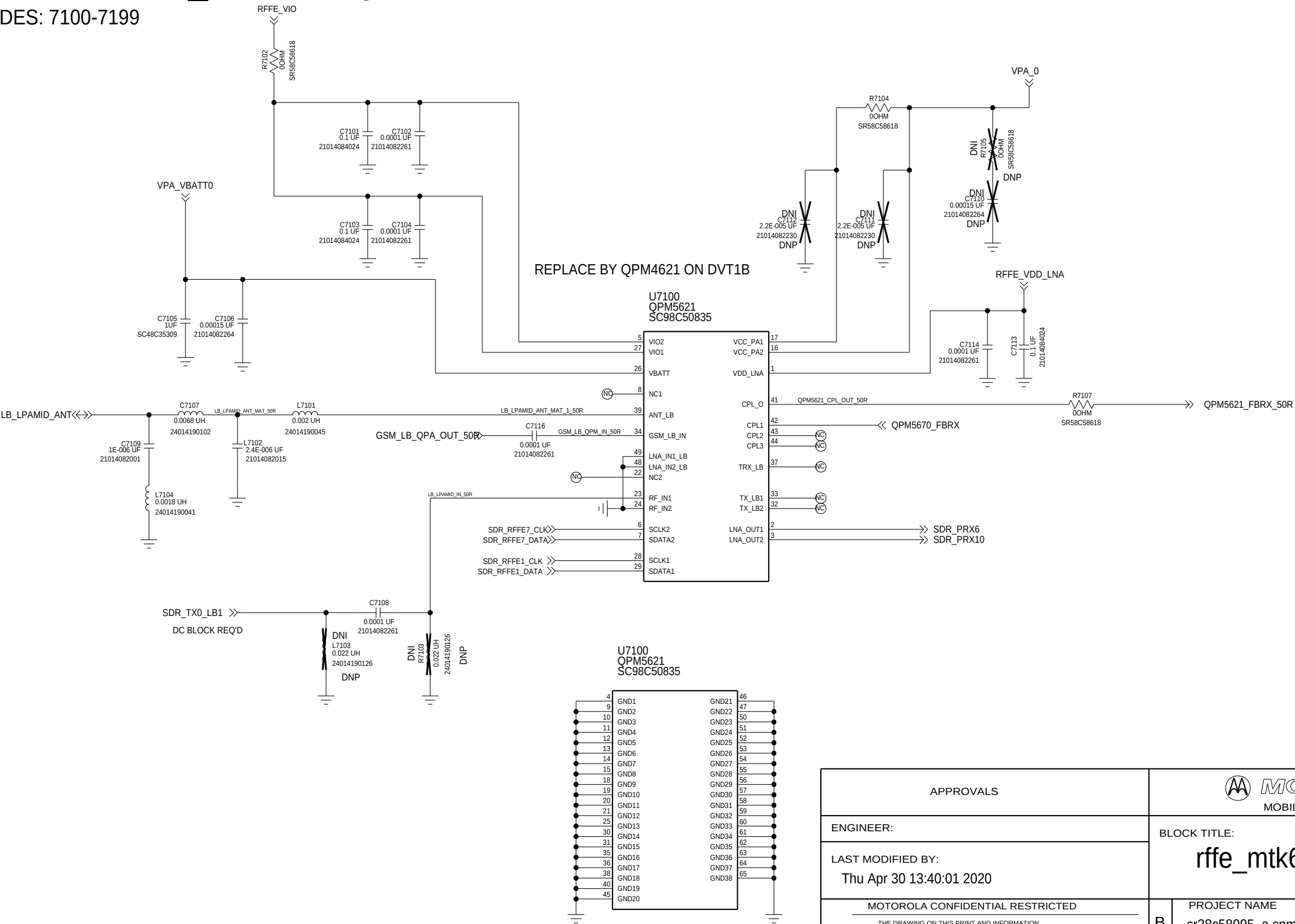
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Apr 30 13:39:59 2020		rffe_mtk6	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	
		B sr28c58995_a.cpm	
		REV 01	
		CAD FILE NAME:	
		PAGE 58 OF 84	

REFDES: 7025-7049

PAGE 59 OF 84

RFFE: TX0-LB_LPAMiD-QPM5621

REFDES: 7100-7199

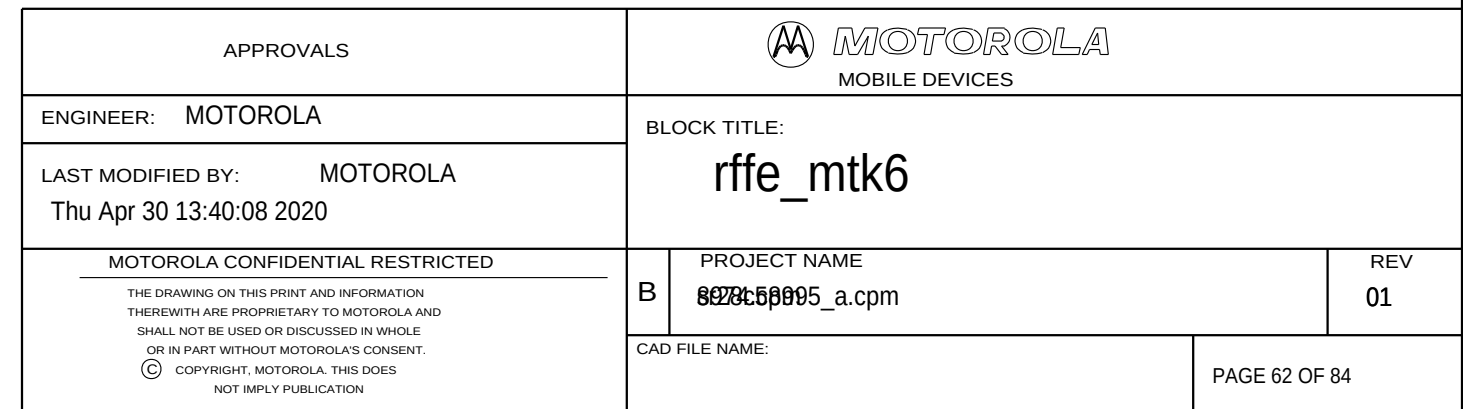


APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE: rffe_mtk6	
LAST MODIFIED BY: Thu Apr 30 13:40:01 2020		PROJECT NAME sr28c58995_a.cpm	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		REV 01	
		PAGE 60 OF 84	

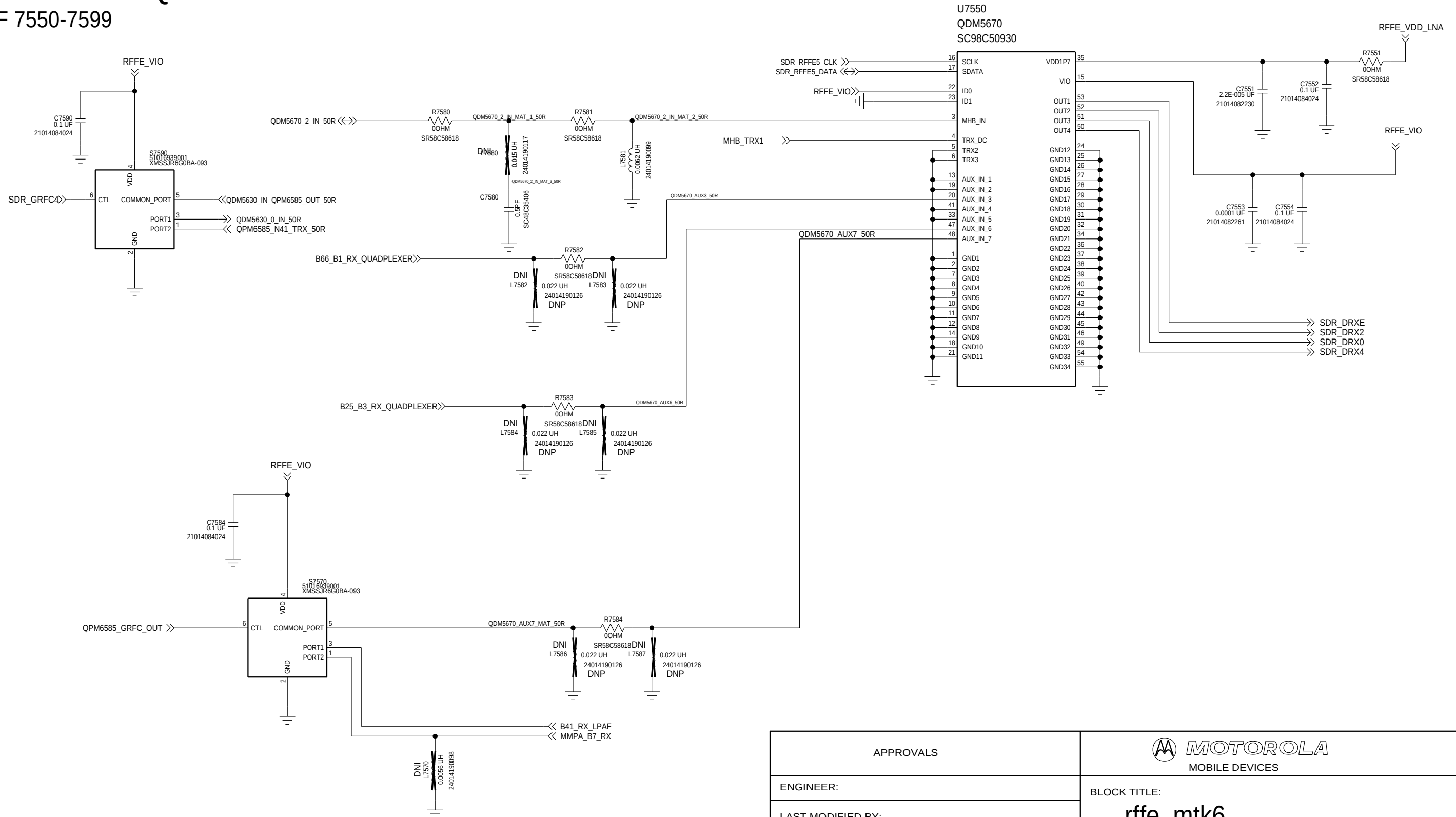
2

A

REF 7600-7699



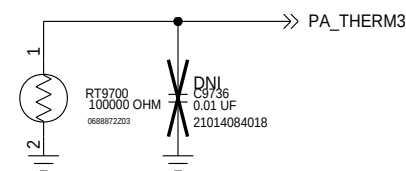
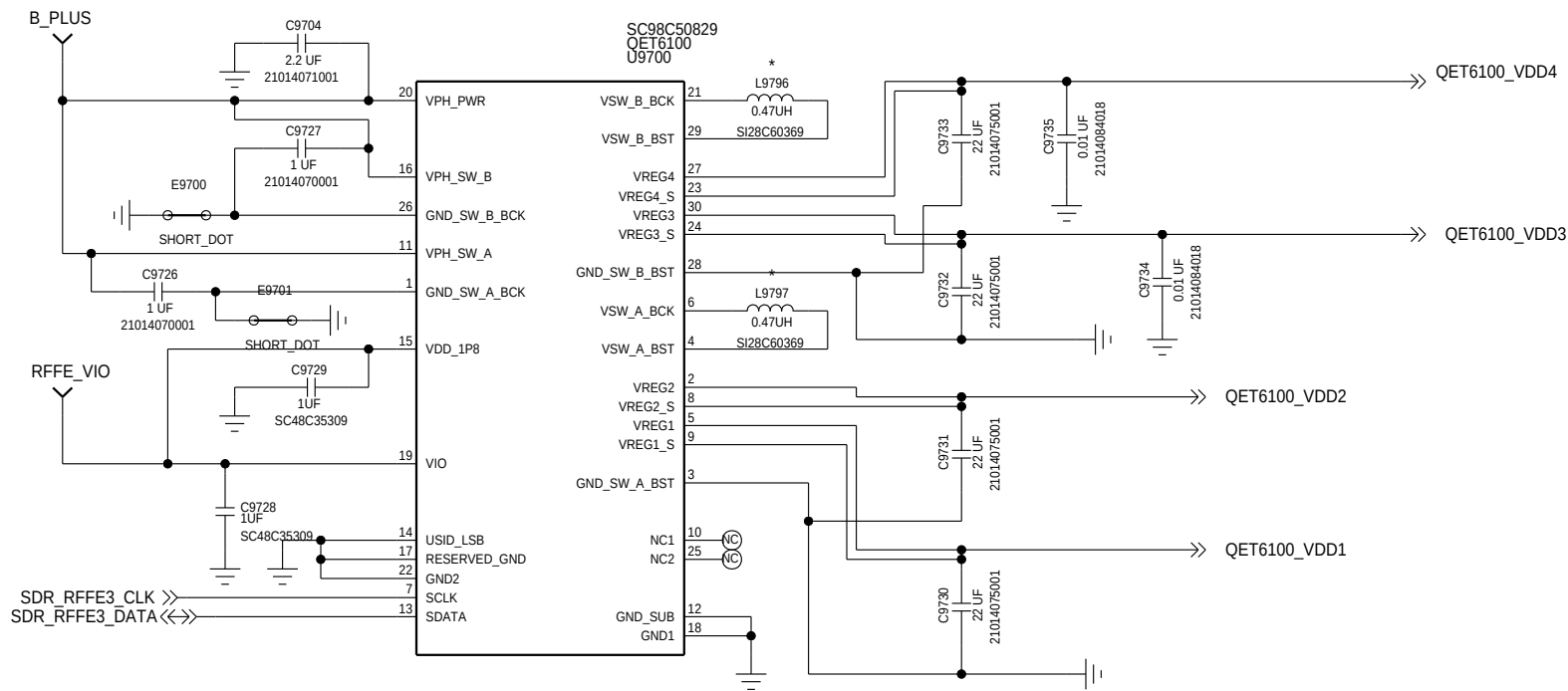
RFFE: MHB-QDM5670#2
REF 7550-7599



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:	BLOCK TITLE: rffe_mtk6		
LAST MODIFIED BY: Thu Apr 30 13:40:14 2020			
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME B sr28c58995_a.cpm	REV 01
		CAD FILE NAME:	PAGE 63 OF 84

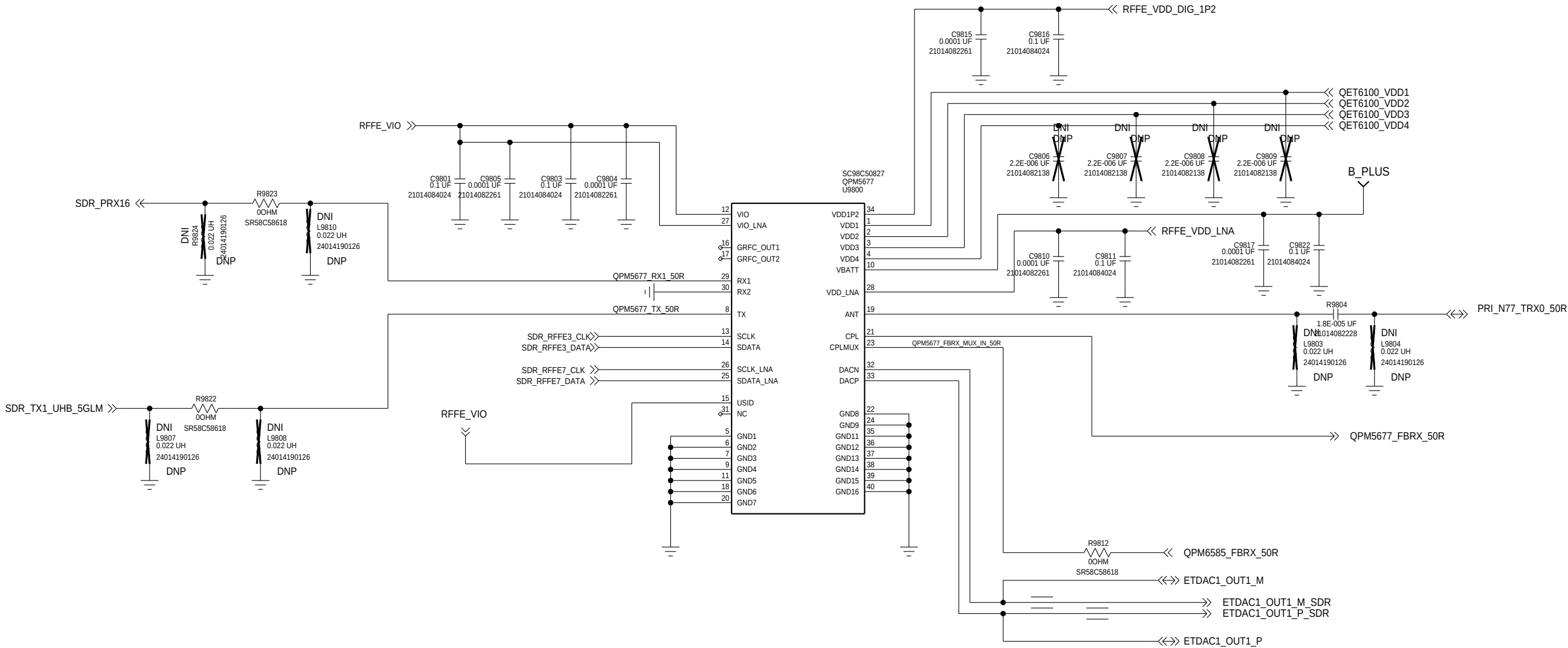
RFFE: SUB6-QET6100

REF 9700-9799



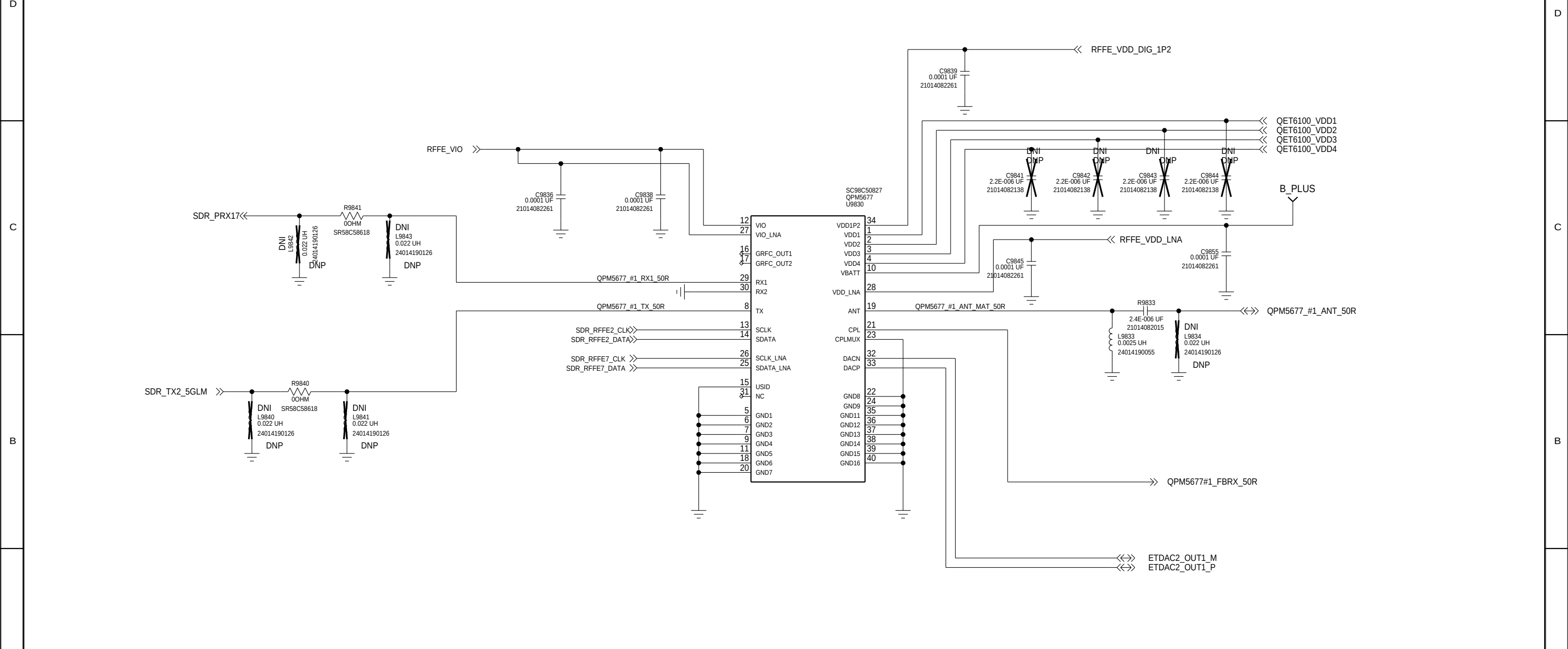
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Apr 30 13:40:17 2020		rffe_mtk6	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	REV
		B 8028150005_a.cpm	01
CAD FILE NAME:		PAGE 64 OF 84	

RFFE: SUB6-QPM5677_#0
REF 9800-9829



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Apr 30 13:40:19 2020		rffe_mtk6	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	
		B 8028-56095_a.cpm	
		CAD FILE NAME:	
		PAGE 65 OF 84	
REV 01			

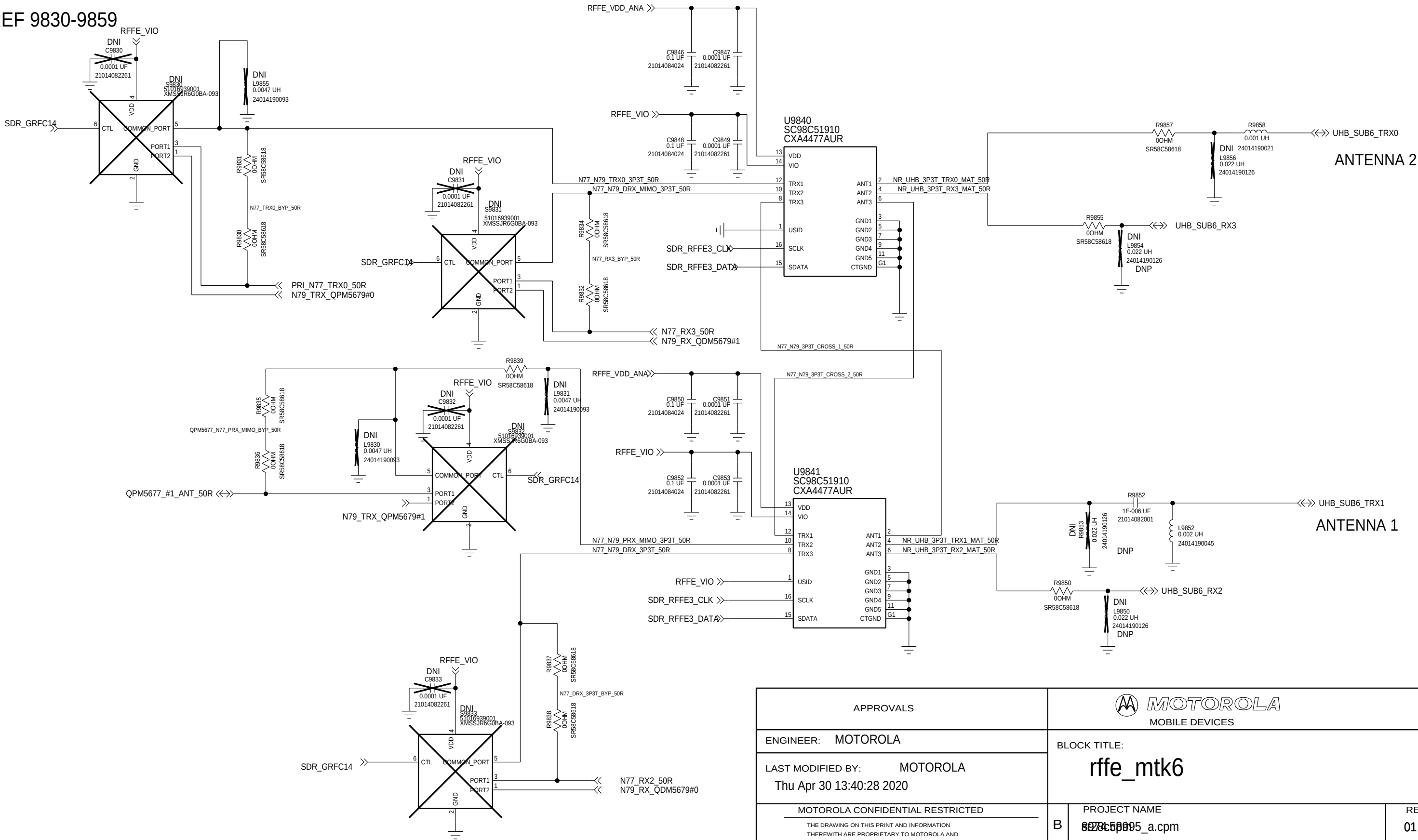
8	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---



A					APPROVALS		 MOTOROLA MOBILE DEVICES		A						
					ENGINEER: MOTOROLA		BLOCK TITLE: rffe_mtk6								
					LAST MODIFIED BY: MOTOROLA Thu Apr 30 13:40:42 2020										
					MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		B	PROJECT NAME sr28c58995_a.cpm		REV 01					
CAD FILE NAME:			PAGE 66 OF 84												
8		7		6		5		4		3		2		1	

RFFE: SUB6-SWITCH

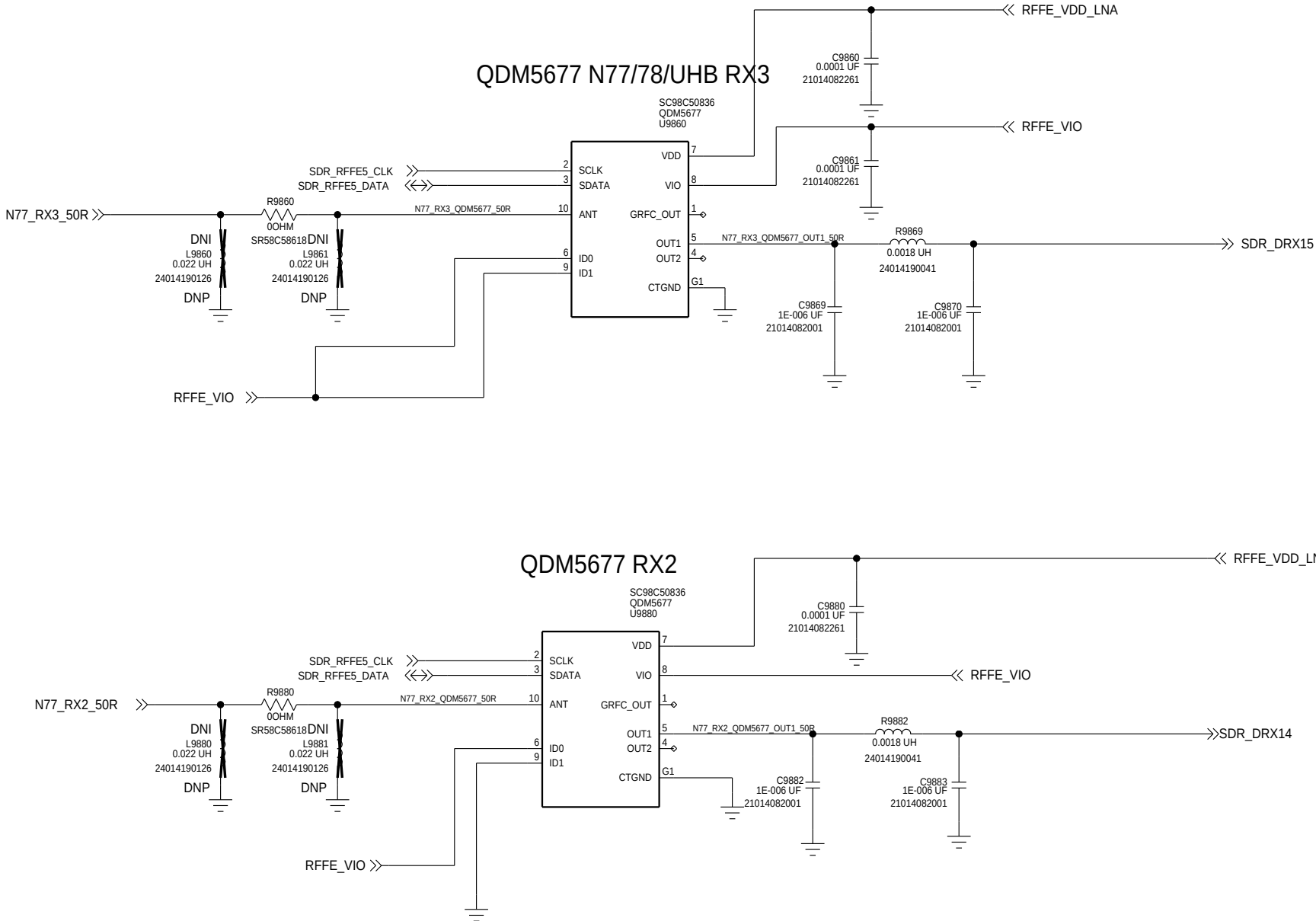
REF 9830-9859



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Apr 30 13:40:28 2020		rffe_mtk6	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	
		B 6278-58095_a.cpm	
		CAD FILE NAME:	
		PAGE 67 OF 84	
		REV 01	

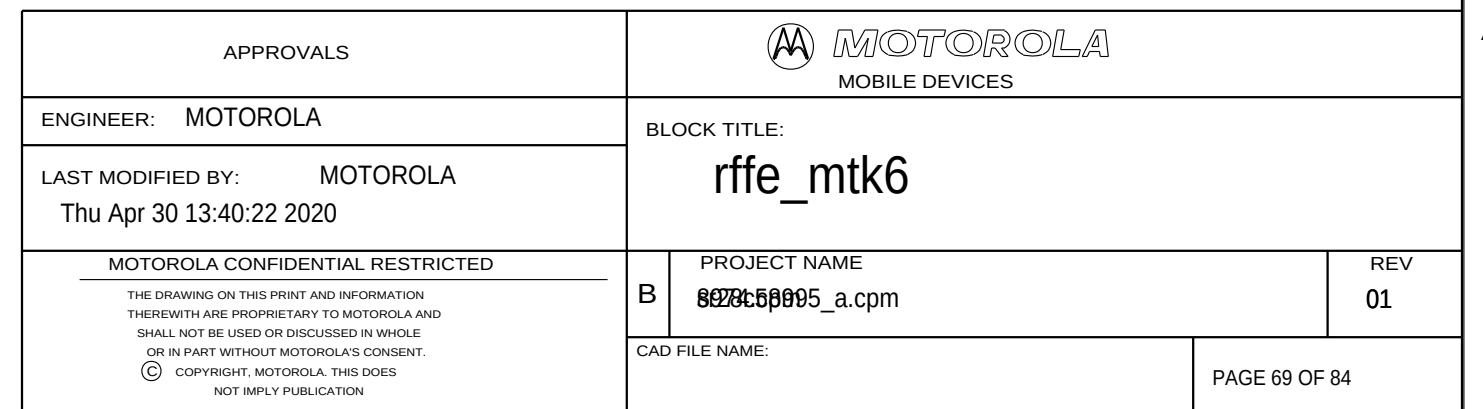
RFFE: SUB6-RX2/3-QDM5677

REF 9860-9899



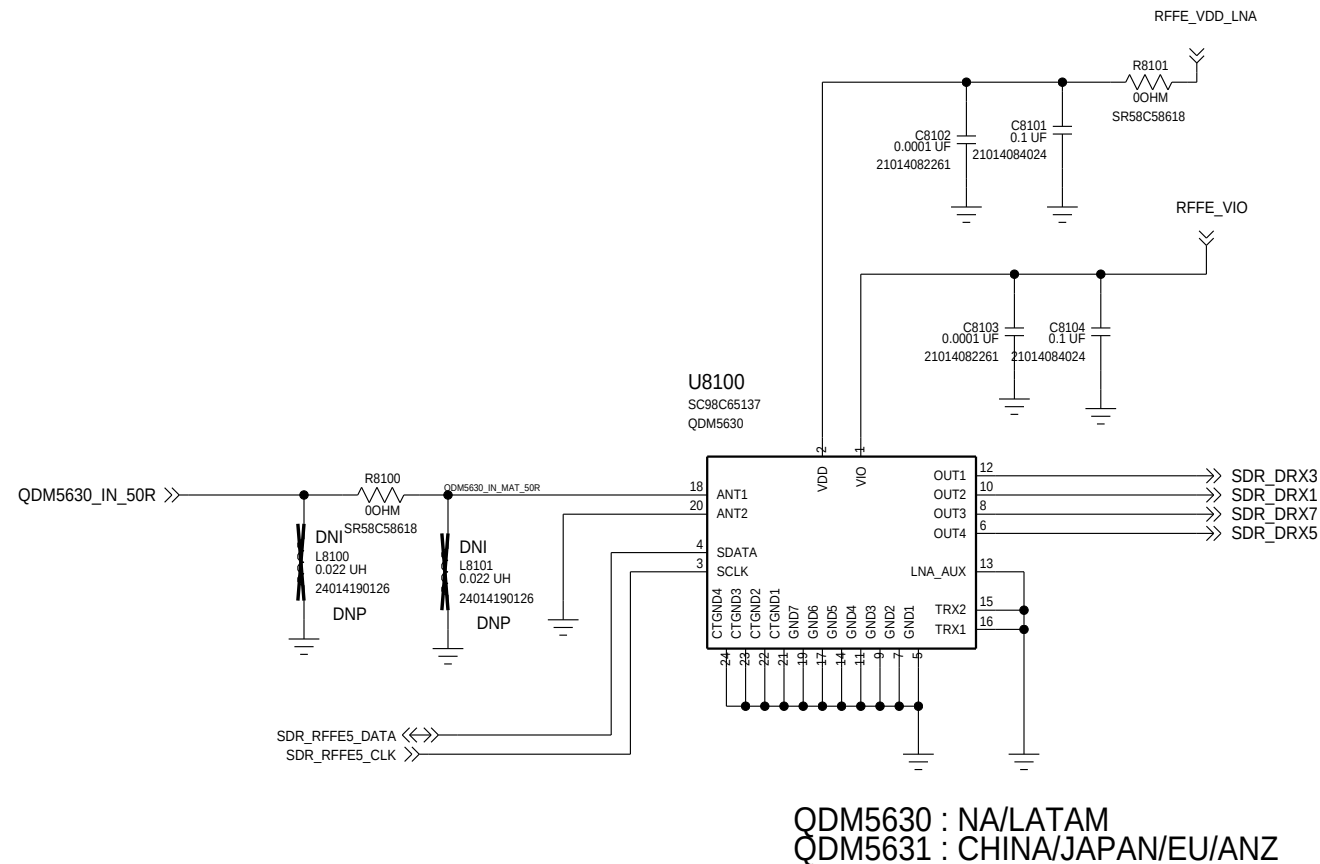
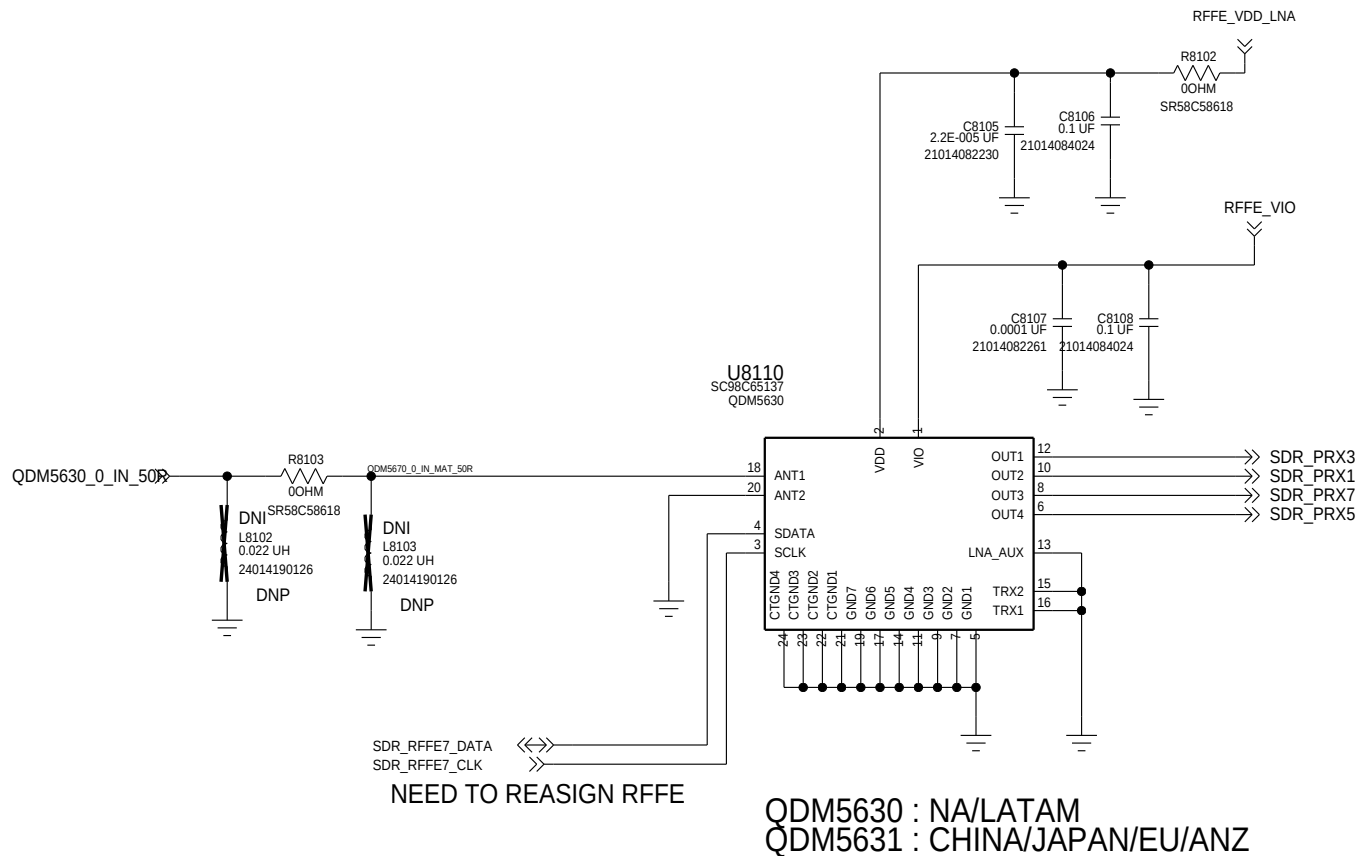
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: rffe_mtk6	
LAST MODIFIED BY: MOTOROLA Thu Apr 30 13:40:39 2020		PROJECT NAME 82315005_a.cpm	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		REV 01	
		CAD FILE NAME: 82315005_a.cpm	
		PAGE 68 OF 84	

REF 9900-9929

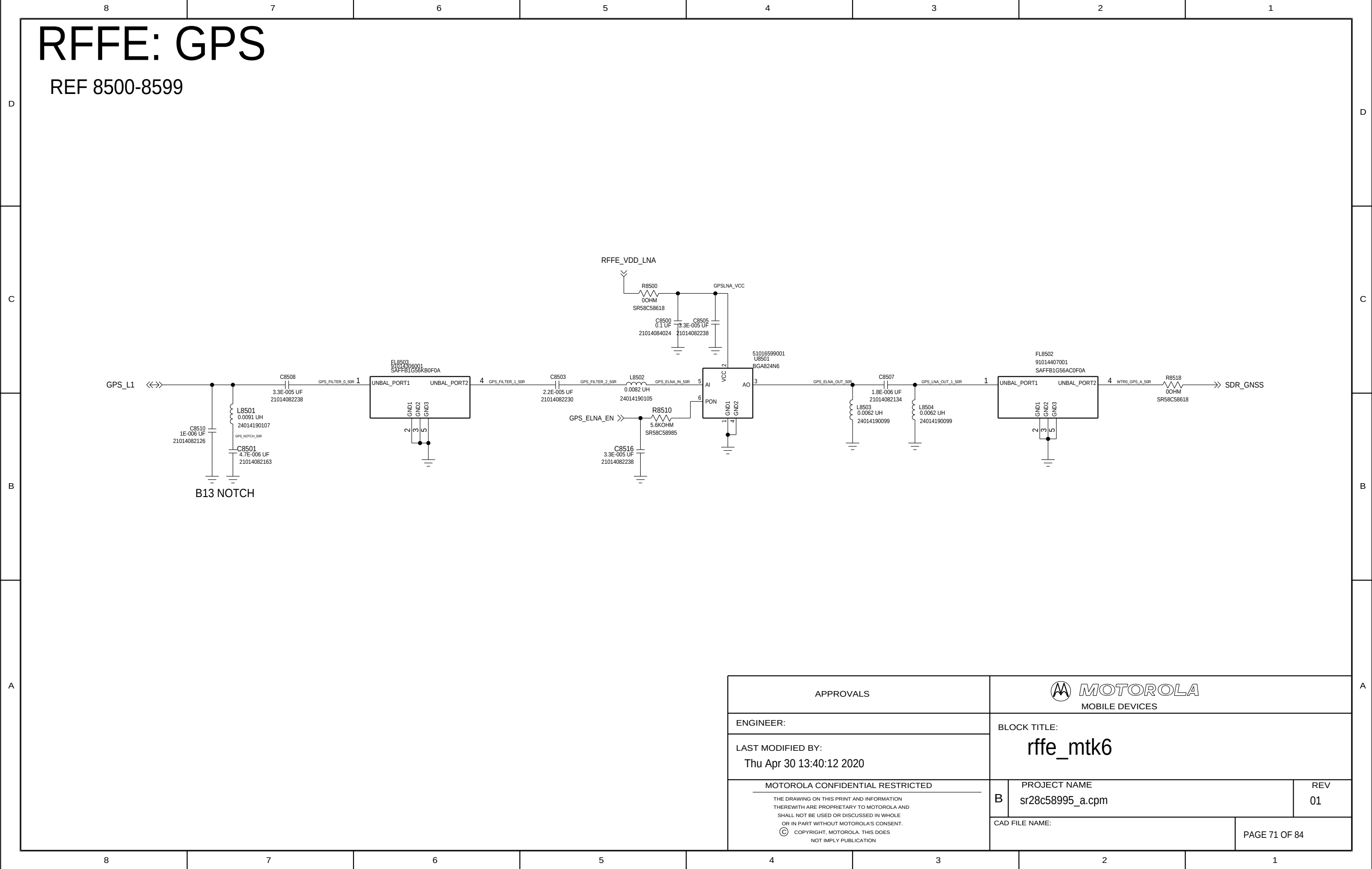


RFFE: SUB6-MIMO-QDM5630/5631

REF 8100-8199



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Apr 30 13:40:25 2020		rffe_mtk6	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	REV
		B 828158095_a.cpm	01
CAD FILE NAME:			PAGE 70 OF 84

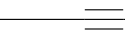
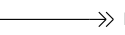
[illegible][illegible][illegible]

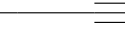
NA RF: PORTS

REFDES: 7000-8199

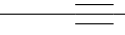
NA RF

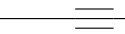
LAA_RX0_LNA_OUT     LAA_RX0_LNA_OUT

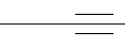
LAA_RX1_LNA_OUT     LAA_RX1_LNA_OUT

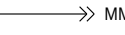
SDR_PRX12     SDR_PRX12

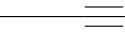
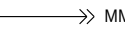
SDR_DRX10     SDR_DRX10

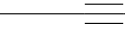
SDR_TX1_MLB     SDR_TX1_MLB

SDR_TX1_MB1     SDR_TX1_MB1

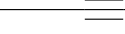
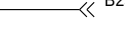
SDR_TX1_HB1     SDR_TX1_HB1

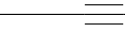
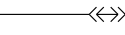
MMPA_B5_TX     MMPA_B5_TX

MMPA_B71_TX     MMPA_B71_TX

MMPA_B7_RX     MMPA_B7_RX

B66_B1_RX_QUADPLEXER     B66_B1_RX_QUADPLEXER

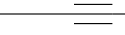
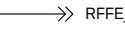
B25_B3_RX_QUADPLEXER     B25_B3_RX_QUADPLEXER

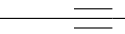
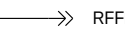
MHB_TRX1     MHB_TRX1

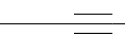
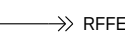
QPA5581_MHB_FBRX     QPA5581_MHB_FBRX

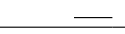
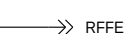
WLAN_PA_EN     WLAN_PA_EN

POWER

VREG_S4A_1P8     RFFE_VIO

VREG_L15A_1P7     RFFE_VDD_LNA

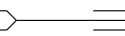
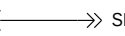
VREG_L15A_1P7     RFFE_VDD_ANA

VREG_L9A_1P2     RFFE_VDD_DIG_1P2

VREG_L10A_3P0     RFFE_VDD_SW_3P0

INTERFACES - GRFC

QPM6585_GRFC_OUT     QPM6585_GRFC_OUT

SDR_GRFC3     SDR_GRFC3

SDR_GRFC5     SDR_GRFC5

INTERFACES - RFFE MIPI

SDR_RFFE2<1..0>         SDR_RFFE2<1..0>

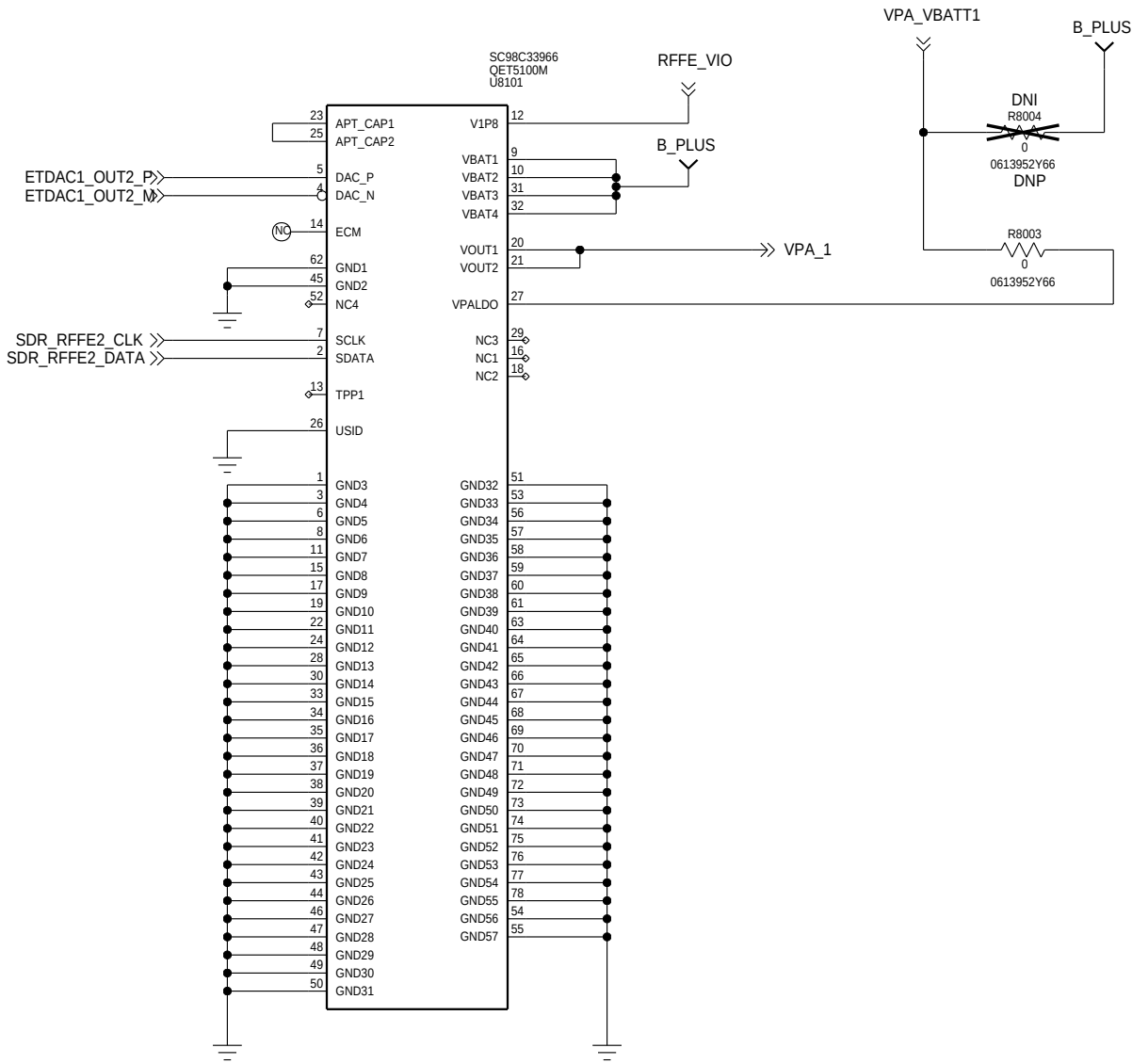
SDR_RFFE7<1..0>         SDR_RFFE7<1..0>

ET DAC

ETDAC1_OUT2 M                             

NA_RF: QET5100_CH1

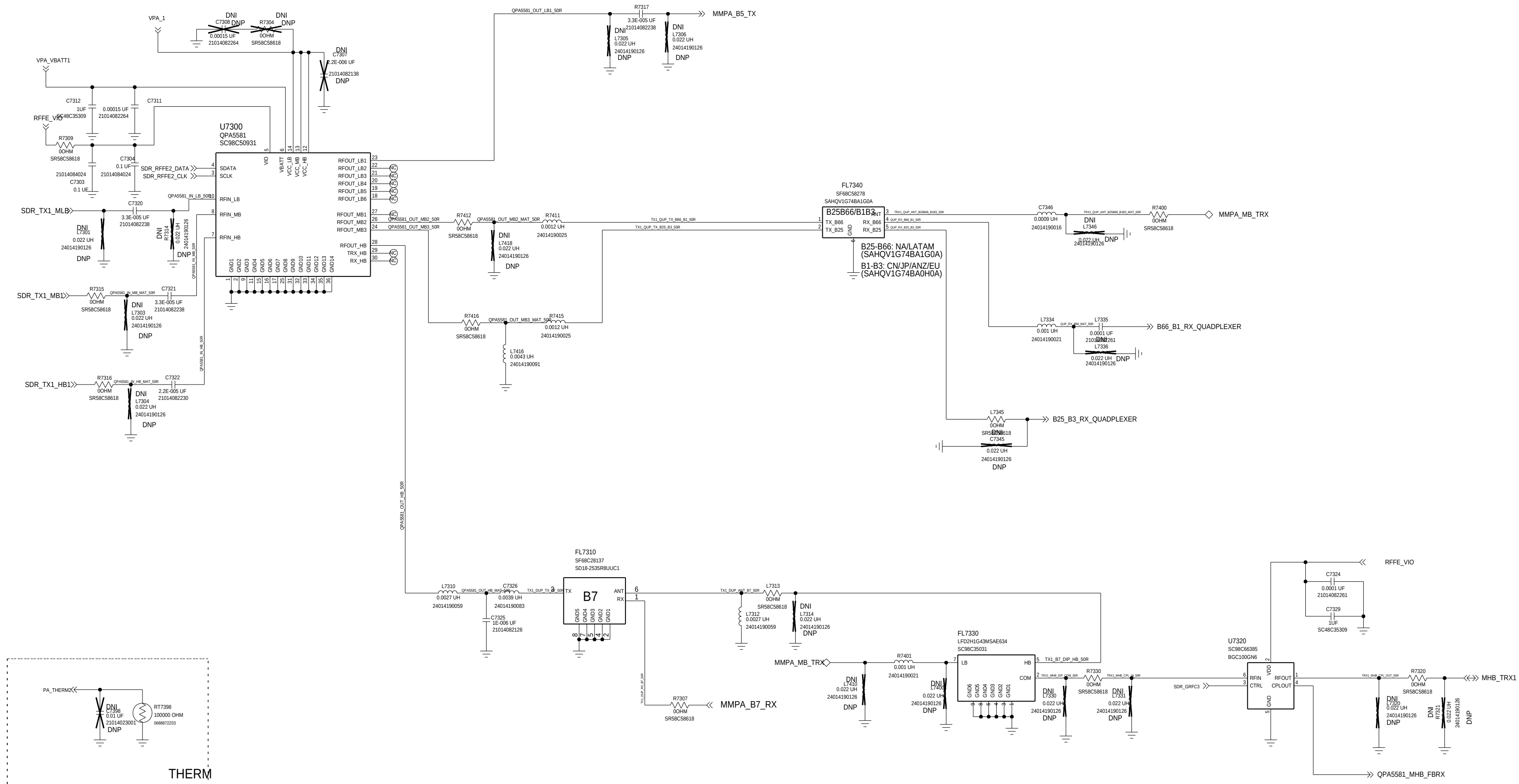
REFDES: 8100-8199



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: Thu Apr 30 13:40:53 2020		na_rf	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	
		B sr28c58995_a.cpm	
		REV 01	
		CAD FILE NAME:	
		PAGE 73 OF 84	

REFDES: 7300-7399

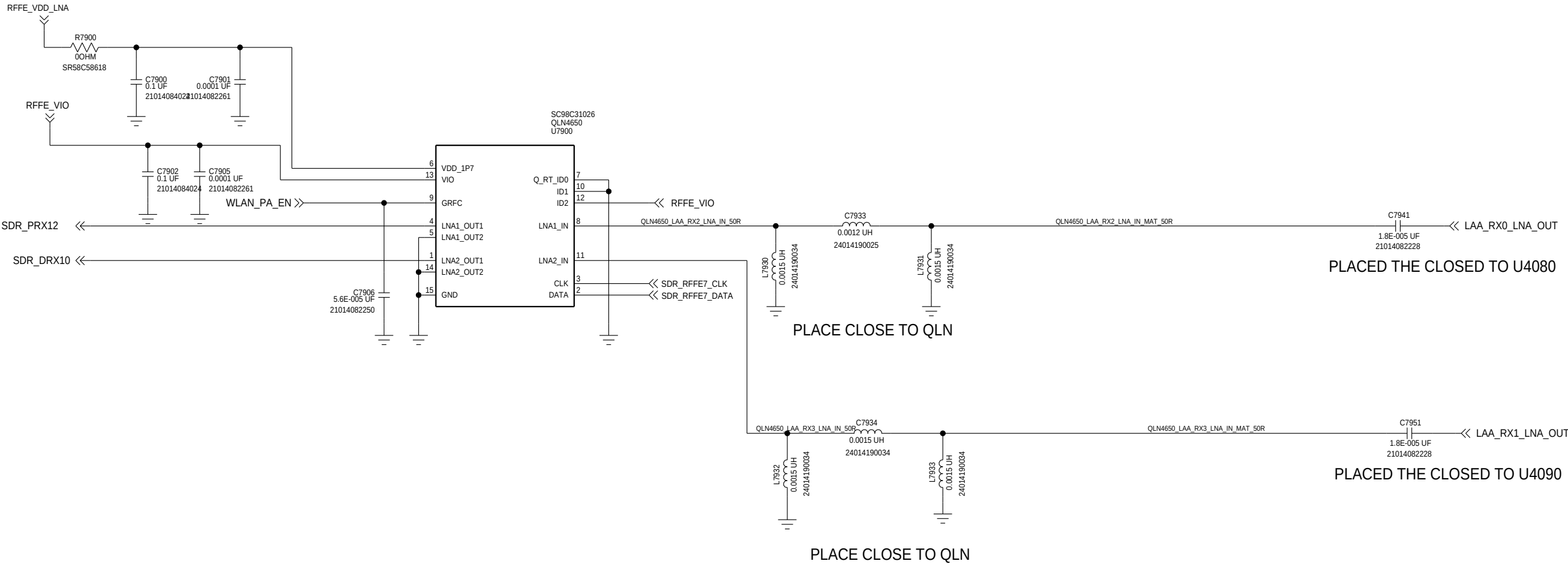
REVISIONS				
ZONE	LTR	DESCRIPTION	DATE	APPR.



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE: na_rf	
LAST MODIFIED BY: Thu Apr 30 13:40:50 2020			
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT.  COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		C PROJECT NAME sr28c58995_a.cpm	REV 01
		CAD FILE NAME:	PAGE 74 of 84

NA_RF: LAA LNA

REF 7900-7999



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: na_rf	
LAST MODIFIED BY: MOTOROLA Thu Apr 30 13:40:47 2020		PROJECT NAME na_rf	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		B	REV 01
		CAD FILE NAME: na_rf_50005_a.cpm	
		PAGE 75 OF 84	

ANTENNAS: PORTS

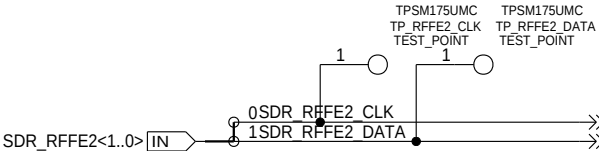
REF 000-499

POWER

VREG_S4A_1P8

TUNER_VIO

INTERFACES - RFFE MIPI



RFFE0_CLK IN RFFE0_CLK

RFFE0_DATA IN RFFE0_DATA

RFFE1_CLK IN RFFE1_CLK

RFFE1_DATA IN RFFE1_DATA

INTERFACES - GPIO

RF_CONN_DET1 OUT RF_CONN_DET1

RF_CONN_DET2 OUT RF_CONN_DET2

RF_CONN_DET3 OUT RF_CONN_DET3

RF_CONN_DET4 OUT RF_CONN_DET4

ANTENNA CONNECTORS

ANT1_LMH_TRX0_UHB_SUB6_TRX1 IO ANT1_LMH_TRX0_UHB_SUB6_TRX1

ANT2_LMH_TRX1_UHB_SUB6_TRX0 IO ANT2_LMH_TRX1_UHB_SUB6_TRX0

ANT3_WL_2G_1_LAA_RX1_5GHZ_WLAN_S1 IO ANT3_WL_2G_1_LAA_RX1_5GHZ_WLAN_S1

ANT4_WL_2G_0_LAA_RX0_5GHZ_WLAN_S0 IO ANT4_WL_2G_0_LAA_RX0_5GHZ_WLAN_S0

ANT5_RX2 IO ANT5_RX2

ANT6_RX3 IO ANT6_RX3

GPS_L1 IO GPS_L1

NFC

NFC_ANT1 IO NFC_ANT1

NFC_ANT2 IO NFC_ANT2

MORTABLE ONLY

ANT5_COAX IO ANT5_COAX

ANT6_COAX IO ANT6_COAX

CAPSENSE

ANT_CSNS_CS2_N IN ANT_CSNS_CS2_N

ANT_CSNS_CS3_N IN ANT_CSNS_CS3_N

REF_CSNS_CS0_N IN REF_CSNS_CS0_N

ANT_CSNS_CS4_N IN ANT_CSNS_CS4_N

ANT_CSNS_CS0_N IN ANT_CSNS_CS0_N

FLIP DETECT

LED_BLUE_SRC IN LED_BLUE_SRC

LED_GREEN_SRC IN LED_GREEN_SRC

LED_SOL_SRC IN LED_SOL_SRC

INTERFACES - FPS

FP_DRDY_N OUT

FP_RST_N IN

FP_SPI_CLK IN

FP_SPI_CS_N IN

FP_SPI_MOSI IN

FP_SPI_MISO OUT

APPROVALS

ENGINEER:

LAST MODIFIED BY:

Thu Apr 30 13:41:10 2020

MOTOROLA CONFIDENTIAL RESTRICTED

THE DRAWING ON THIS PRINT AND INFORMATION
THEREWITH ARE PROPRIETARY TO MOTOROLA AND
SHALL NOT BE USED OR DISCUSSED IN WHOLE
OR IN PART WITHOUT MOTOROLA'S CONSENT.
© COPYRIGHT, MOTOROLA. THIS DOES
NOT IMPLY PUBLICATION



MOTOROLA

MOBILE DEVICES

BLOCK TITLE:

antennas

PROJECT NAME

sr28c58995_a.cpm

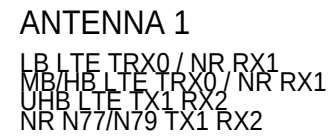
REV

01

CAD FILE NAME:

PAGE 76 OF 84

REF 000-199



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:	BLOCK TITLE: antennas		
LAST MODIFIED BY: Thu Apr 30 13:41:04 2020			
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME B sr28c58995_a.cpm	REV 01
		CAD FILE NAME:	PAGE 77 OF 84

8	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---

D



B

A

5

(

3 |

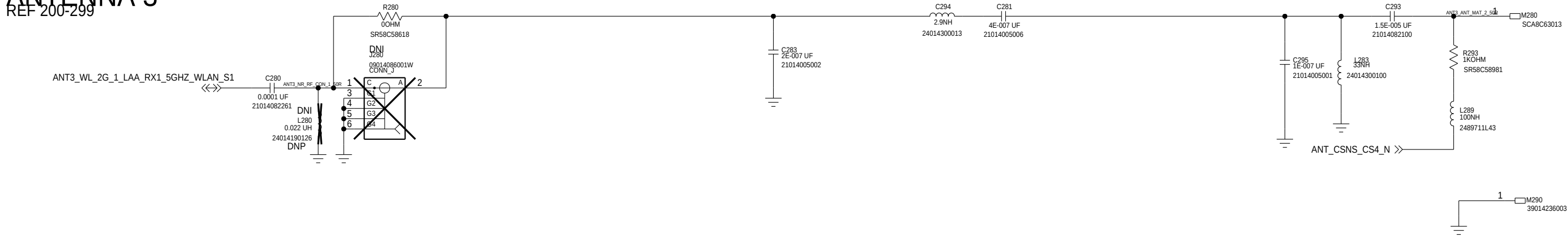
A |

8	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---

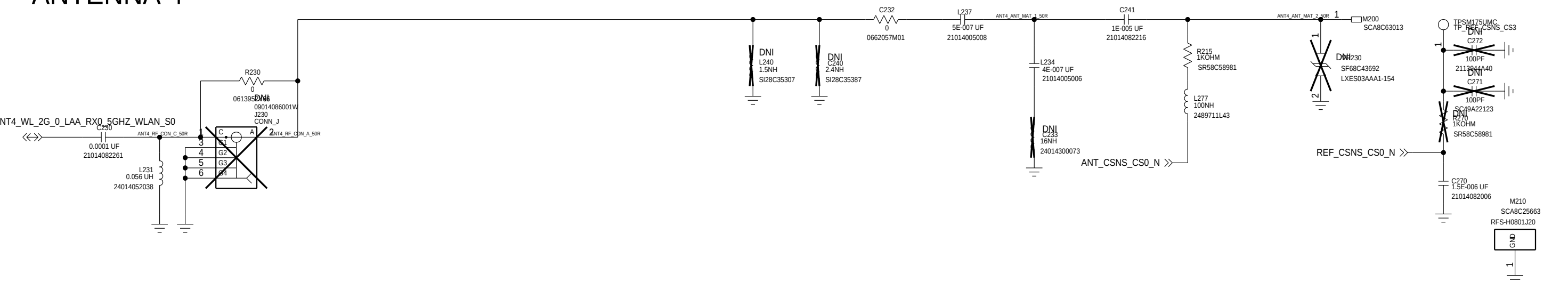
ANTENNAS: ANT3, ANT4

ANTENNA 3

REF 200-299



ANTENNA 4

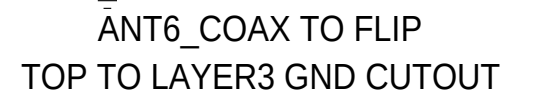


APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: Mon Jun 15 11:03:05 2020		antennas	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME B sr28c58995_a.cpm	REV 01
		CAD FILE NAME:	
		PAGE 79 OF 84	

REF 300-499



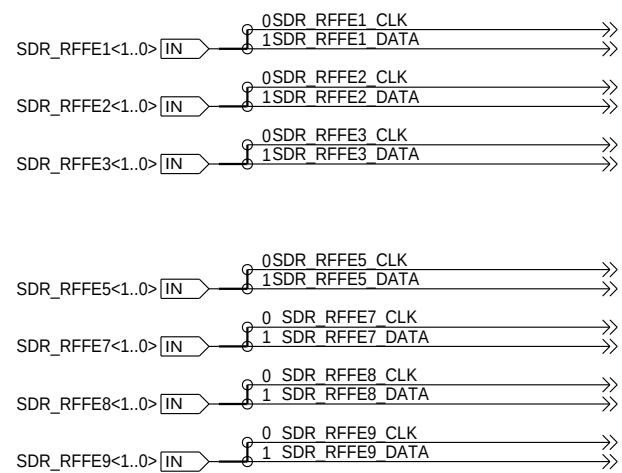
09014246001W 22

A

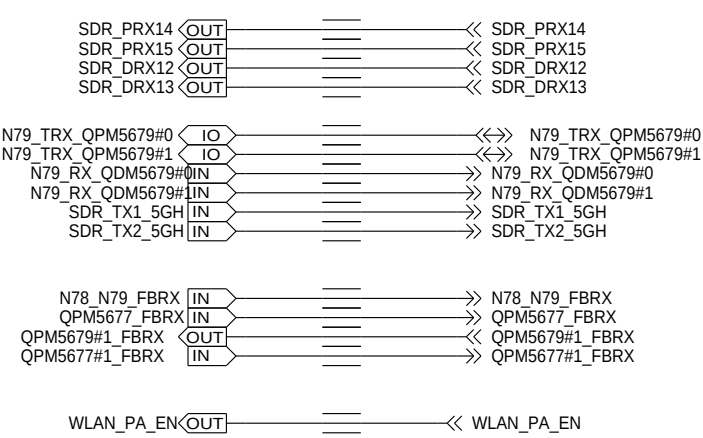
CHINA RF: PORTS

REFDES: 7000-8199

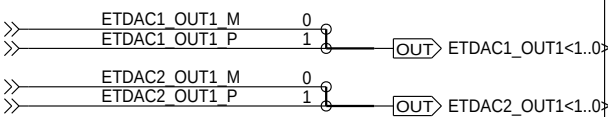
INTERFACES - RFFE MIPI



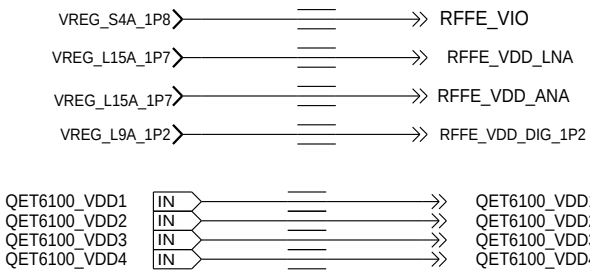
CHINA - RF



ET DAC



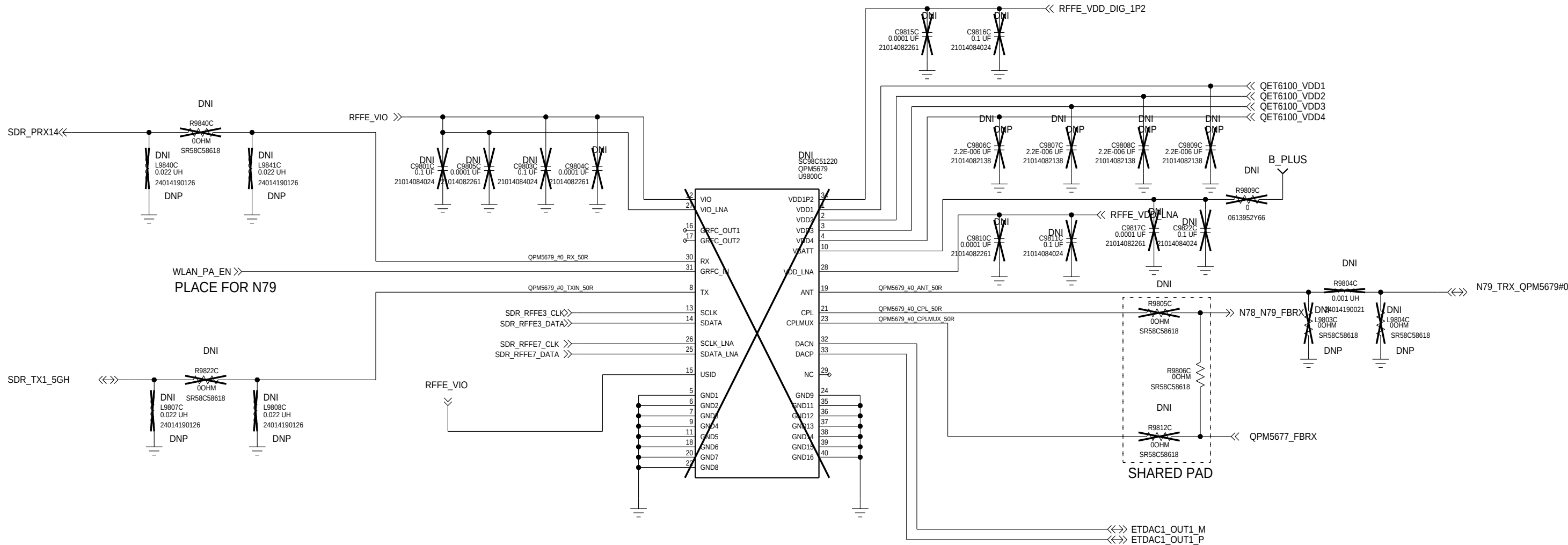
POWER



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Wed Apr 15 10:38:43 2020		china_rf	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	
		B sr28c58995_a.cpm	
		REV 01	
		CAD FILE NAME:	
		PAGE 81 OF 84	

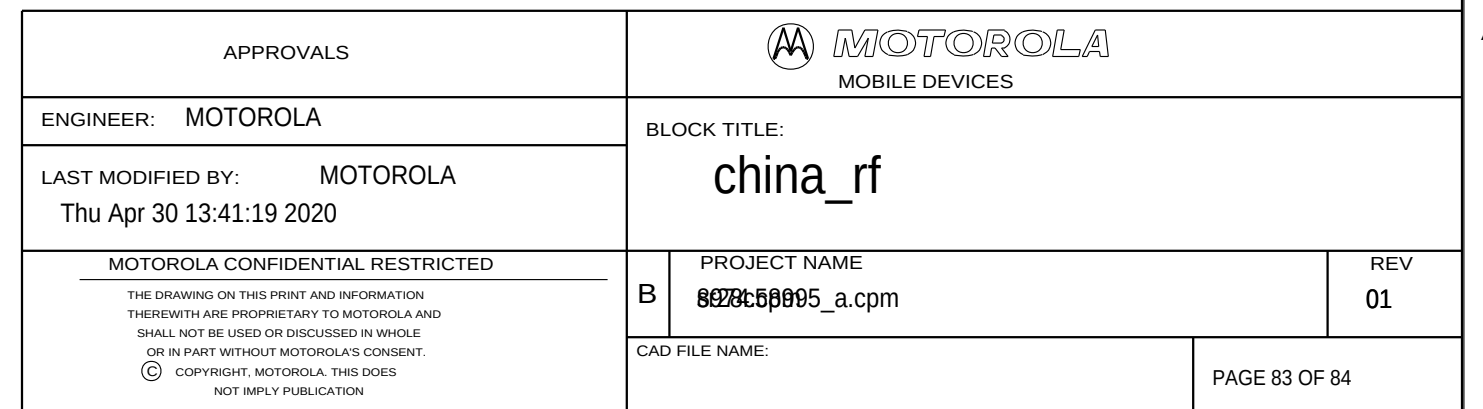
CHINA_RF: SUB6-QPM5679#0

REF 9830-9859

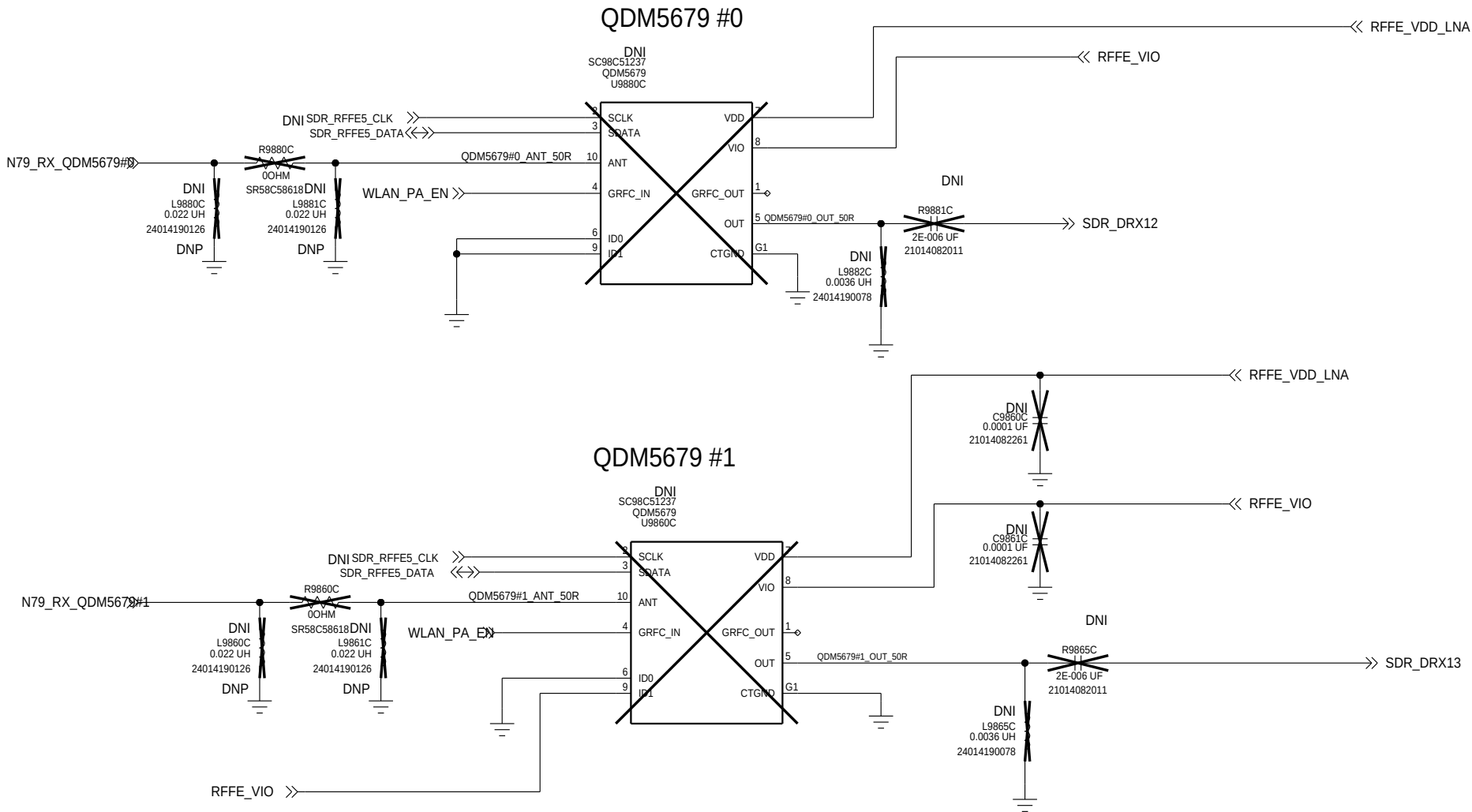


APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Apr 30 13:41:16 2020		china_rf	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	REV
		B 8028-56095_a.cpm	01
CAD FILE NAME:			PAGE 82 OF 84

REF 9830-9859



CHINA_RF: SUB6-QDM5679#0/#1



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Thu Apr 30 13:41:14 2020		china_rf	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	
		B 8020-5679-005_a.cpm	
		CAD FILE NAME:	
		PAGE 84 OF 84	
REV 01			